TRIBHUVAN UNIVERSITY

ASIAN SCHOOL OF MANAGEMENT & TECHNOLOGY

Final Year Project Report

On

"Futsal Bookings and Management Platform"

Submitted To :

Office of Dean

Institute of Science and Technology

Under the supervision of

Mr. Surya Bam

In partial fulfillment of the requirement for the bachelor's degree in Computer Science and

Information Technology

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Shrawan, 2083

SUPERVISOR’S RECOMMENDATION

It is my pleasure to recommend that a report on “Futsal Bookings and Management Platform” has been prepared under my supervision by Aashish Rijal, Dipesh Devkota, and Kedar Dhungana in partial fulfillment of the requirement of the degree of Bachelor of Science in Computer Science and Information Technology (BSc.CSIT). Their report is satisfactory to process for the future evaluation.

…………………………………

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CERTIFICATE OF APPROVAL

This is to certify that this report has been read and recommended to the Department of Computer Science and Information Technology for acceptance of report entitled "Futsal Bookings and Management Platform" submitted by Aashish Rijal, Dipesh Devkota and Kedar Dhungana in partial fulfillment for the degree of Bachelor of Science in Computer Science and Information Technology (BSc.CSIT), Institute of Science and Technology, Tribhuvan University.

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ACKNOWLEDGEMENT

We would like to express our honest gratitude to all those who contributed to this project. Our sincere thanks go to our supervisor and coordinator **Mr. Surya Bam**, who provided us with precious guidance and support throughout the project. Their invaluable expertise, encouragement, and support have been a constant source of inspiration. We would also like to thank Asian School of management and technology College for providing us with the resources and support needed to complete this project. Finally, we would like to extend our gratitude to our families and friends for their support and encouragement throughout the completion of the project.

With respect,

Aashish Rijal

Dipesh Devkota

Kedar Dhungana

ABSTRACT

The Futsal Booking and Management System is a web-based multi-tenant SaaS platform designed to digitize and streamline futsal court booking and management in Nepal. Addressing the widespread reliance on manual booking methods, such as phone calls and handwritten registers the system provides a centralized platform where users can search, book, and review futsal centers, while owners manage schedules, pricing, and analytics through a dedicated dashboard. Built using React, Node.js, and MySQL, the platform incorporates machine learning features including Logistic Regression for sentiment analysis, and the Haversine formula for location-based court recommendations. Developed following the Agile methodology, the system aims to eliminate double bookings, improve operational efficiency, and deliver an intelligent, scalable solution for futsal facility management.

Keywords: SaaS, Multi-tenant, Sentiment Analysis, Node Js, React Js, Centralized, Logistic Regression, Agile.

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LIST OF ABBREVIATIONS

API	Application Programming Interface
AWS	Amazon Web Services
DB	Database
HLD	High Level Design
JWT	JSON Web Token
ML	Machine Learning
MySQL	My Structured Query Language
NLP	Natural Language Processing
Oauth	Open Authorization
OTP	One Time Password
PWA	Progressive Web Application
Redis	Remote Dictionary Server
SaaS	Software as a Service
TF-IDF	Term Frequency - Inverse Document Frequency
URL	Uniform Resource Locator
VPS	Virtual Private Server

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CHAPTER 1
INTRODUCTION

1.1 Introduction

The Futsal bookings and management platform is an innovative web-based application designed to facilitate real-time futsal court booking and management, with a primary focus on digitizing the operations of futsal centers across Nepal. It allows users to discover and connect with futsal facilities across different locations, creating opportunities for seamless court reservations, live slot availability checks, and community engagement through ratings and reviews. The platform is designed to be intuitive and user-friendly, enabling players to easily search for courts, book sessions, and manage their reservations through a centralized digital interface.

For players, this platform offers features such as AI-powered Sentiment Analysis of reviews , personalized court recommendations based on location, and booking history tracking, allowing users to monitor their activity and make informed decisions. The platform also supports social interaction through community ratings, reviews, and a discussion forum, fostering a collaborative and trustworthy sports community.

From the administrative perspective, Futsal Bookings and management system provides a robust backend system for futsal owners to manage courts, time slots, pricing, and customer interactions. Administrators can oversee platform-wide activity, monitor tenant usage, and ensure smooth operation of the system. The system leverages modern technologies such as the MERN stack, Redis, and Docker to provide reliable, scalable, and secure booking capabilities within a multi-tenant SaaS architecture.

One of the key advantages of Futsal bookings and management platform is its emphasis on intelligent features and data-driven decision support, allowing futsal owners to access analytics dashboards with revenue trends, peak usage times, and customer engagement insights.

Moreover, Futsal Bookings and Management is developed with future scalability in mind, enabling integration with additional features such as mobile application support, online payment gateways, tournament management, and expanded AI capabilities for enhanced user experience.

1.2 Problem Statement

The growing demand for futsal facilities in urban areas of Nepal has highlighted several challenges for both players and facility operators. Traditional booking methods, such as phone calls, handwritten registers, and social media messages, often fail to provide real-time availability updates and lead to frequent double bookings and scheduling conflicts.

Players frequently struggle to find available courts at convenient times, and opportunities to compare multiple futsal centers based on location, pricing, and user reviews are extremely limited. The absence of a centralized digital platform makes the process of discovering and booking a suitable futsal court time-consuming and inefficient.

From the facility management perspective, futsal owners face challenges in tracking bookings, managing schedules, and maintaining accurate records of revenue and customer interactions. The lack of digital tools also limits their ability to make data-driven decisions, as insights such as peak usage times and revenue trends are not readily available.

1.3 Objectives

- To develop a centralized online platform for futsal discovery and real-time court booking, ensuring convenience and accessibility for users.
- To implement a multi-tenant SaaS architecture that allows multiple futsal facilities to operate independently within a single scalable and secure system.
- To enable community features including verified ratings, reviews, and a discussion forum to enhance user interaction and trust.

1.4 Scope and Limitation

The scope of the Futsal Booking and Management System are as follows:

- Enabling users to search futsal centers by location, availability, and pricing.
- Providing real-time court booking and slot conflict prevention.
- Managing reservation history, cancellations, and booking confirmations.
- Allowing futsal owners to manage courts, time slots, pricing, and customer interactions through a dedicated dashboard.
- Providing an admin panel for platform-wide operations, tenant management, and subscription plans.
- Incorporating community features such as ratings, reviews, and discussion forums.
- Integrating a location-based recommendation engine to suggest nearby futsal centers based on user proximity and preferences.

Every system has errors so this system also has limitations which are as follows:

- Dependence on internet connectivity, the system may not function properly without a stable internet connection.
- Mobile application support is limited in the initial phase, as the platform is primarily designed for desktop and web browsers.
- AI-powered features are based on available training datasets and may not capture every nuance of user sentiment accurately.
- The system's effectiveness depends on futsal owners actively maintaining up-to-date court information and availability schedules.
- Requires user training for effective usage of owner dashboards and admin features.
- Currently limited to futsal facility management; support for other sports may be added in future updates.

1.5 Development Methodology

The Futsal bookings and management platform is developed using the **Agile Development Methodology**, emphasizing flexibility, collaboration, and iterative progress. The development process is organized into sprints, where each sprint focuses on delivering key functionalities while ensuring a user-centric design.

Iteration 1: The first phase establishes core infrastructure, including environment setup, database schema design, and wireframe creation. A minimum viable product (MVP) is developed covering user registration, login, and basic futsal listing.

Iteration 2: This iteration focuses on the core booking engine, including slot availability checks, conflict prevention algorithms, booking confirmation, and cancellation management.

Iteration 3: The third iteration integrates the futsal owner dashboard with court management, pricing configuration, schedule management, and booking approval workflows.

Iteration 4: In this phase, community features including ratings, reviews, and discussion forums are implemented alongside notification systems and user profile management.

Iteration 5: The final iteration introduces advanced AI and ML components including sentiment analysis and location-based recommendations. The system is fully tested for performance, security, and scalability.

By following this iterative Agile methodology, futsal management and bookings platform ensures continuous improvement, rapid delivery of features, and adaptability to feedback throughout the development process.

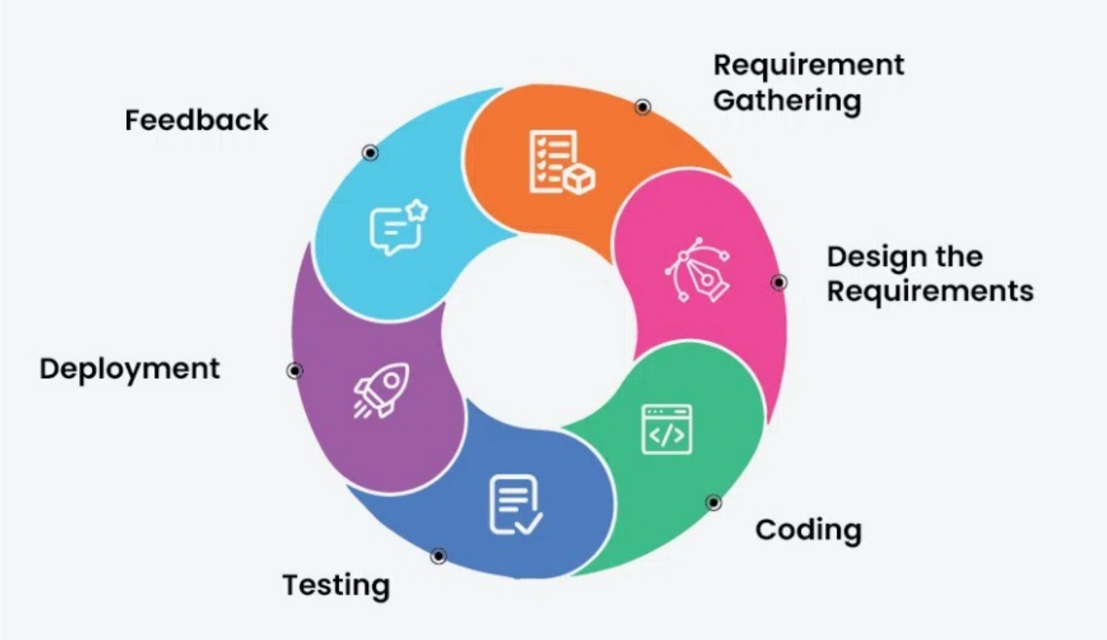


Figure 1.1 Agile Development model

1.6 Report Organization

The Report Organization describes how the study is structured and conducted. This report is divided into six chapters, each addressing different aspects of the study. Chapter One presents the introduction, statement of the problem, objectives of the study, scope and limitations, development methodology and organization of the study. Chapter Two focuses on the background study and review of related literature, providing the necessary theoretical foundation. Chapter Three discusses system analysis, including requirement analysis, feasibility study, data modeling, and process modeling of the system. Chapter Four deals with the design of the system, particularly the system architecture and its components. Chapter Five explains the implementation phase, the algorithms used, and the tools applied, along with the testing of the system both as individual units and as a 3complete system. Finally, Chapter Six provides the conclusions drawn from the study, highlights its limitations and offers recommendations for future improvements.

CHAPTER 2

BACKGROUND STUDY AND LITERATURE REVIEW

2.1 Background Study

The need for an Online Futsal Booking and Management System arises from the limitations of traditional manual booking methods such as phone calls, handwritten registers, and social media messages. These older approaches often cause double bookings, scheduling conflicts, missing records, and difficulties in tracking court availability and customer interactions. As the number of futsal centers in Nepal continues to grow rapidly, managing bookings manually becomes increasingly inefficient and unreliable.

A digital futsal booking platform addresses these issues by automating the entire reservation process. Players can search for available courts online, make bookings in real time, and receive instant confirmation, while the system prevents slot conflicts through automated time-slot validation. Futsal owners benefit from dedicated dashboards that streamline court management, pricing configuration, and customer communication, reducing administrative workload and minimizing human error.

Additionally, the platform integrates machine learning components to enhance user experience through sentiment analysis of reviews and location-based court recommendations. Overall, the Futsal system enhances booking efficiency, data accuracy, and operational transparency for both players and futsal facility operators.

2.2 Literature Review

The Futsal Booking and Management System is a web-based SaaS application designed to automate the process of discovering, booking, and managing futsal courts. It enables players to search for futsal centers by location and availability, make real-time reservations, and manage their booking history, while futsal owners can manage courts, schedules, pricing, and customer interactions through a dedicated management dashboard. The system maintains a centralized multi-tenant database that stores each facility's booking records, customer data, and operational details, ensuring that all information remains accurate, up-to-date, and securely isolated per tenant [1].

Traditionally, futsal centers managed bookings through phone calls, paper registers, and messaging apps, which were inefficient, time-consuming, and prone to double bookings, data loss, and poor record management [2]. With advancements in web technologies and digital platforms, automated booking systems have replaced manual processes, improving reliability, scheduling clarity, and operational transparency across the sports facility industry [3]. A study by Suhaimi and Sahid developed a futsal court booking system to replace manual records with a database-driven platform, demonstrating measurable improvements in booking visibility and electronic data management through a structured booking interface built using the prototyping model [1]. Similarly, Fauzi et al. presented a web-based futsal field rental information system aimed at improving business opportunities and operational efficiency in the sports rental sector, highlighting that website-based booking platforms help replace handwritten rental processes while increasing accessibility and customer reach [2].

Industry analysis further supports the effectiveness of specialized sports booking software, reporting measurable reductions in scheduling conflicts and improved operational transparency when centralized booking systems are deployed [3]. In addition, mobile-first and Progressive Web Application (PWA) approaches are recommended for booking platforms because they provide a responsive, app-like user experience across devices without the high cost of native mobile app development [4]. Modern system design literature recommends multi-tenant SaaS architecture for shared booking platforms serving multiple independent facility owners with proper data isolation, ensuring scalability and security across all tenants [5].

Machine learning techniques are increasingly applied to process user reviews and support decision-making in digital platforms. Term Frequency-Inverse Document Frequency (TF-IDF) is a widely used text feature extraction technique that identifies the most important terms in a document relative to a collection of documents, enabling automatic review summarization and smarter content ranking [6]. Recent studies utilize Logistic Regression as a robust supervised learning algorithm for sentiment analysis across various domains, classifying user feedback into binary outcomes such as positive or negative sentiment based on extracted TF-IDF feature vectors [7]. Furthermore, location-based recommendation methods using coordinate-based distance computation such as the Haversine formula are used to rank nearby services based on user proximity and preference patterns, improving the relevance of court suggestions presented to users [6][7].

The Futsal Bookings and management system is developed following the Agile Software Development Life Cycle (SDLC), which supports iterative development, continuous feedback, and flexible requirement handling throughout the project lifecycle [8]. Each sprint delivers a working version of the system with improved features and refinements, ensuring that modules such as booking management, owner dashboards, and machine learning components are developed and tested incrementally [8].

In conclusion, the AllFutsal system serves as a secure, scalable, and intelligent platform that digitizes traditional futsal booking processes and brings efficiency to sports facility operations. It not only enhances scheduling accuracy, transparency, and operational management for futsal owners but also provides AI-driven insights and personalized recommendations for players, addressing the clear gap in existing manual and partially digital solutions.

CHAPTER 3

SYSTEM ANALYSIS

3.1 System Analysis

System analysis involves studying the system in detail to understand how it works and what improvements are needed. It focuses on identifying problems, gathering data, and separating the system into smaller parts to make it easier to study. By analyzing each part and how they connect, we can clearly see what the system is expected to do. In this way, system analysis acts as a method of problem solving, helping to check whether each component works properly and contributes toward the overall goal of the system.

3.1.1 Requirement Analysis

To develop the futsal bookings and management, we identified both functional and non-functional requirements that the system must fulfill.

3.1.1.1 Functional Requirement

Functional requirement are the features and functions that the Futsal Bookings and management System must provide.

- **User Registration and Authentication:** Only registered users can access the system. Users can create and manage their profiles, while futsal owners and administrators have extended privileges to manage courts, bookings, and platform-wide operations respectively.
- **Futsal Discovery and Search:** Users can search for futsal centers by location, price, and available time slots, and view detailed information including court facilities, pricing, and community reviews before making a booking decision.
- **Real-Time Court Booking:** Users can book available futsal courts online, receive instant booking confirmation and notifications, and manage their reservation history.

Bookings can also be cancelled or rescheduled based on the platform's defined rules and policies.

- **Ratings, Reviews and Recommendations:** Users can submit ratings and reviews after playing, and the system provides personalized futsal court recommendations based on the user's location and preferences using a location-based recommendation engine.
- **Futsal Owner Management:** Futsal owners can register and manage their futsal center profile, add and manage courts, set available time slots and pricing, approve or manage incoming booking requests, view booking schedules and customer details, manage offers and subscription plans, view an analytics dashboard for bookings and revenue insights, and respond to customer reviews and queries.
- **Admin Control Panel:** Administrators can manage all futsal tenants on the platform, manage user accounts and permissions, monitor overall system usage and generate reports, manage subscription plans and packages, and handle complaints and system-level moderation to ensure smooth platform operation.
- **Community Forum:** Registered users and futsal owners can create, view, and participate in forum discussions related to futsal topics such as match arrangements, court reviews, team formations, and general sports discussions, fostering a collaborative community environment.
- **Notification and Confirmation System:** The system automatically sends booking confirmations, status updates, and relevant alerts to users, futsal owners, and administrators based on actions performed within the platform.

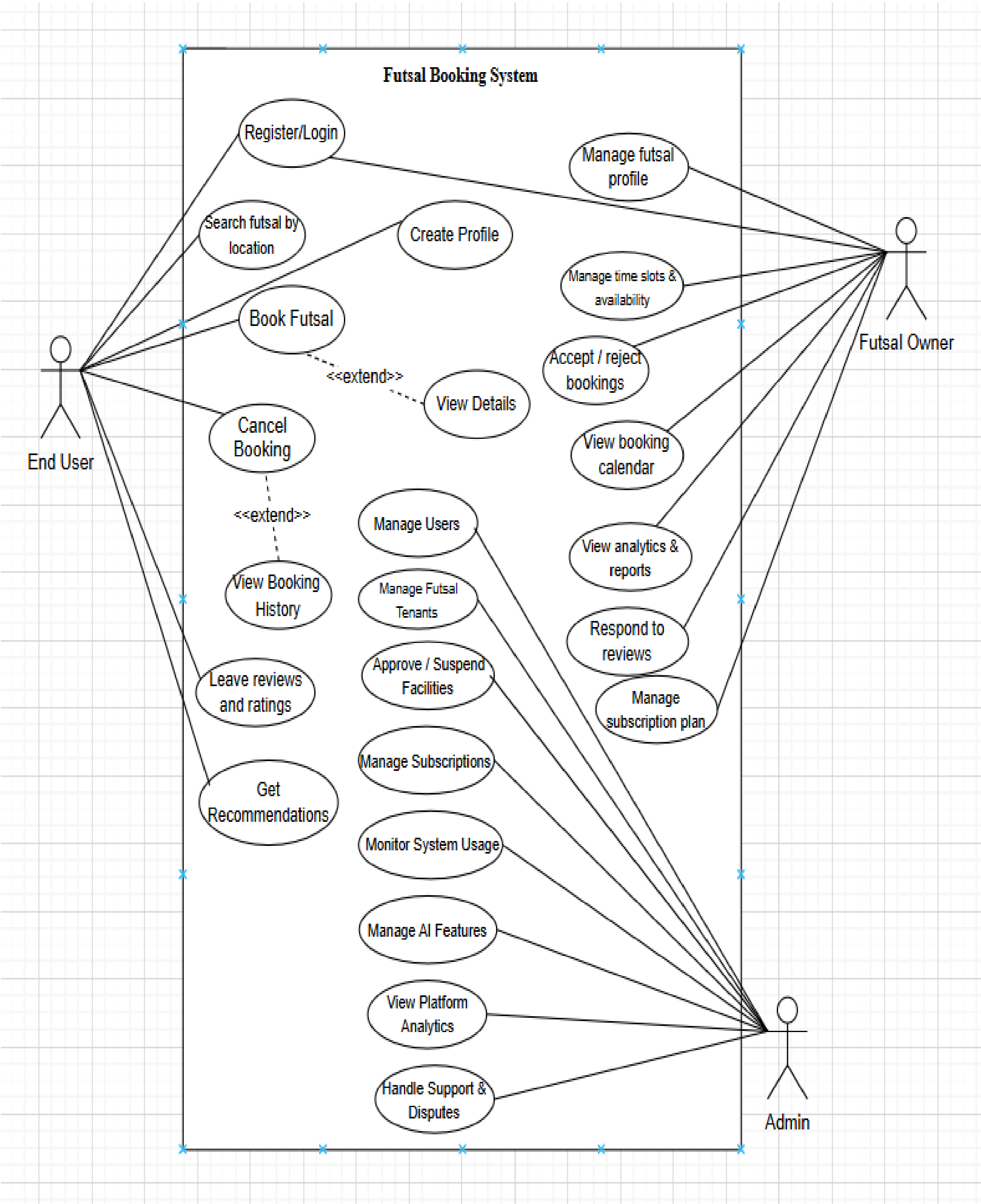


Figure 3.1 : Use Case Diagram of Futsal Bookings Management System

3.1.1.2 Non-Functional Requirements

Non-functional requirements are the requirements that specifies criteria that can be used to judge the operation of a system, rather than specific behaviors. They are contrasted with functional requirements that define specific behavior or functions. The non- functional requirements of the system are:

- **Usability**: The interface should be simple, clean and easy to use for all users (admin, teacher and student).
- **Reliability**: System is accurate, as we don't use many components and thus provides the user specified results in specific time duration.
- **Performance**: System should handle multiple user requests simultaneously without slowing down.
- **Scalability**: The system should be designed so it can accommodate more users or institutions in the future without major redesign.
- **Security**: All sensitive data must be securely encrypted during storage and transmission. Only authorized users can access certain features, and all communication must be encrypted using HTTPS.
- **Maintainability**: The system should be easy to update and maintain, with a clear and modular architecture that supports future enhancements and feature upgrades.

3.1.1.3 Dataset Description

For this project, a custom synthetic dataset was created to test and validate the Futsal Booking and Management System workflow, particularly for sentiment analysis of user reviews and TF-IDF based review summarization. The dataset contains 1200 records with 5 attributes, designed to simulate real-world futsal court user review submissions. The dataset was synthetically generated using a Python script due to the unavailability of domain-specific public futsal review datasets on platforms such as Kaggle and Hugging Face. In addition, the review scenarios and rating patterns were prepared based on discussions among the three project team members, who are regular futsal players and were further validated by the project supervisor to ensure realistic and practical review data representation.

id	user_id	rating	review	sentiment_label
1	461	3	Well Spacious court and proper lighting setup at night session	Positive
2	301	2	To be honest Facilities were not clean at all but during our regular game	Negative
3	197	3	Frankly speaking Spacious court and proper lighting setup at night session	Positive
4	152	5	Overall Very professional staff and management during our match time	Positive
5	443	3	To be honest Facilities were outdated and not maintained we played in the evening	Negative
6	427	4	Actually The court was spacious and well-maintained, providing ample room for playe	Positive
7	318	1	To be honest Not satisfied with the overall experience during our match time	Negative
8	404	4	The best place to play futsal in town but staff was rude and unhelpful while playing wi	Positive
9	383	4	From what I saw The turf was in excellent condition and very safe but poor lighting an	Positive
10	272	4	From my experience The booking process was seamless and efficient, making it easy	Positive
11	496	2	To be honest Too much delay in booking confirmation while playing with friends	Negative
12	387	3	Actually The booking process was seamless and efficient, making it easy to reserve a	Positive
13	188	5	Well during peak hours	Positive
14	136	1	horrible	Negative
15	482	3	Based on my visit Too much delay in booking confirmation on a rainy day	Negative
16	168	3	Well The court was overcrowded and chaotic but the price was reasonable for the qu	Negative
17	155	5	From my experience The best place to play futsal in town during weekend booking	Positive
18	253	4	Overall Loved the turf and lighting setup while playing with friends	Positive
19	70	3	not satisfied	Negative
20	80	5	In my opinion Affordable price with great quality during our regular game	Positive
21	17	2	Well Had to wait for a long time to get the court ready during weekend booking	Negative
22	51	4	From my experience Spacious court and proper lighting setup while trying out the new	Positive
23	496	1	In my opinion Not satisfied with the overall experience but booking system was fast a	Negative
24	340	3	never again	Positive
25	65	3	Well Great futsal court with excellent facilities but the worst futsal court i've ever been	Positive
26	169	1	From my experience Not worth the price during weekend booking	Negative
27	67	1	Court was dirty and poorly maintained while testing the lighting	Negative

Figure 3.2 : Dataset records

The attributes include both categorical and numerical features describing each review record. The attributes are as follows:

- **id:** Unique identifier for each user review record.
- **user_id:** Unique identifier for the user who submitted the review.
- **rating:** Numerical rating provided by the user on a scale of 1 to 5, where 1–2 indicates a negative experience and 4–5 indicates a positive experience.

- **review:** The textual content of the user's review submitted after playing, used as input for TF-IDF feature extraction and sentiment classification.
- **sentiment_score:** A numerical probability score between 0.00 and 1.00 representing the confidence of the sentiment prediction, where values closer to 1.00 indicate stronger positive sentiment.

For sentiment classification, the sentiment category was derived from the rating values during preprocessing, where lower ratings represented negative sentiment and higher ratings represented positive sentiment. The review text data was preprocessed by removing stop words and applying tokenization before TF-IDF vectorization to make the textual data suitable for machine learning analysis.

This dataset was synthetically prepared to simulate realistic futsal user review scenarios, providing a practical basis for testing the TF-IDF Review Feature Extraction Algorithm and the Logistic Regression Sentiment Classification Algorithm implemented within the Futsal Bookings and Management platform.

3.1.2 Feasibility Analysis

The proposed project's viability is determined by the feasibility study:

I) Technical Feasibility

The futsal booking and management platform is a multi-tenant SaaS web application for managing court bookings, owner operations, and AI-based review analysis across multiple futsal facilities. It uses algorithms for real-time booking validation, location-based court recommendations, and machine learning for review sentiment analysis. Built with React, Node.js, Express.js, MySQL, and Scikit-learn, the system can be developed using freely available tools and existing web development knowledge without requiring specialized hardware.

ii) Operational Feasibility

The futsal booking and management system enables players to book courts in real time while allowing futsal owners to manage courts, schedules, pricing, and bookings through a web-based dashboard. It is accessible through any standard web browser, requires no additional software installation, and features a user-friendly interface that needs minimal training. Futsal owners only need to register their facilities and keep court availability updated.

iii) Economic Feasibility

The futsal booking and management system can be developed at minimal cost using free and open-source technologies such as the MERN stack, Scikit-learn, and NLTK. Development and deployment can be carried out using local environments and free-tier cloud services, requiring no significant financial investment.

iv) Schedule Feasibility

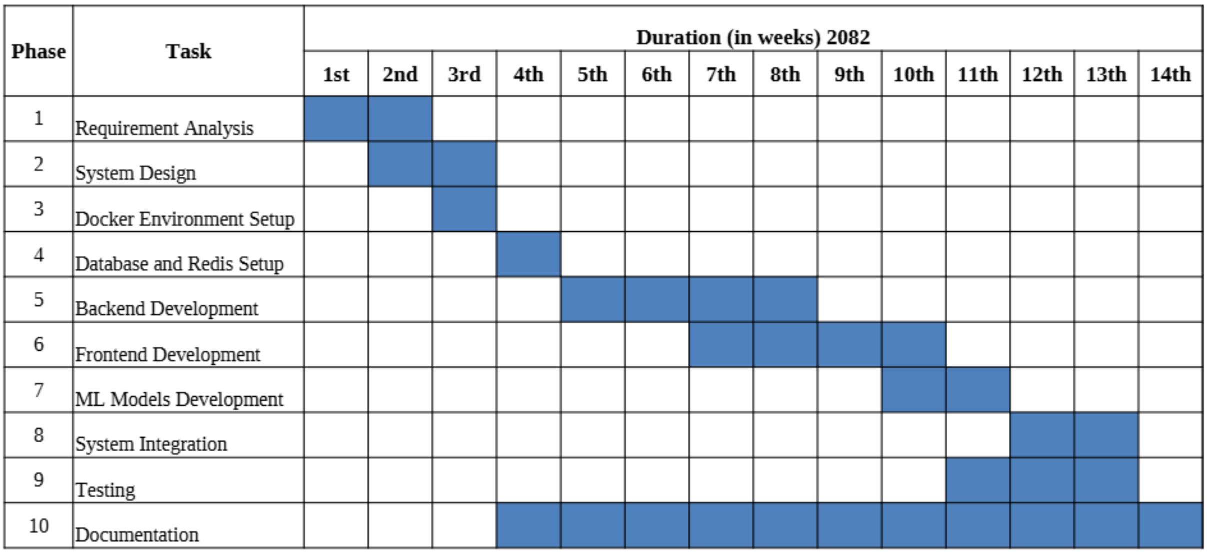


Figure 3.3 : Gantt Chart

The Futsal Booking and Management project is scheduled to be completed within 14 weeks through phases including requirement analysis, system design, environment setup, database configuration, backend and frontend development, machine learning integration, system testing, and documentation. An Agile development approach is followed, allowing overlapping tasks to ensure timely project completion.

3.1.3 Analysis

3.1.3.1 ER Diagram

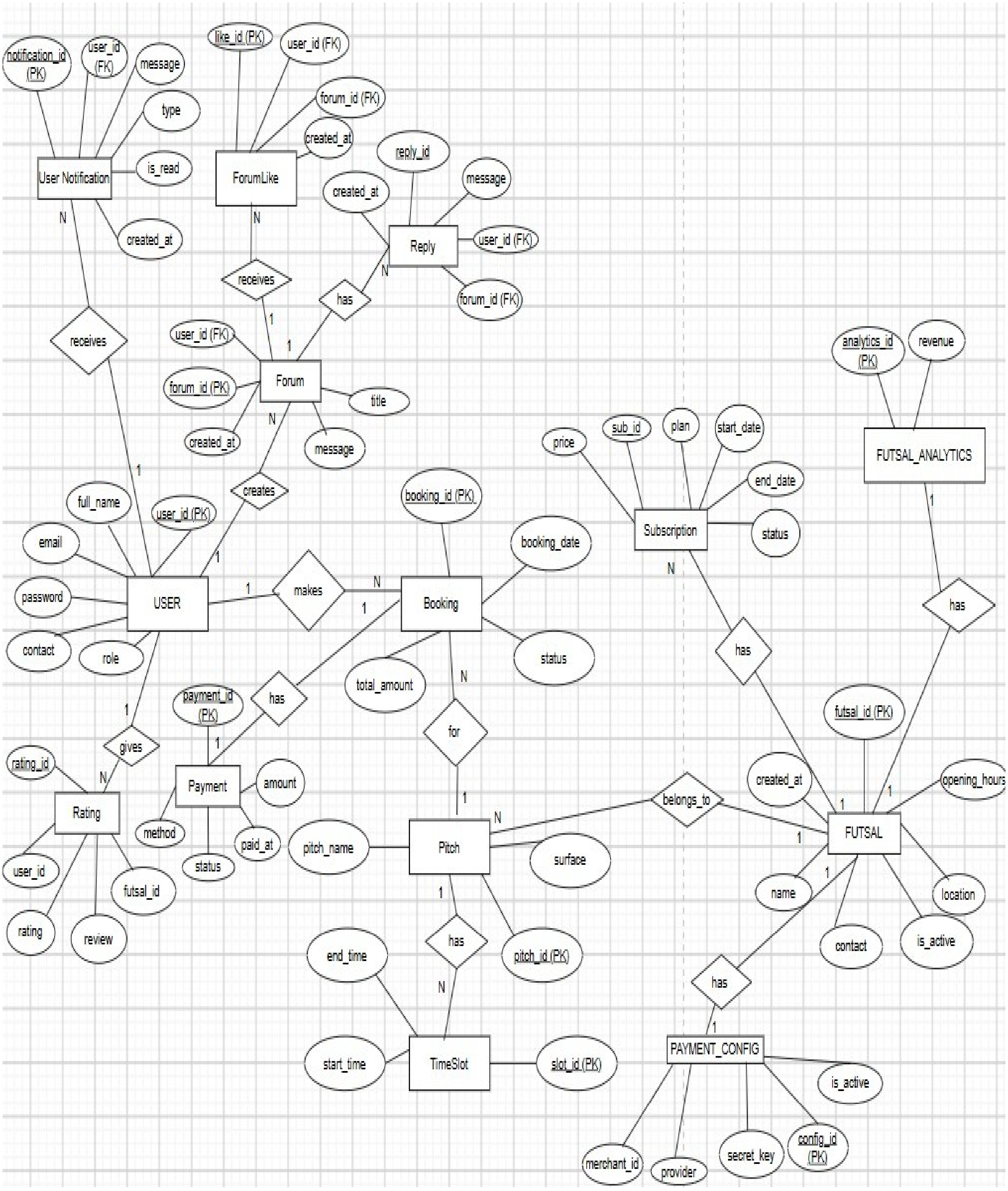


Figure 3.4 : ER Diagram

3.1.3.2 DFD level-0 Diagram

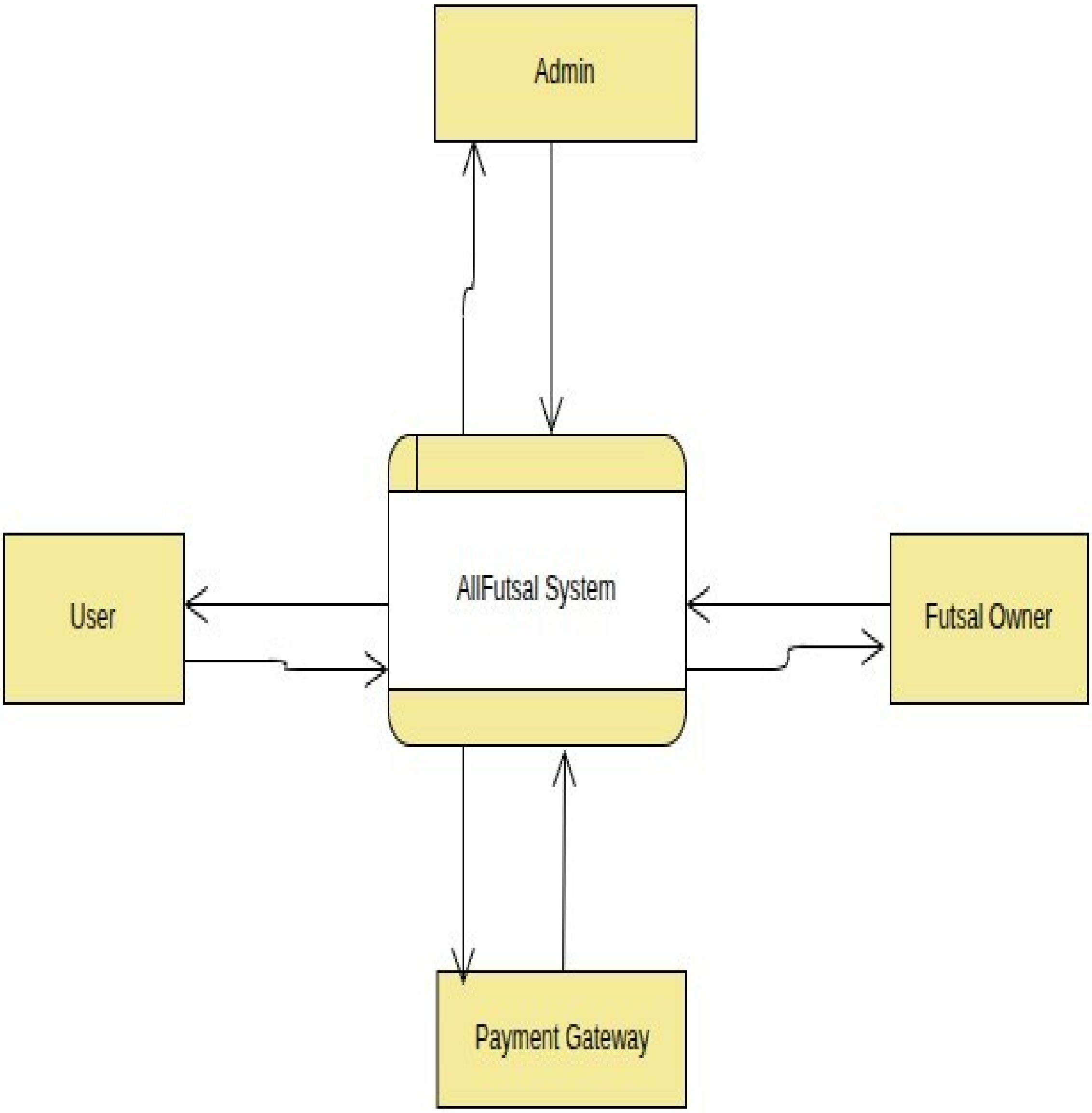


Figure 3.5 : DFD level-0 Diagram

3.1.3.3 DFD level-1 Diagram

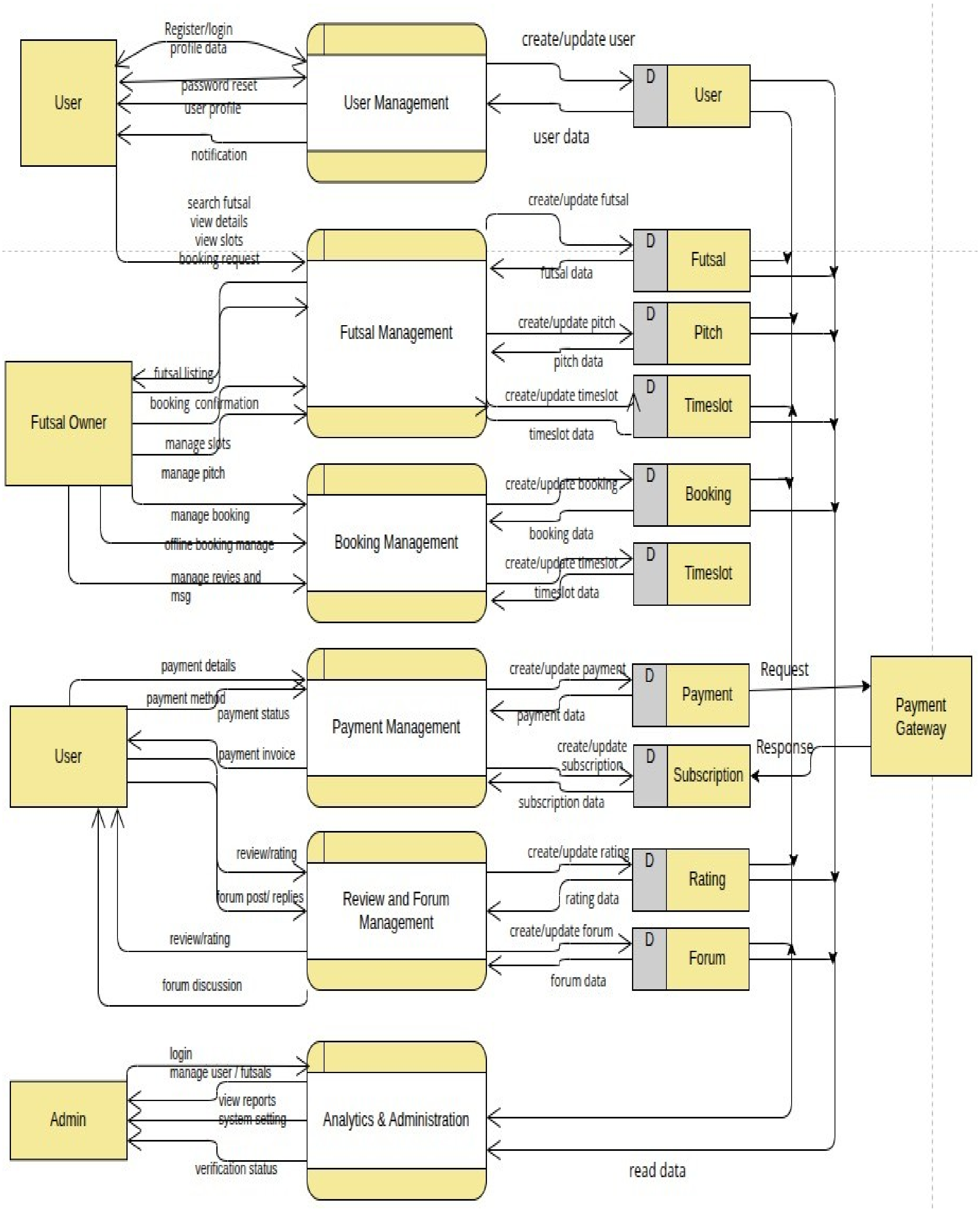


Figure 3.6 : DFD level-1 Diagram

3.1.3.4 DFD level-2 process 1 Diagram

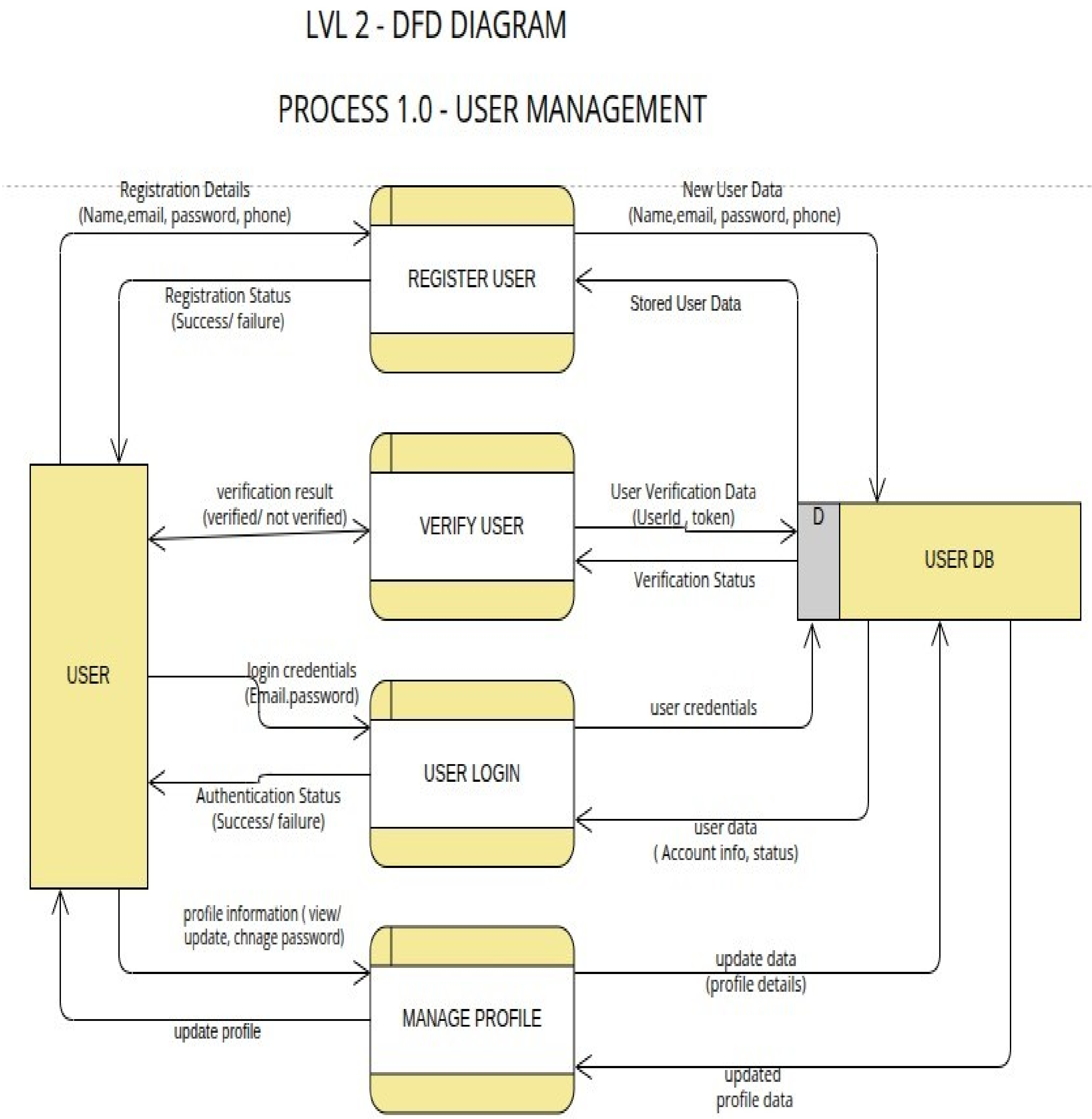


Figure 3.7 : DFD level-2 process 1 Diagram

3.1.3.5 DFD level-2 process 2 Diagram

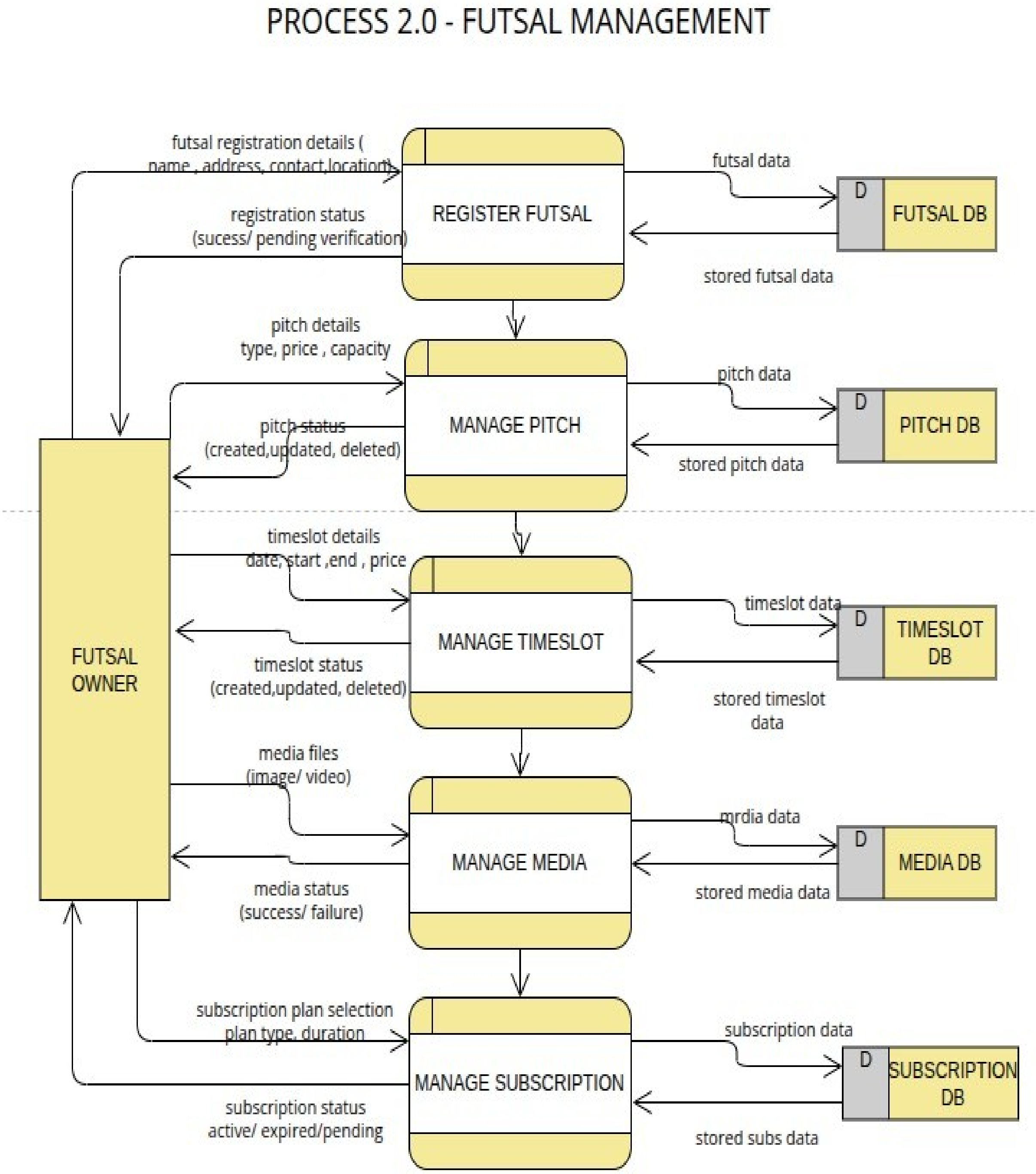


Figure 3.8 : DFD level-2 process 2 Diagram

3.1.3.6 DFD level-2 process 3 Diagram

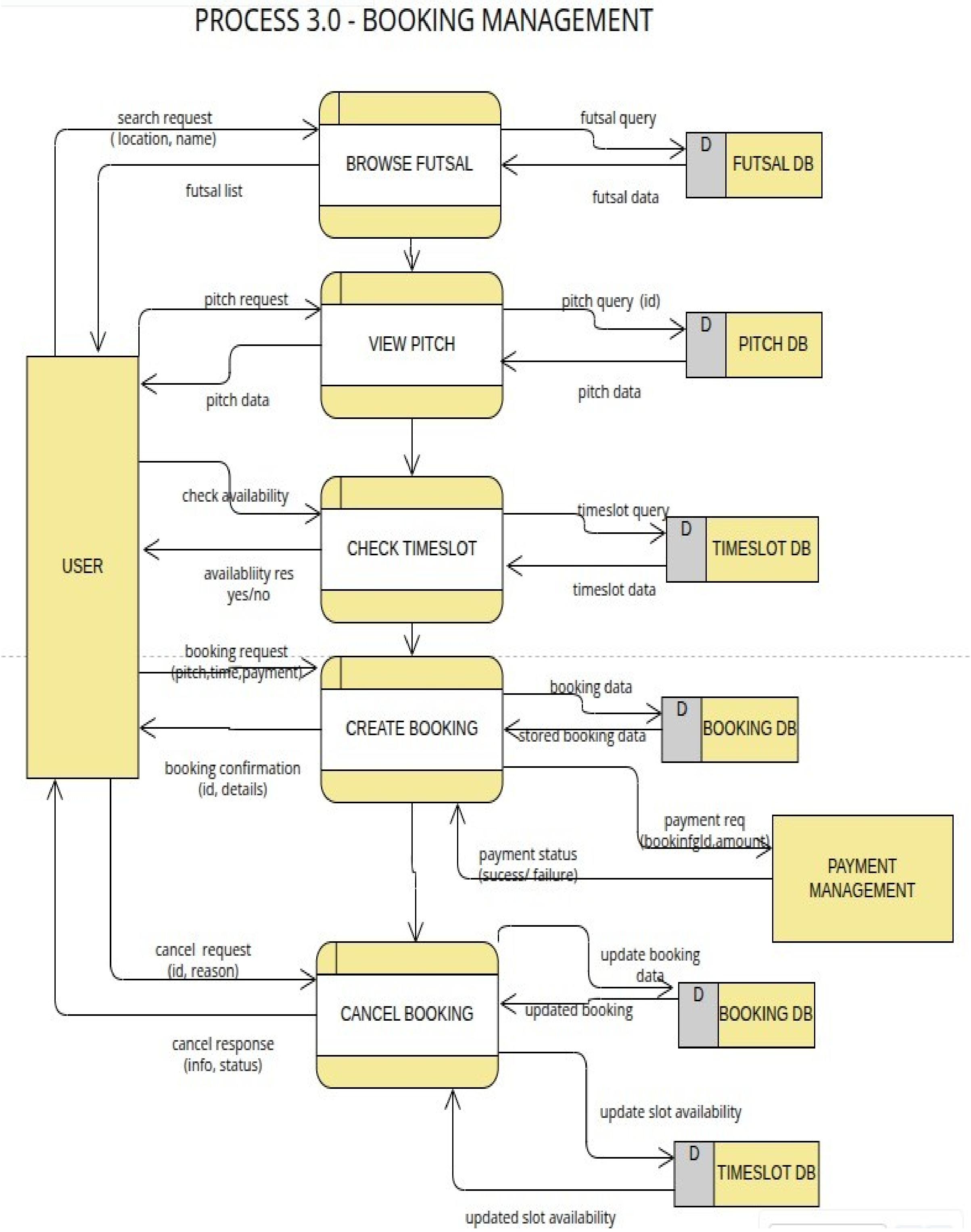


Figure 3.9 : DFD level-2 process 3 Diagram

3.1.3.7 DFD level-2 process 4 Diagram

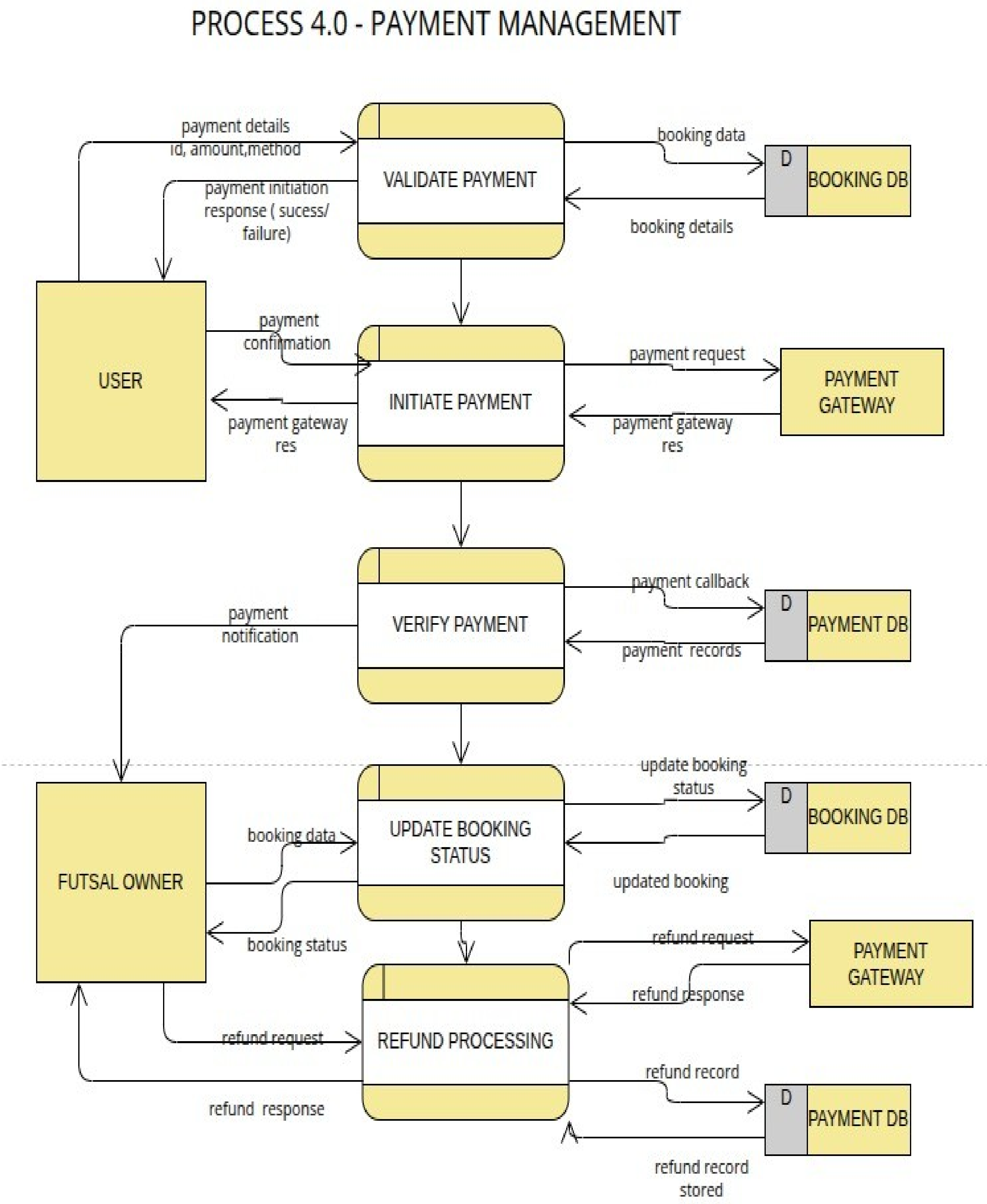


Figure 3.10 : DFD level-2 process 4 Diagram

3.1.3.8 DFD level-2 process 5 Diagram

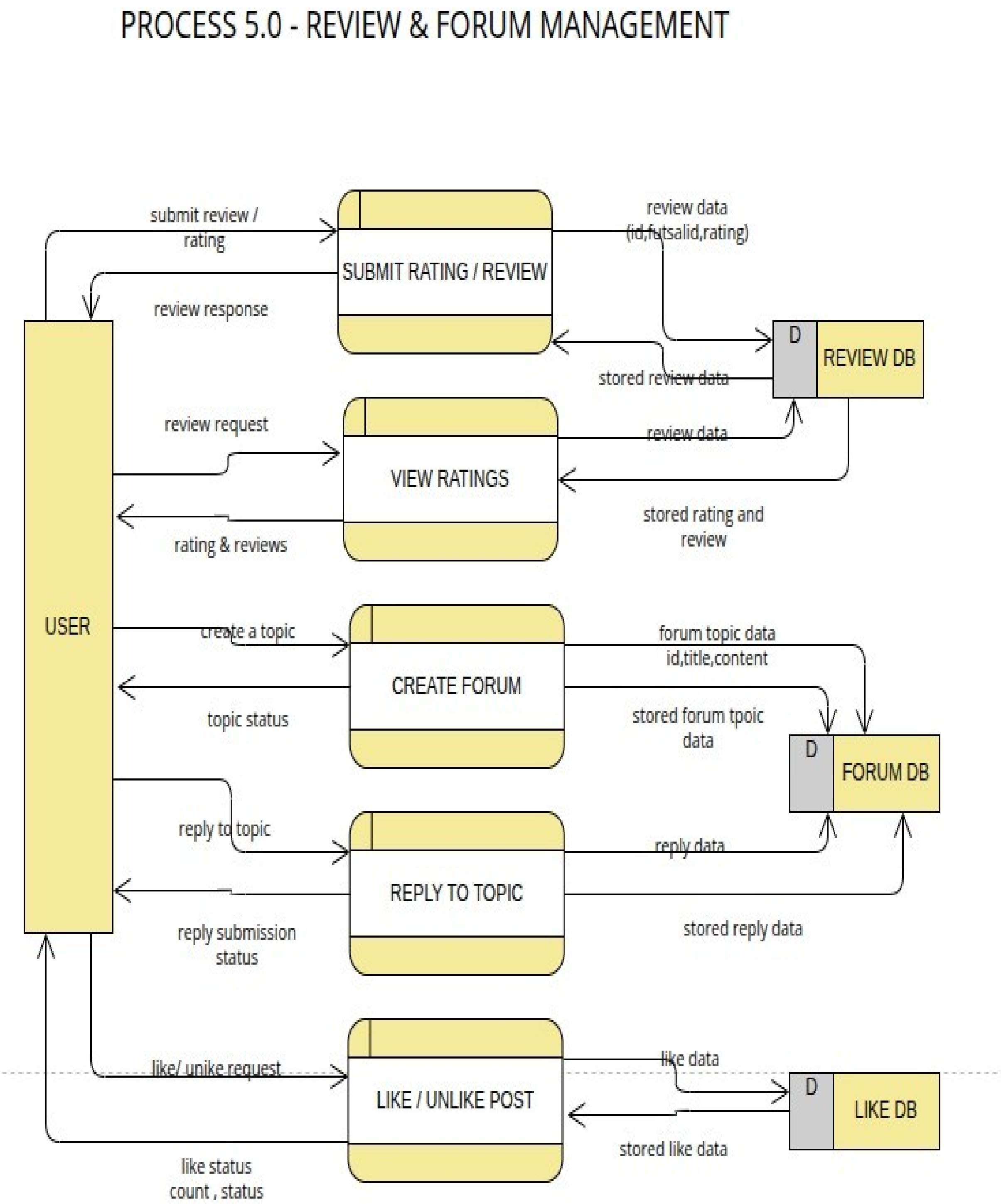


Figure 3.11 : DFD level-2 process 5 Diagram

3.1.3.9 DFD level-2 process 6 Diagram

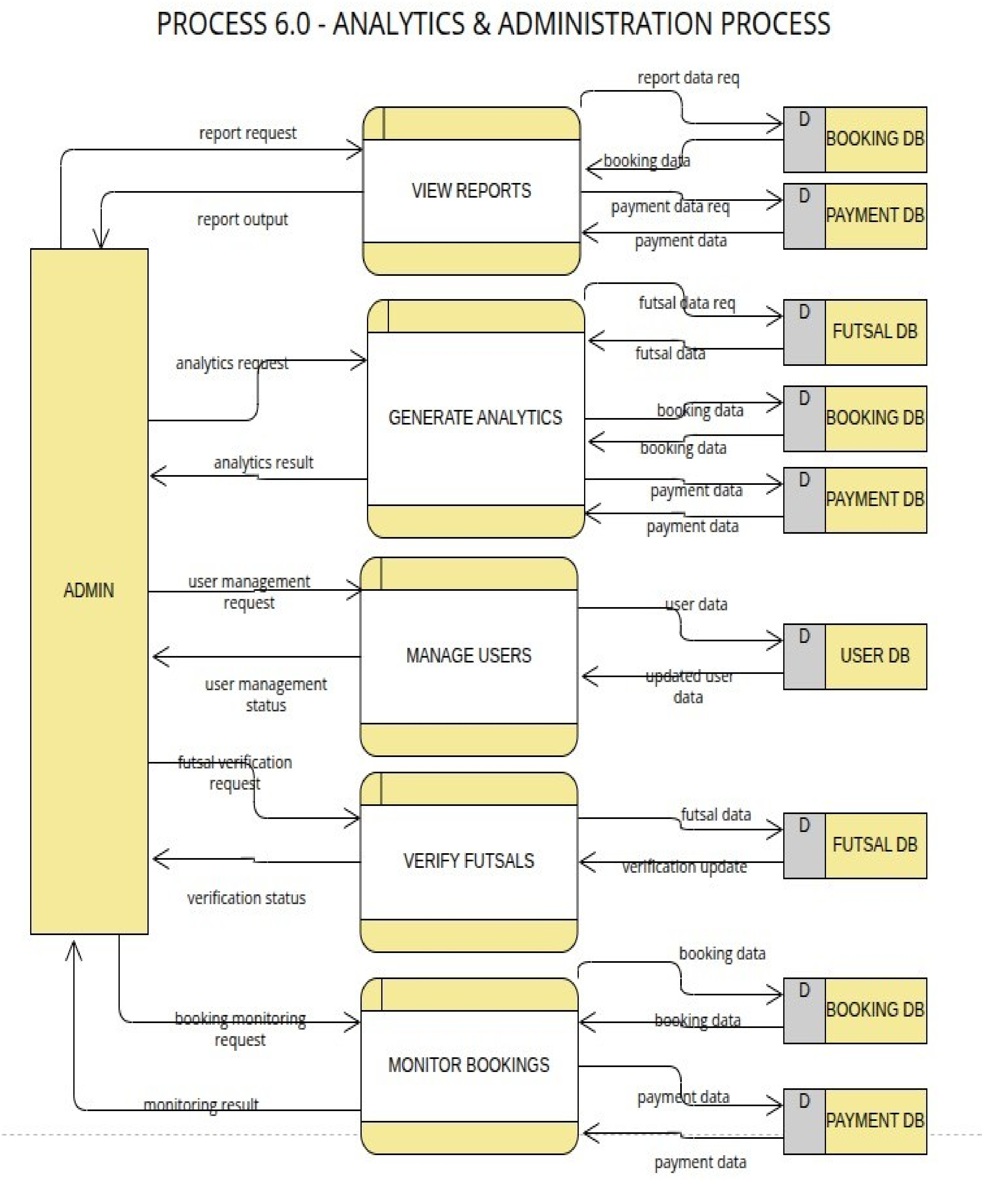


Figure 3.12: DFD level-2 process 6 Diagram

3.1.3.5 Sequence Diagram

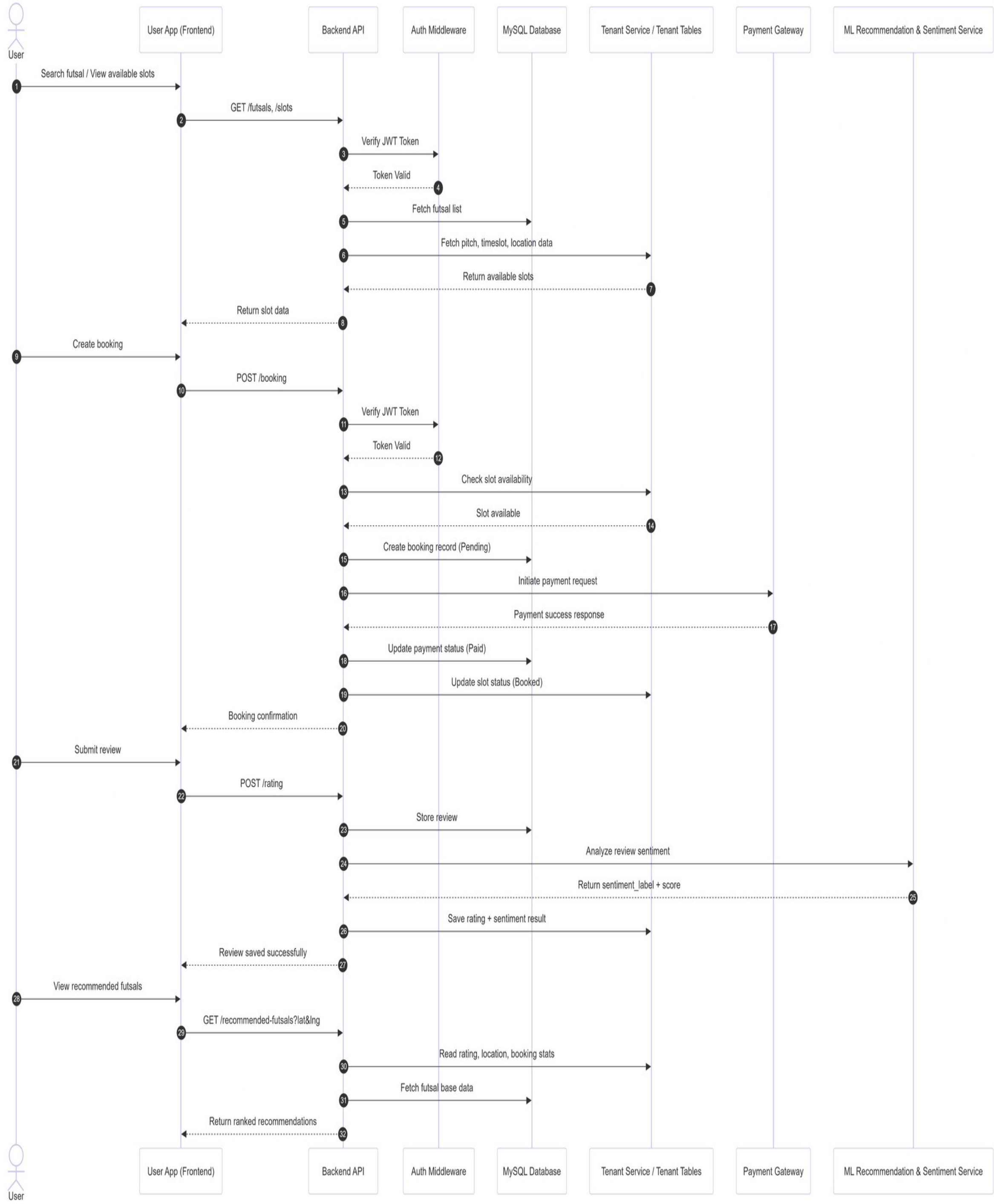


Fig 3.13: Sequence Diagram of Futsal Bookings and Management System

CHAPTER FOUR

SYSTEM DESIGN

4.1 Design

System Design includes the means and methodologies to improve the management and control of the software development process. It includes structuring and simplifying the process using several diagrams and standardizing the development process by specifying the required activities and techniques to be implemented.

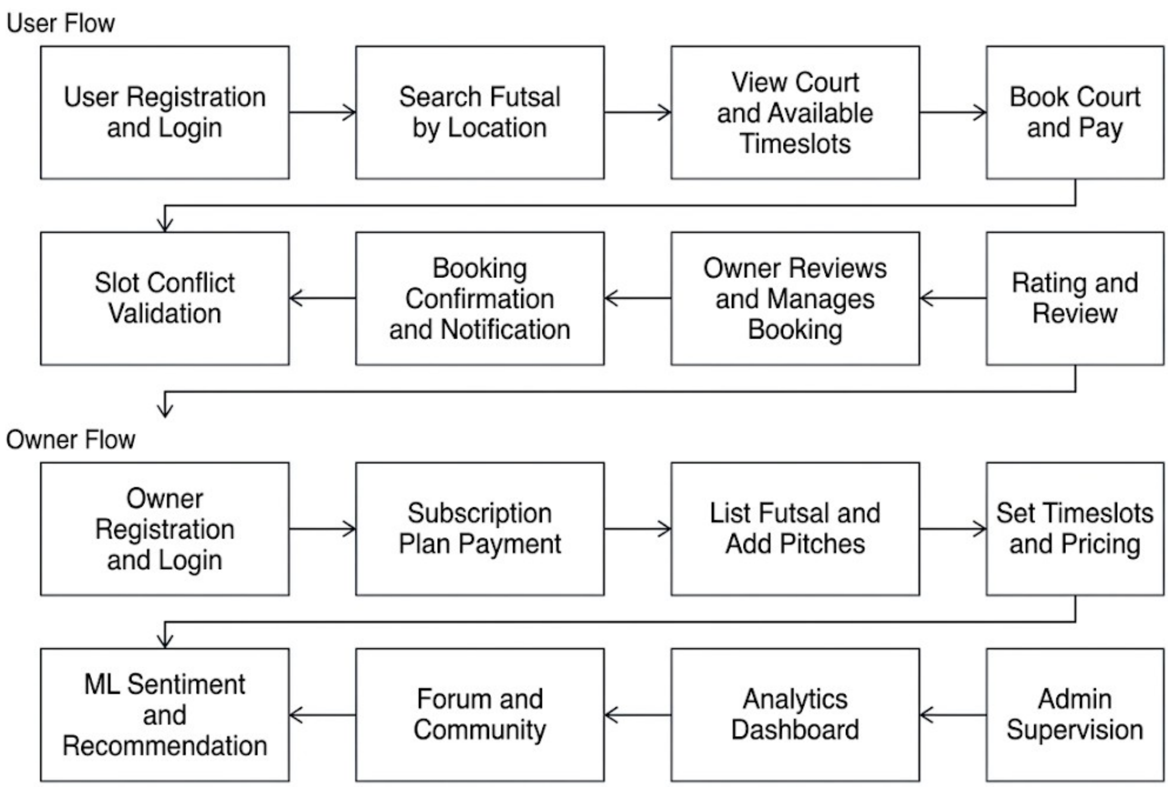


Fig 4.1: High-Level System Design of Futsal Bookings and Management System

The system architecture is shown in Figure 4.1. The procedures involved are:

User Registration and Login

Players register or log in using their credentials or Google OAuth to access the platform. The system validates user information, securely stores account details, and grants access to the user dashboard. From there, users can search futsals, manage bookings, and view booking history.

Search Futsal by Location

Users search for nearby futsal centers by entering a location or using their current location. The system applies the Haversine distance algorithm to recommend nearby venues and allows filtering by price, availability, and ratings. This helps users quickly find suitable courts.

View Court and Available Timeslots

Users can view detailed information about a futsal center, including facilities, pricing, and available courts. Real-time timeslot availability is displayed to ensure booking accuracy. Users can then choose their preferred court and schedule.

Book Court and Pay

After selecting a court and timeslot, users confirm their booking and complete the payment process. The system records the reservation details and updates the booking status. A successful payment finalizes the reservation.

Slot Conflict Validation

Before confirming a booking, the system checks for overlapping reservations using the Slot Conflict Prevention Algorithm. If a conflict exists, the user is prompted to choose another timeslot. Otherwise, the booking is successfully confirmed.

Booking Confirmation and Notification

Once the booking is completed, the system updates the database and confirms the reservation. Notifications containing booking details are sent to both the user and the futsal owner. This ensures all parties receive instant confirmation.

Rating and Review

After completing a booking, users can submit ratings and written reviews for the futsal center. Reviews are analyzed using TF-IDF and Logistic Regression to determine sentiment. The results help improve recommendations and provide useful feedback for future users.

Owner Registration and Login

Futsal owners register their business accounts by providing center information and contact details. The system creates a dedicated tenant account within the SaaS architecture. Owners can then access their management dashboard.

Subscription Plan Payment

Owners select and pay for a subscription plan to activate their futsal listing. The chosen plan determines the features and services available to them. Once payment is verified, the account becomes fully active.

List Futsal and Add Pitches

Owners create futsal center profiles by adding descriptions, facilities, and images. They can register multiple courts with individual specifications. This information becomes visible to users searching for futsal centers.

Set Timeslots and Pricing

Owners define available booking times and pricing for each court. They can update schedules, block unavailable slots, or change prices through the dashboard. The system synchronizes these changes for real-time availability.

ML Sentiment and Recommendation

The platform analyzes user reviews using machine learning to generate sentiment scores. It also combines review data with location information to recommend suitable futsal centers. This improves user experience and discovery.

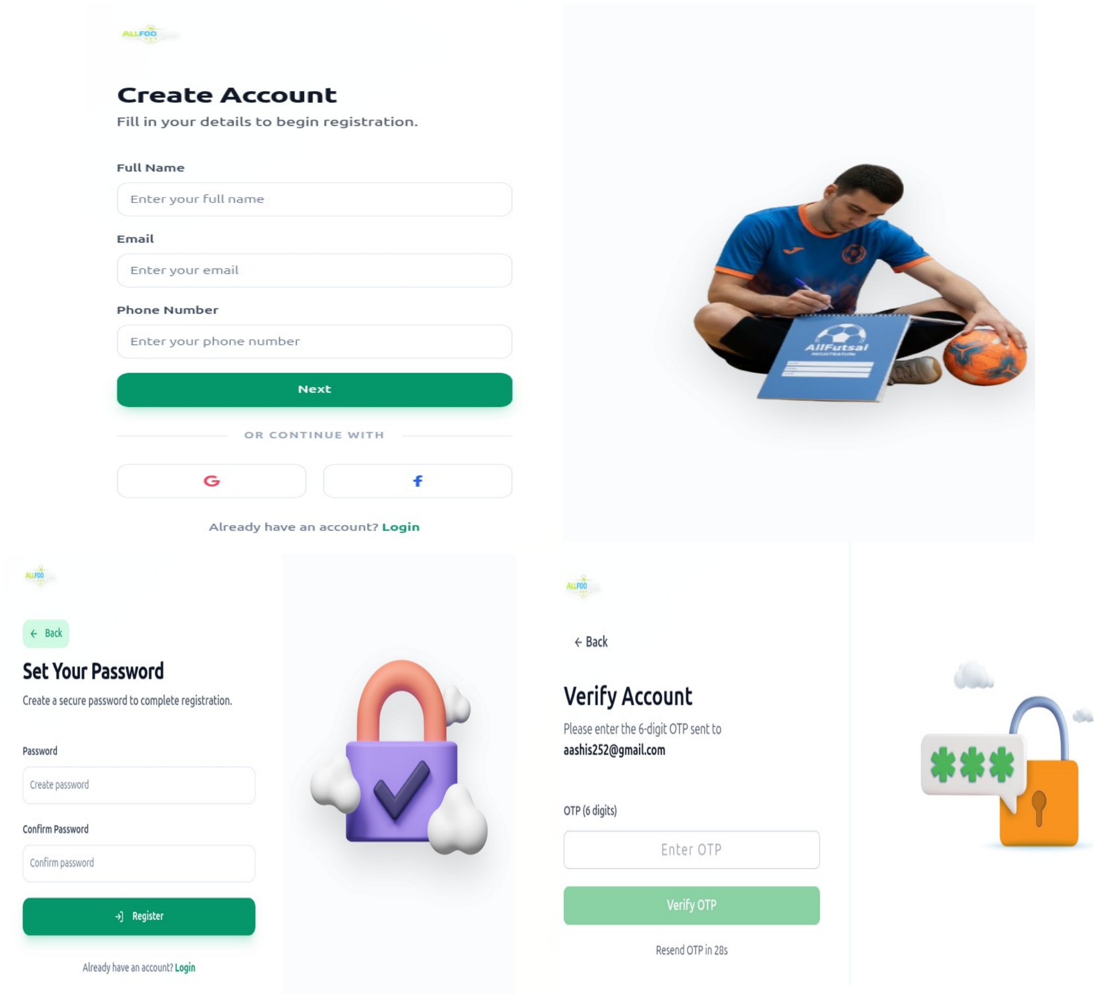
Forum and Community

The platform includes a discussion forum where users and owners can share posts and interact. Community discussions support match arrangements, reviews, and general futsal topics. Administrators monitor content to maintain a healthy environment.

Analytics Dashboard

Owners access an analytics dashboard showing booking trends, revenue, occupancy, and customer activity. These insights help improve business decisions and operational planning. The dashboard provides a clear overview of performance.

4.2 Interface Design



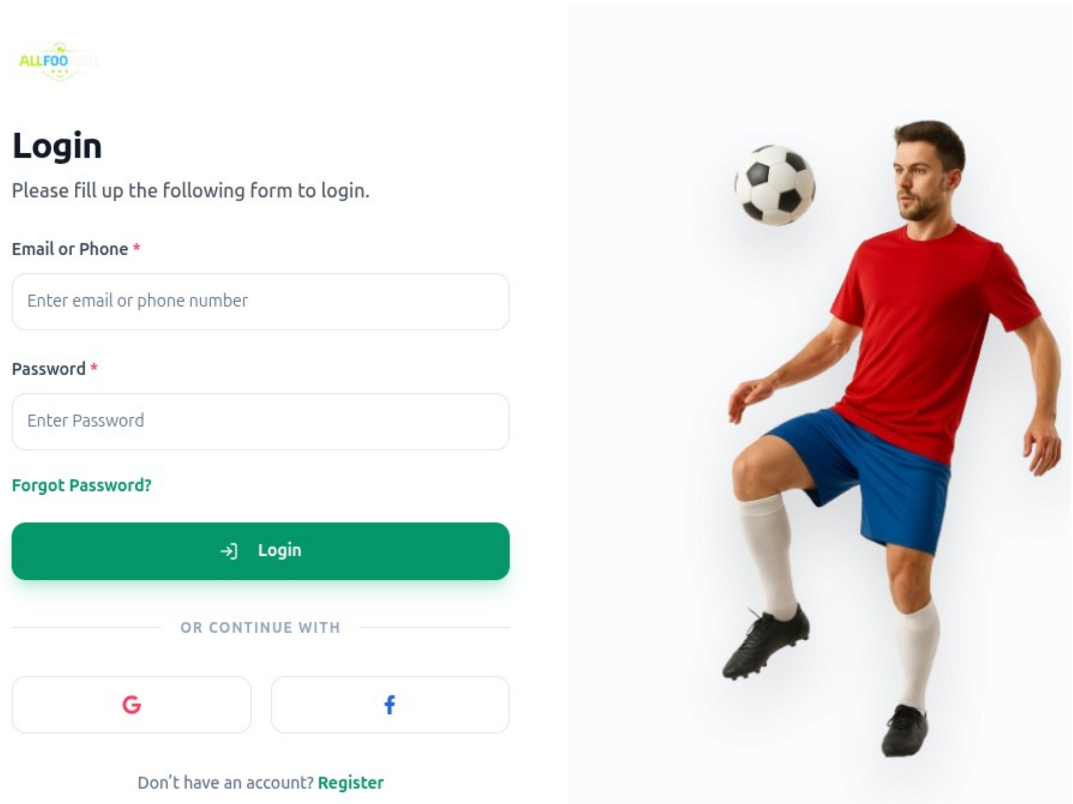


Figure 4.2: Registration and Login Page Design

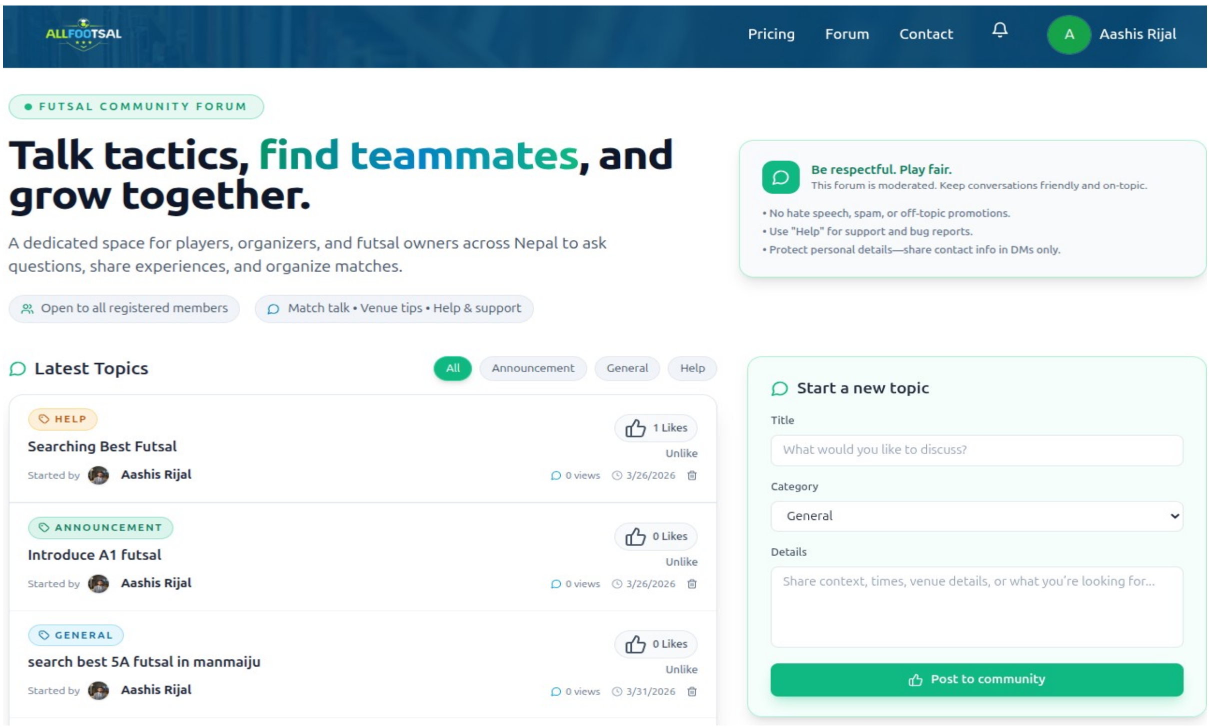


Figure 4.3 : Forum Page Design

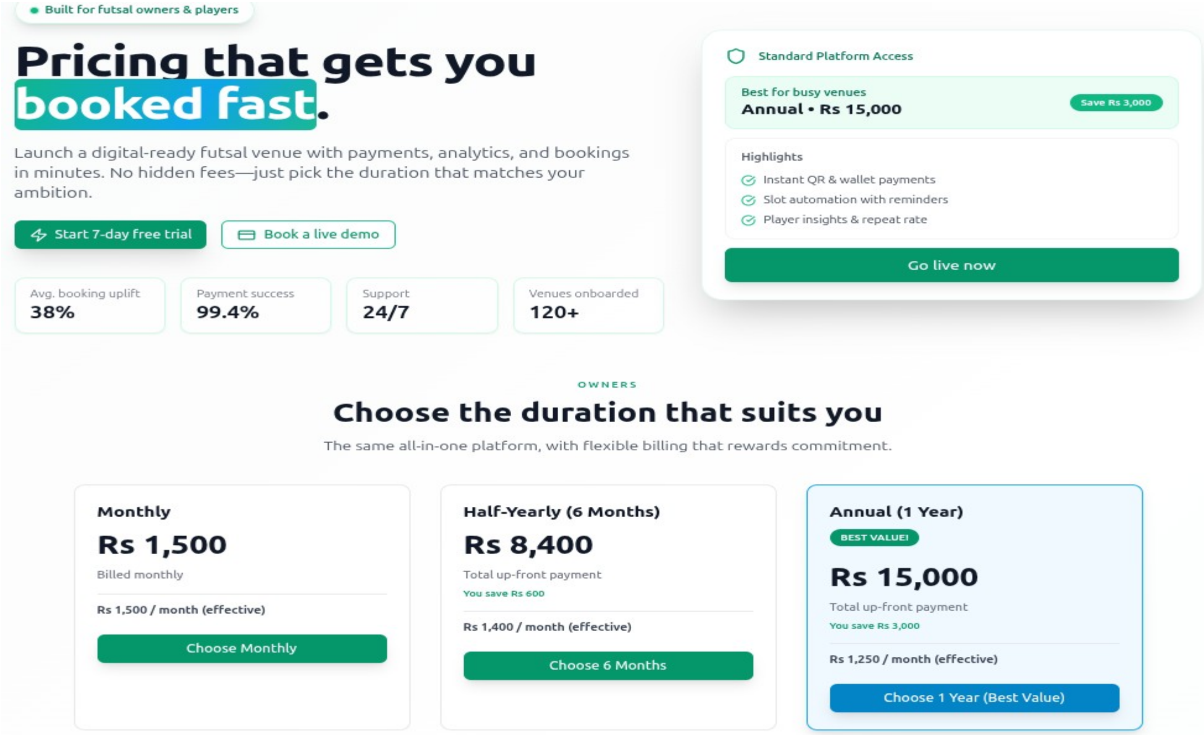
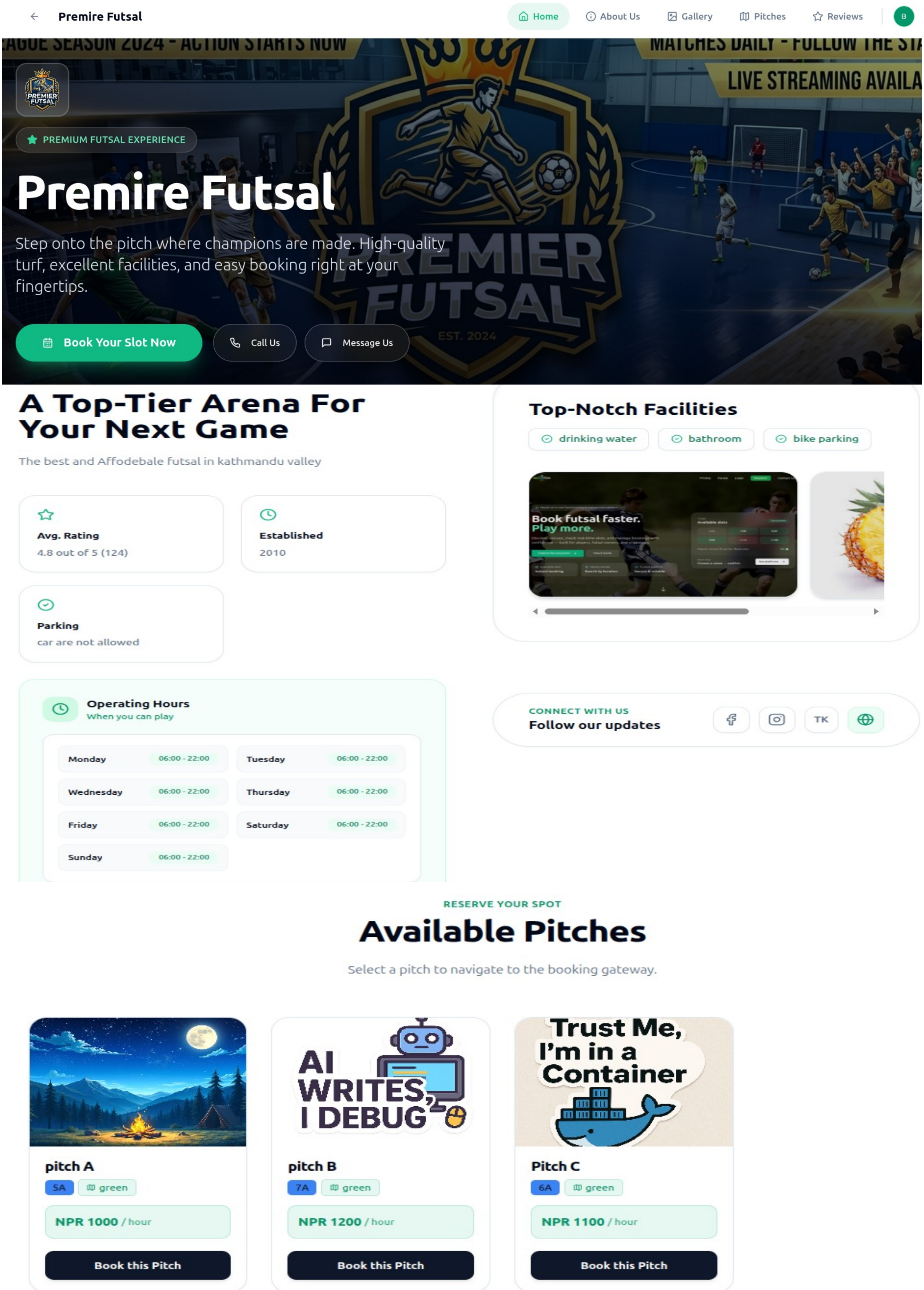


Figure 4.4: Pricing Page Design

Figure 4.5: Futsal listing Page Design


Premire Futsal
Home
About Us
Gallery
Pitches
Reviews
B
LEAGUE SEASON 2024 - ACTION STARTS NOW
MATCHES DAILY - FOLLOW THE STA
LIVE STREAMING AVAILA
PREMIER FUTSAL
PREMIUM FUTSAL EXPERIENCE
Premire Futsal
Step onto the pitch where champions are made. High-quality turf, excellent facilities, and easy booking right at your fingertips.
PREMIER FUTSAL
EST. 2024
Book Your Slot Now
Call Us
Message Us
A Top-Tier Arena For Your Next Game
The best and Affodebale futsal in kathmandu valley
Avg. Rating
4.8 out of 5 (124)
Established
2010
Parking
car are not allowed
Top-Notch Facilities
drinking water
bathroom
bike parking
Book futsal faster.
Play more.
Operating Hours
When you can play
Monday 06:00 - 22:00
Tuesday 06:00 - 22:00
Wednesday 06:00 - 22:00
Thursday 06:00 - 22:00
Friday 06:00 - 22:00
Saturday 06:00 - 22:00
Sunday 06:00 - 22:00
CONNECT WITH US
Follow our updates
TK
RESERVE YOUR SPOT
Available Pitches
Select a pitch to navigate to the booking gateway.
AI WRITES, I DEBUG
Trust Me, I'm in a Container
pitch A
5A
green
NPR 1000 / hour
Book this Pitch
pitch B
7A
green
NPR 1200 / hour
Book this Pitch
Pitch C
6A
green
NPR 1100 / hour
Book this Pitch

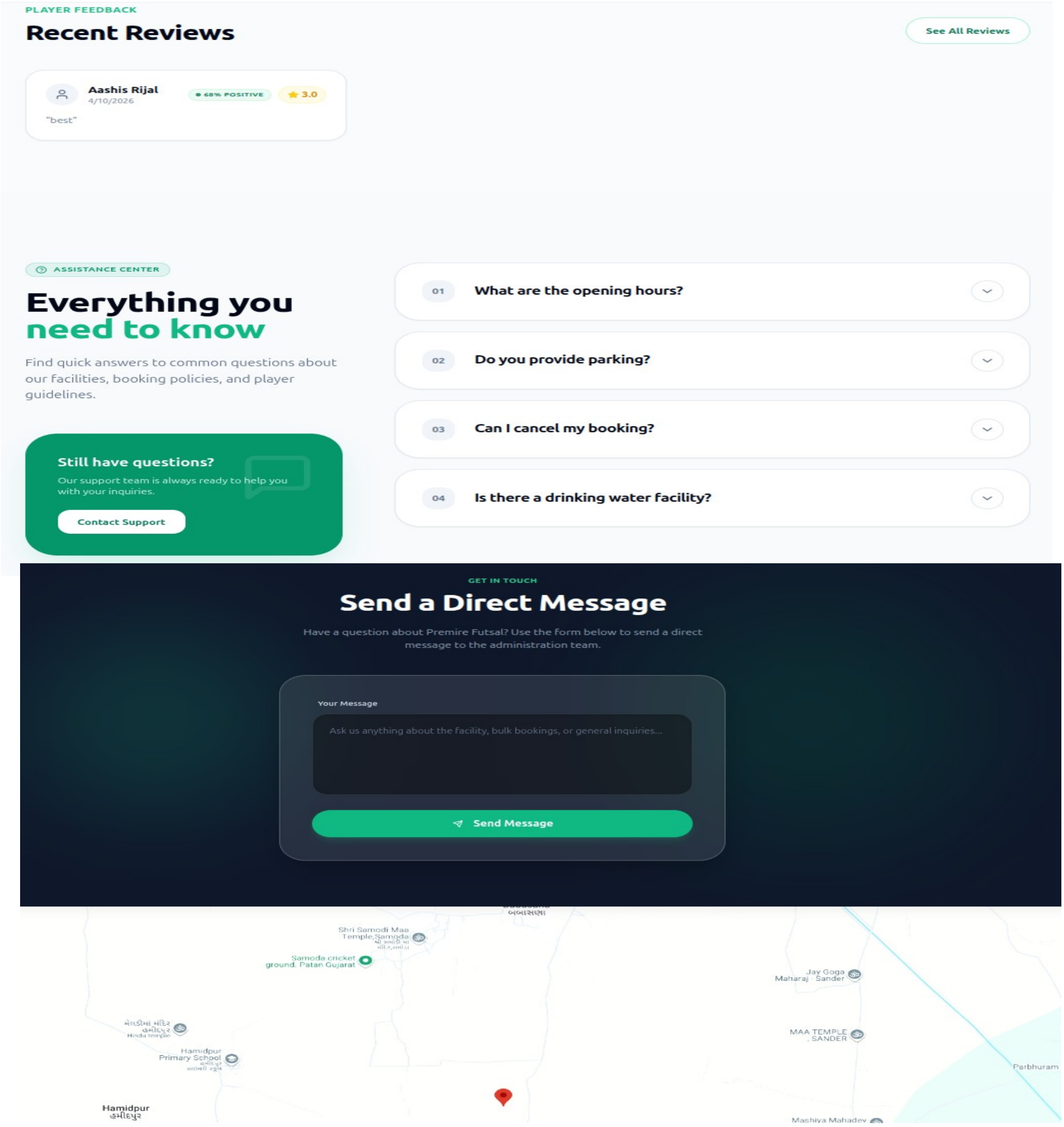


Figure 4.6 : Futsal Page Design

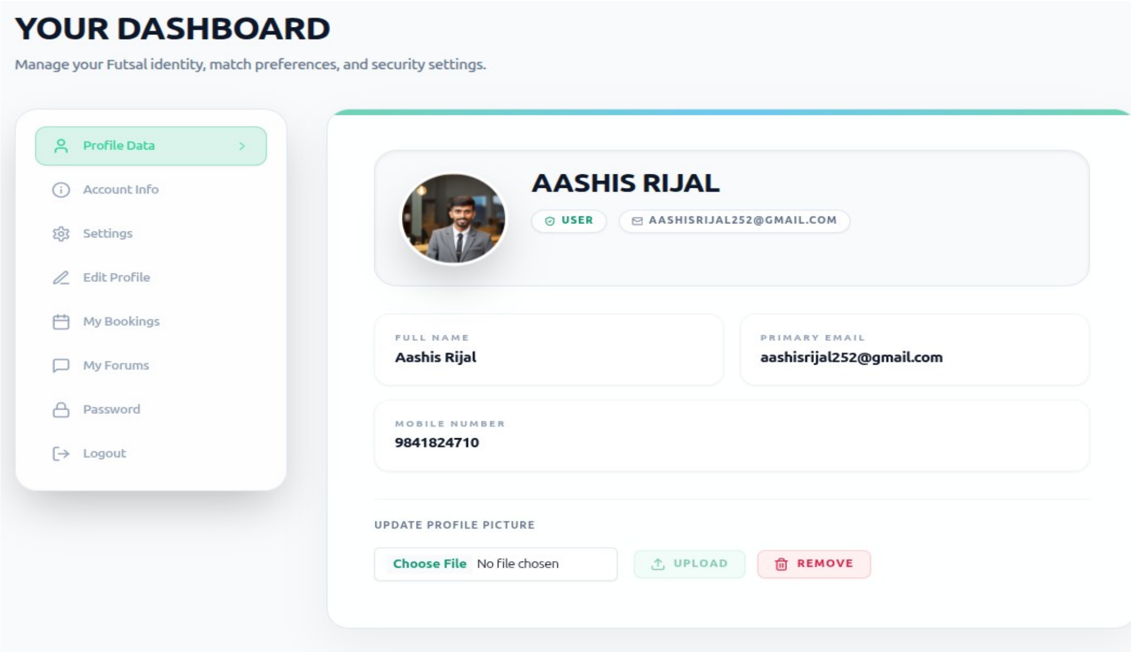


Figure 4.7 : User Profile page Design

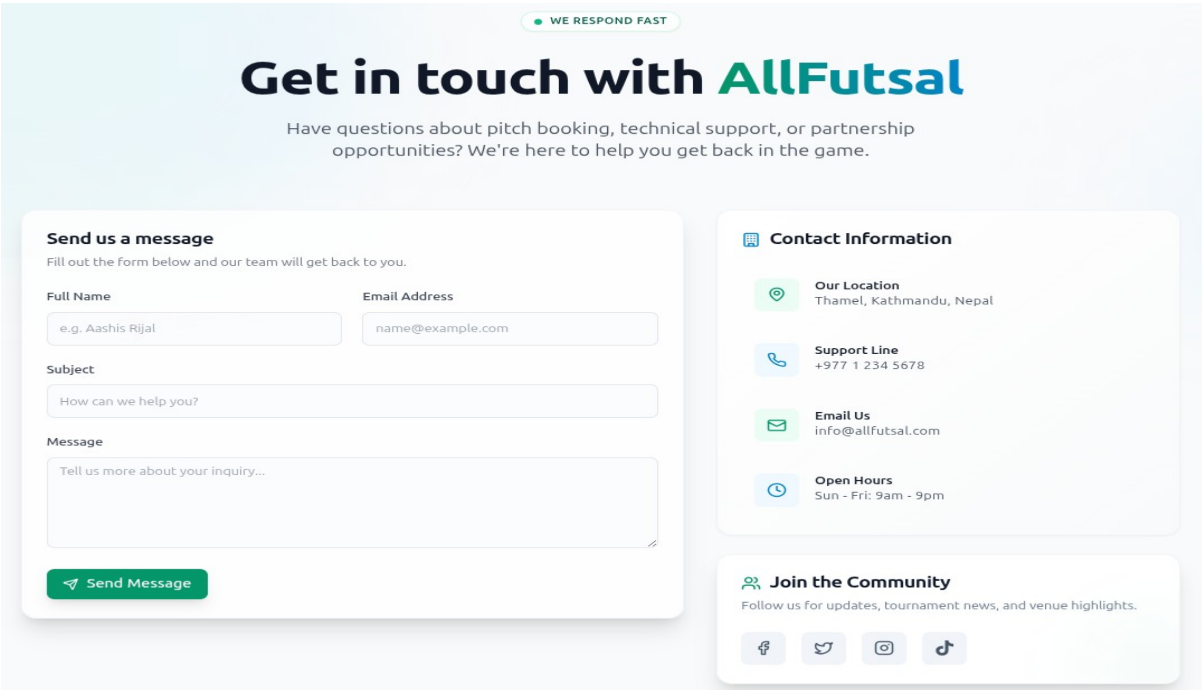


Figure 4.8 Contact page Design

4.3 Description of Algorithm

4.3.1 Location Distance Algorithm (Nearest Futsal Recommendation)

The Location Distance Algorithm is used to recommend nearby futsal centers to users based on their current geographical position. The system calculates the real-world distance between the user's location and each registered futsal center using the Haversine formula, which accurately computes the great-circle distance between two points on the surface of the Earth given their latitude and longitude coordinates.

The working flow of the algorithm can be described as follows:

User Location Detection: The system reads the user's current latitude and longitude coordinates, either through browser geolocation or a manually entered location search.

Futsal Center Coordinates Retrieval: The system retrieves the registered latitude and longitude coordinates of all active futsal centers stored in the database.

Distance Calculation: For each futsal center, the system computes the geographical distance from the user's location using the Haversine formula.

Sorting and Ranking: All futsal centers are sorted in ascending order of calculated distance, placing the nearest futsal centers at the top of the recommendation list.

Result Display: The sorted list of nearby futsal centers is returned to the user interface, where courts are displayed with their calculated distance from the user's current location, allowing users to make informed decisions based on proximity.

Mathematical Representation (Haversine Formula):

$$d = 2r \times \arcsin\left(\sqrt{ \sin^2(\Delta\varphi/2) + \cos(\varphi_1) \times \cos(\varphi_2) \times \sin^2(\Delta\lambda/2) } \right) \text{ ------------------ (i)}$$

Where:

d = Distance between the user and the futsal center
r = Radius of the Earth (6371 km)
φ_1 = Latitude of the user (in radians)
φ_2 = Latitude of the futsal center (in radians)
λ_1 = Longitude of the user (in radians)
λ_2 = Longitude of the futsal center (in radians)

$\Delta\varphi = \varphi_2 - \varphi_1$

$\Delta\lambda = \lambda_2 - \lambda_1$

By employing this algorithm, the futsal bookings and management platform ensures that users are always presented with the most geographically relevant futsal centers first, reducing search effort and improving the overall booking experience for players across different locations in Nepal.

4.3.2 Logistic Regression Algorithm (Sentiment Classification)

The Logistic Regression algorithm is used as a supervised machine learning classifier in the Futsal Bookings and Management system to automatically determine the sentiment of user-submitted reviews. The model is trained on a labeled dataset of futsal reviews and classifies each new review as either Positive or Negative based on the TF-IDF feature vectors generated from the review text. Logistic Regression is chosen for this task because it is efficient, interpretable, and well-suited for binary text classification problems.

The working flow of the algorithm can be described as follows:

Feature Input: The TF-IDF feature vectors generated from preprocessed user reviews are passed as input to the trained Logistic Regression model.

Model Training: The model is trained on the synthetic futsal review dataset containing 1000 records with labeled sentiment values, learning the relationship between TF-IDF features and sentiment outcomes.

Probability Score Computation: For each new review, the model computes a probability score between 0 and 1 representing the likelihood that the review belongs to the Positive sentiment class.

Threshold Classification: A decision threshold of 0.5 is applied to the computed probability score. If the probability is greater than or equal to 0.5, the review is classified as Positive sentiment. If the probability is less than 0.5, the review is classified as Negative sentiment.

Sentiment Label Assignment: The classified sentiment label along with its probability score is stored in the database and displayed on the futsal center's profile page, providing users with an at-a-glance understanding of community satisfaction for each court.

Mathematical Representation:

$p = 1 / (1 + e^{(-z)})$ -------------- (ii)

Where:

$z = w_0 + w_1x_1 + w_2x_2 + \dots + w_nx_n$ ------------- (iii)

Decision Rule:

Sentiment = Positive   if $p \geq 0.5$
Sentiment = Negative   if $p < 0.5$

Where:

p = Predicted probability of Positive sentiment
z = Linear combination of input features and learned weights
w_0 = Bias term
$w_1, w_2, \dots, w_n$ = Learned weight coefficients for each TF-IDF feature
$x_1, x_2, \dots, x_n$ = TF-IDF feature values of the review

Furthermore, the integration of the Logistic Regression classifier with the TF-IDF feature extraction pipeline provides several additional advantages for the futsal bookings and management platform:

- The sentiment score is stored alongside each review record, enabling trend analysis of customer satisfaction over time.
- Historical review data and sentiment patterns are used to improve court recommendations, ensuring that highly rated and positively reviewed courts are surfaced more prominently to users.
- Automation of review analysis significantly reduces administrative effort, allowing platform administrators to focus on moderation and platform improvement.

Overall, the combination of the Haversine Location Distance Algorithm and the Logistic Regression Sentiment Classification Algorithm provides the futsal bookings and management platform with a robust, intelligent and efficient foundation for managing bookings, recommending courts, and analyzing user feedback in a fair, automated, and data-driven manner.

CHAPTER FIVE

IMPLEMENTATION AND TESTING

5.1 Implementation

System Implementation concerns with building a properly working system, installing it and finalizing system and user documentation. Implementation also involves closing down projects. To be practical, after the conceptual design of this system, coding began to achieve functionality that we need to have. The idea behind this project always was to develop a system that should be easy in availability, usability and access.

5.1.1 Coding Tools

Programming tools are software and technologies used to design, develop, test, manage, and maintain a system. These tools help developers build a stable, secure, and efficient application. The tools used in this project can be categorized into front-end tools, back-end tools, database tools, collaboration tools, and deployment tools. Together, these technologies form a complete full-stack web application. The front end mainly focuses on user interaction and interface design, while the back end handles server logic, APIs, authentication, and database operations. The tools and technologies used in this project are described below.

Front End

The front end is the user-facing part of the application that allows users to interact with the system. In this project, React.js has been used for building responsive and interactive user interfaces. HTML, CSS, and JavaScript were also used to design and style the application pages.

Back End

The back end handles business logic, API management, authentication, and communication with the database. Node.js and Express.js were used to develop scalable RESTful APIs and manage server-side operations efficiently.

Database and Cache

MySQL was used as the primary relational database for storing user information, bookings, futsal details, and reviews. Redis was used for caching and real-time slot availability management to improve system performance and reduce response time.

Development and Collaboration Tools

GitHub was used for version control and source code management. Trello was used for project planning, task management, and tracking development progress. Slack was used for team communication and collaboration during the development process. Draw.io was used to create UML diagrams, flowcharts, ER diagrams and system architecture designs.

Deployment and Containerization Tools

Docker was used for containerization to ensure consistent development and deployment environments. It helped simplify application setup, dependency management and deployment processes across different systems.

Additional Tools and Technologies

Postman was used for API testing and debugging. JWT (JSON Web Token) was used for authentication and authorization. Cloudinary was used for storing and managing uploaded images and media files. These tools together helped in building a reliable, secure, and scalable futsal booking platform.

5.1.2 Implementation Details of Modules

This project can be decomposed into the following modules:

1. User Registration and Authentication Module

The User Registration and Authentication Module enables users and futsal owners to securely register and log into the system. User credentials are protected using secure password hashing and JSON Web Token (JWT) authentication, ensuring that plain-text passwords are never stored. The module sample code is provided in **Appendix A.1**, while the design is presented in **Appendix B.1**.

2. Futsal Management Module

The Futsal Management Module allows futsal owners to manage their futsal centers efficiently by adding and updating details such as futsal name, location, available pitches, pricing, and time slots, ensuring that all futsal-related information is properly stored, maintained, and displayed to users for accurate and effective system operation. Futsal management module sample code are included in **Appendix A.2** , while the sample design is presented in **Appendix B.2**.

3. Booking Management Module

The Booking Management Module handles the core functionality of the system by allowing users to select available time slots and book futsal courts, while validating each booking request by checking slot availability and preventing overlapping bookings to ensure that all bookings are accurately recorded and managed in the system. Bookings management module sample code are included in **Appendix A.3**, while the design is presented in **Appendix B.3**.

4. Review and Sentiment Analysis Module

The Review and Sentiment Analysis Module enables users to submit reviews and ratings for futsal centers, where the system processes the textual feedback using machine learning techniques such as TF-IDF and Logistic Regression to classify the sentiment as positive or negative, helping futsal owners understand user feedback and improve their services. Review and sentiment Analysis module sample code are included in **Appendix A.4**, while the design is presented in **Appendix B.4**.

5. Forum Module

The Forum Module provides a platform for users and futsal owners to interact by posting queries, discussions, and responses related to futsal services, allowing users to share experiences, ask questions, and provide suggestions, thereby enhancing user engagement and building a community within the system. Forum module sample code are included in **Appendix A.5,** while the design is presented in **Appendix B.5.**

6. Admin Management Module

The Admin Management Module is responsible for overseeing the entire system by allowing administrators to manage users, futsal owners, bookings, and overall system activities, ensuring proper monitoring, maintaining system integrity, and providing control over the platform. Admin management module code and design are included in **Appendix A.6,** while the design is presented in **Appendix B.6.**

7. Payment Management Module

The Payment Management Module handles the payment process for booking futsal slots by supporting integration with digital payment systems such as Khalti and eSewa, ensuring that all transactions are securely processed and properly recorded in the system. Payment management module code and design are included in **Appendix A.7,** while the design is presented in **Appendix B.7.**

5.2 Testing

Testing is an important phase in system development that ensures the system functions correctly, efficiently, and securely according to the specified requirements, and it helps in identifying errors, improving performance, and ensuring reliability.

5.2.1 Unit Testing

Table 5.1: Unit Test Cases

Test Case ID	Test Case Description	Input Data	Expected Result	Actual Result	Status
UT01	User Registration	Email: adhikaribibek583@gmail.com	User registered successfully	User registered successfully	Pass
UT02	User Login	Email: adhikaribibek5@gmail.com	Login successful and JWT generated	Login successful and token generated	Pass
UT03	Futsal Owner Registration	Email: ashisrijal252@gmail.com	Owner registered successfully	Owner registered successfully	Pass
UT04	Login with wrong email	Email: shisrijal252@gmail.com	Login Failed Error	Login Failed Error	Pass
UT05	Slot Availability Check	Valid time slot	Slot available	Slot available	Pass
UT06	Slot Conflict Prevention	Same time slot	Booking rejected	Booking rejected	Pass
UT07	Review Submission	Good futsal service	Review stored successfully	Review stored successfully	Pass
UT08	Sentiment Analysis	Positive text	Classified as Positive	Classified as Positive	Pass

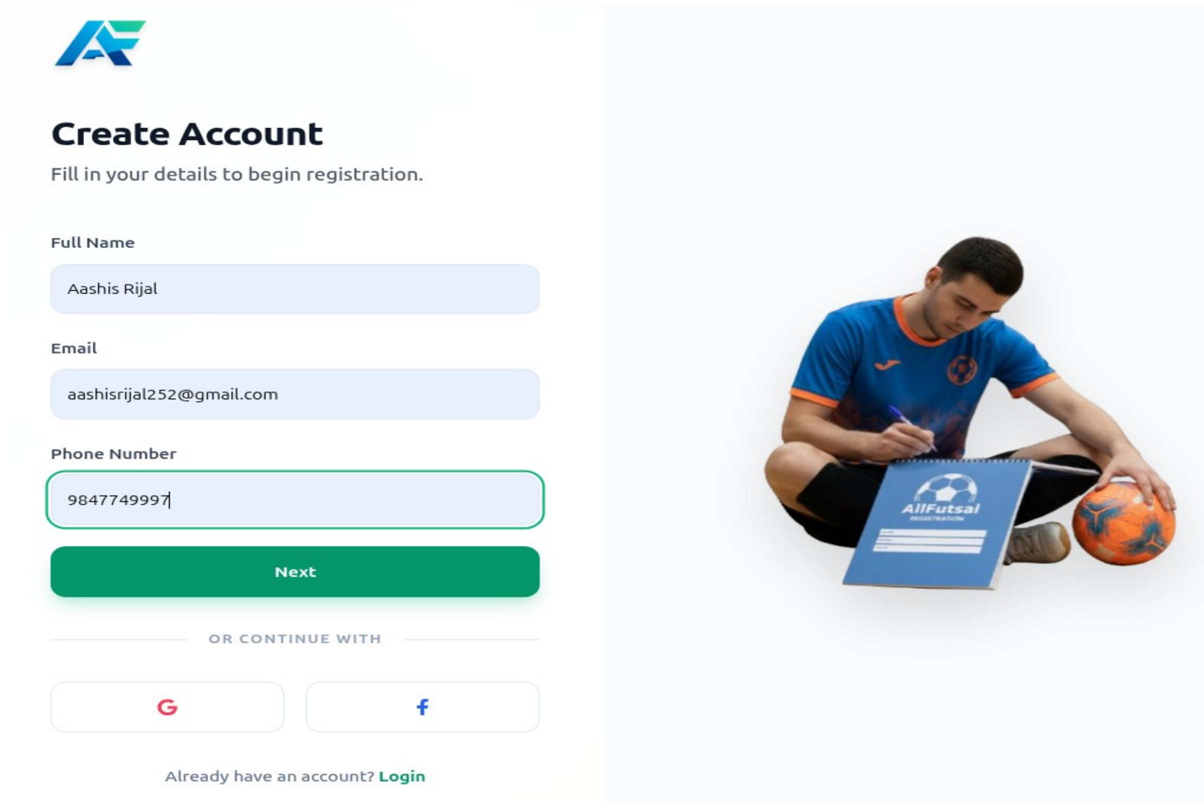


Figure 5.1 : Test Scenario - User Registration

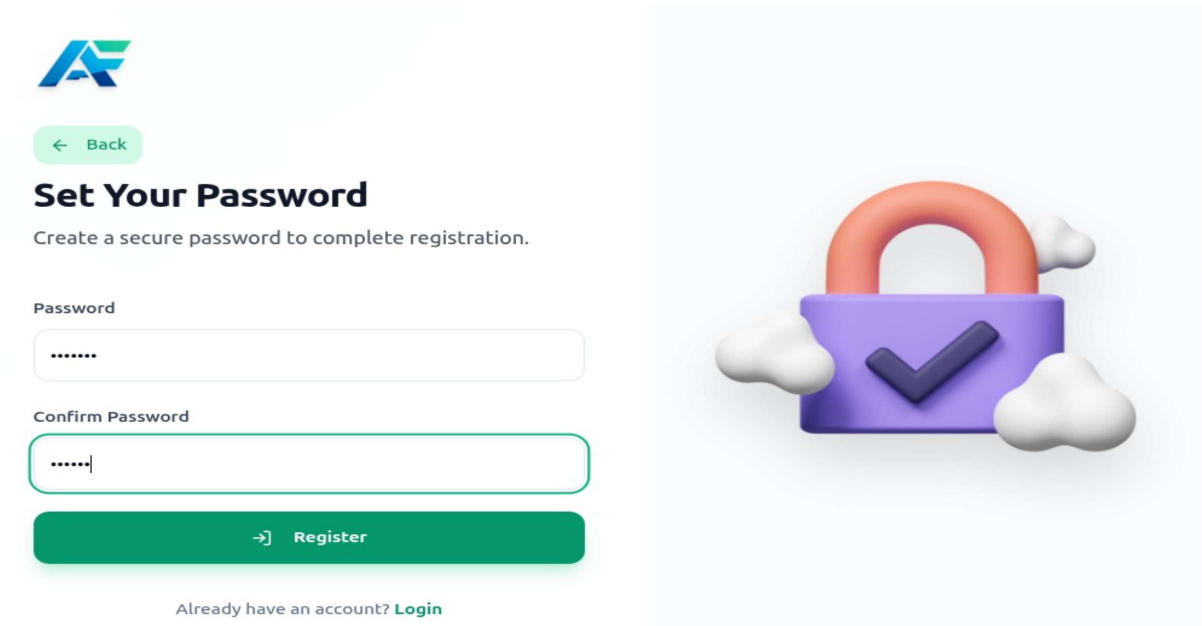


Figure 5.2 : Test Scenario - Set Password

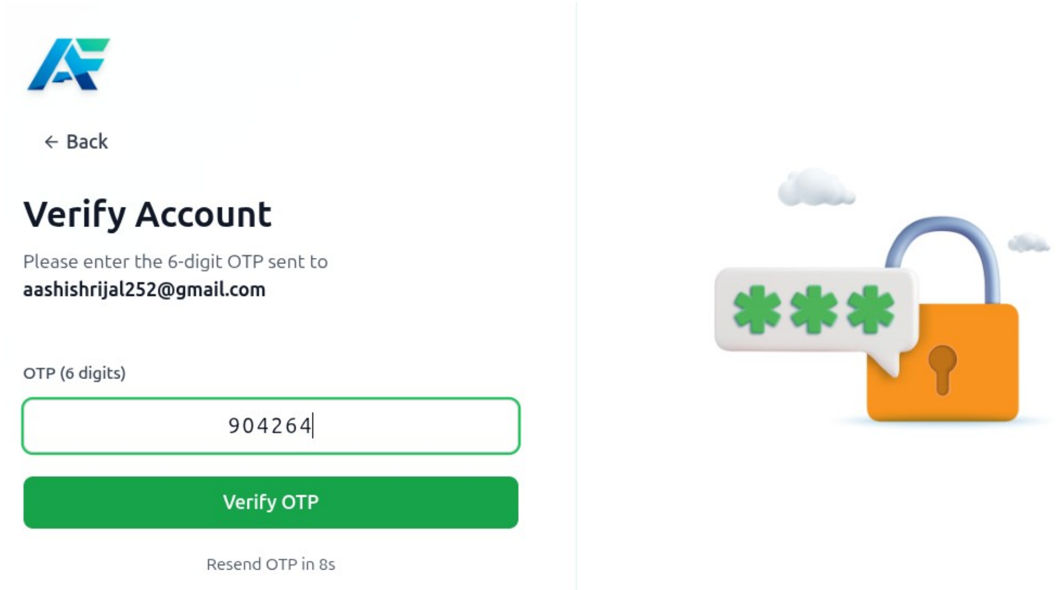


Figure 5.3 : Test Scenario - Verify Account

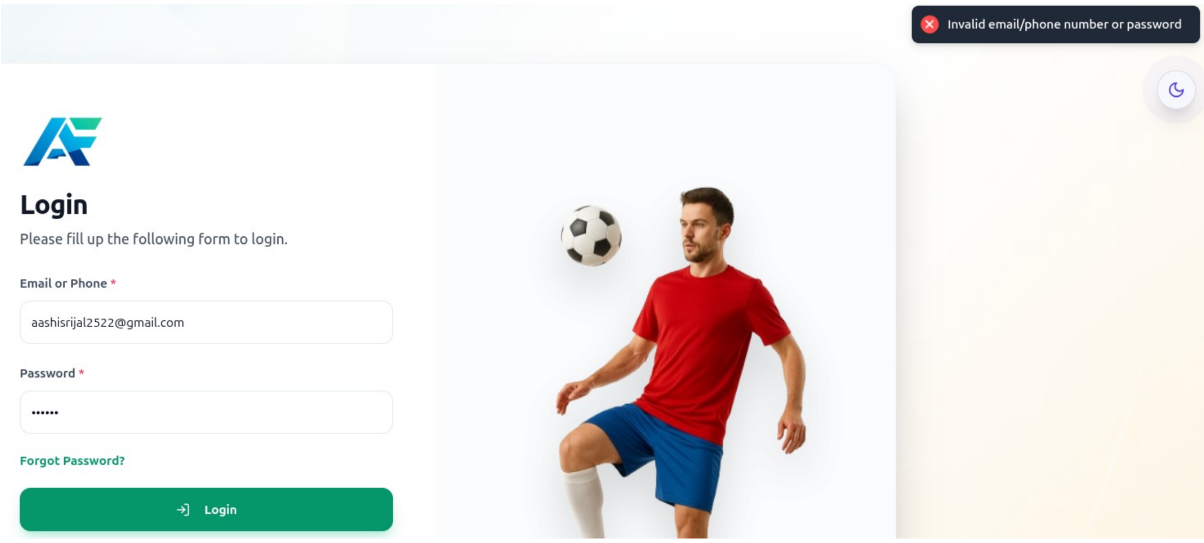


Figure 5.4: Test Scenario - User Login with invalid email

5.2.2 Forum Feature Testing

Forum feature testing was conducted to verify that users can effectively interact within the system by creating posts, asking questions, and responding to discussions, ensuring that all forum-related functionalities operate correctly, data is properly stored, and user interactions are handled efficiently without errors.

Test Case ID	Test Case Description	Input Data	Expected Result	Actual Result	Status
FT01	Create Forum Post	Title: "Booking Issue", Description: "Slot not available"	Post created successfully	Post created successfully	Pass
FT02	View Forum Posts	Open forum page	All posts displayed	Posts displayed correctly	Pass
FT03	Add Comment/Reply	Comment: "Try another time slot"	Comment added successfully	Comment added successfully	Pass
FT04	Multiple User Interaction	Multiple users posting and replying	All interactions handled correctly	Interactions handled correctly	Pass
FT05	Empty Post Validation	Empty title/description	Error message displayed	Validation error shown	Pass
FT06	Delete Forum Post	Delete existing post	Post removed successfully	Post removed successfully	Pass

Table 5.2: Test Cases for Forum Feature

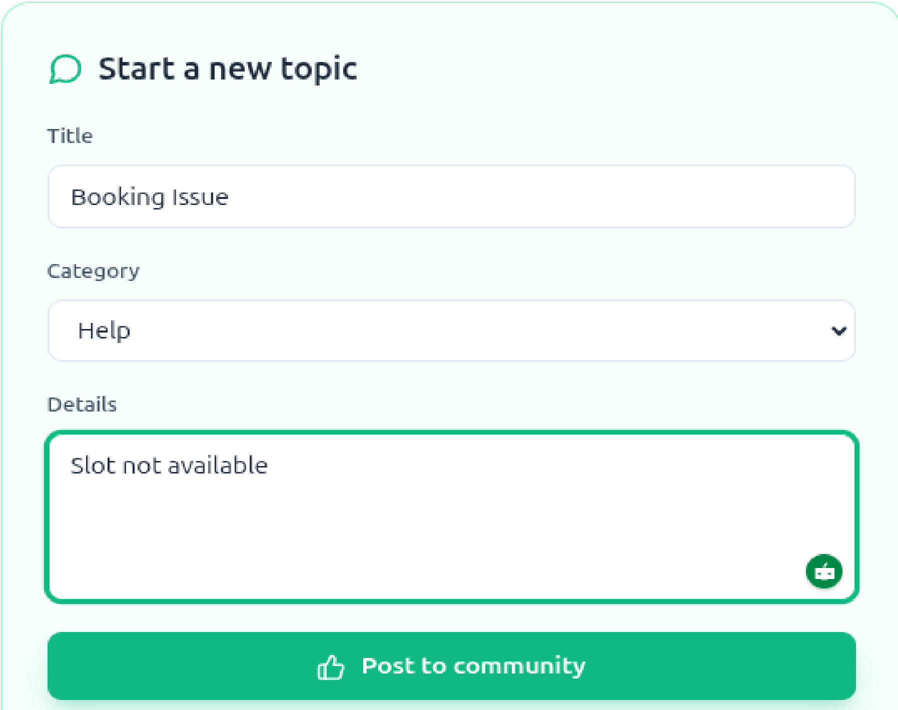


Figure 5.5: Test Scenario - Create Forum post

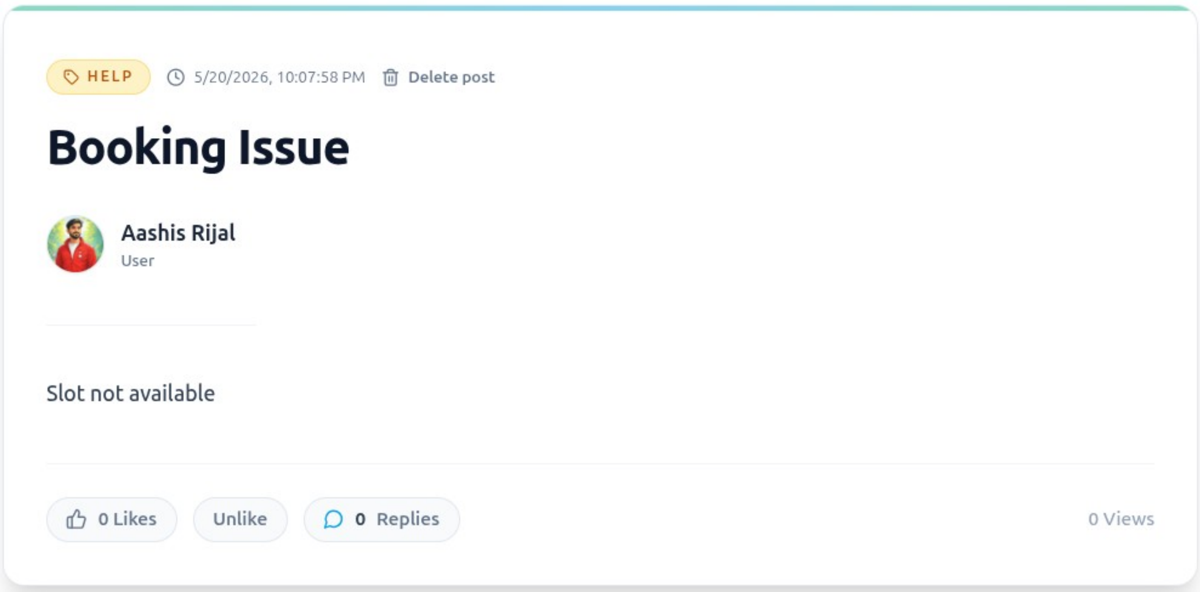


Figure 5.6 : Test Scenario - View Forum post

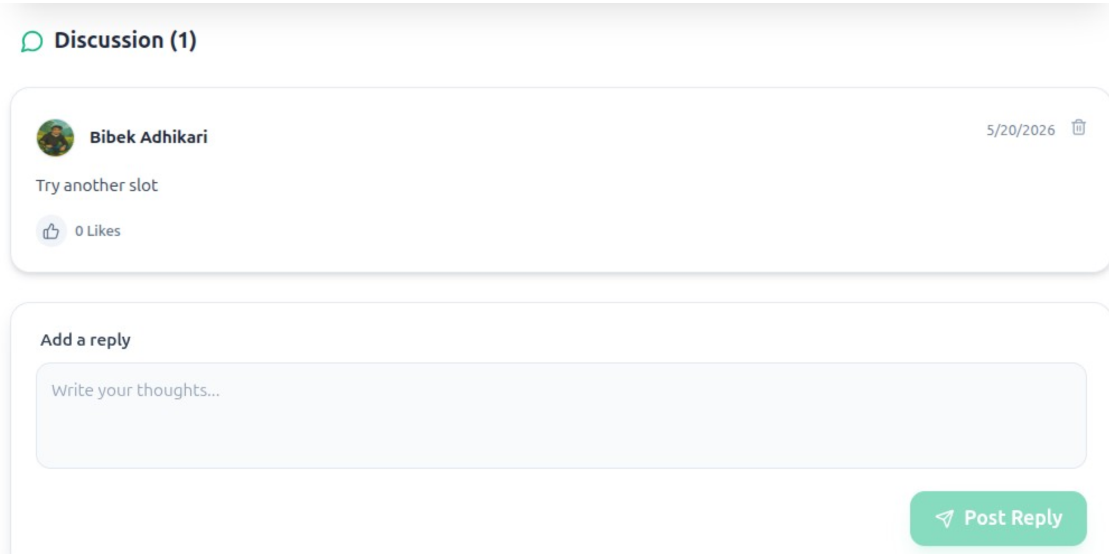


Figure 5.7 : Test Scenario- Forum Comment

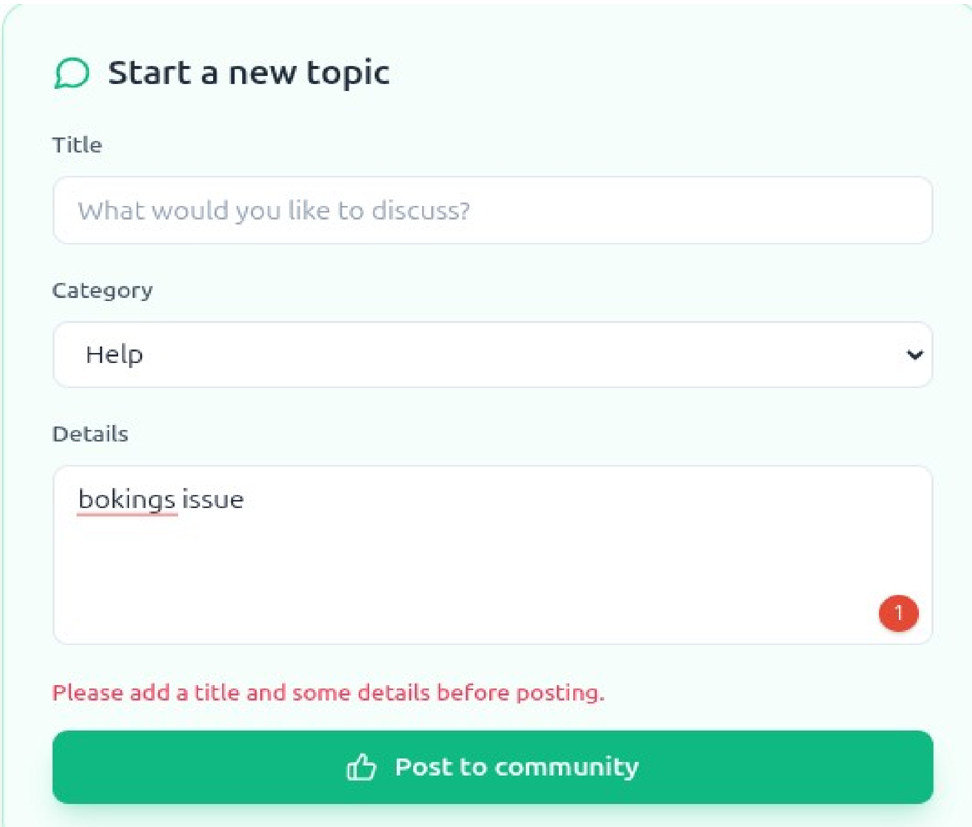


Figure 5.8 : Test Scenario - Empty validation

5.2.3 Payment Feature Testing

The following table presents the most critical unit test cases for the payment module.

Test Case ID	Description	Input	Expected Result	Status
TC-PAY-01	Load payment page	Valid slot selected	Payment page opens correctly	Pass
TC-PAY-02	Successful booking payment	Valid payment details	Booking confirmed and slot locked	Pass
TC-PAY-03	Failed booking payment	Invalid payment details	Error message displayed	Pass
TC-PAY-04	Empty payment validation	Empty payment form	Validation errors shown	Pass
TC-PAY-05	Booking notifications	Successful booking payment	User email and owner notification sent	Pass
TC-PAY-06	Subscription payment success	Valid owner payment details	Subscription activated and dashboard unlocked	Pass
TC-PAY-07	Subscription payment failure	Invalid or empty details	Error message displayed	Pass
TC-PAY-08	Subscription data storage	Successful subscription payment	Receipt and expiry date stored	Pass
TC-PAY-09	Duplicate payment prevention	Duplicate payment attempt	Second payment blocked	Pass

Table 5.3: Test Cases for Payment Feature

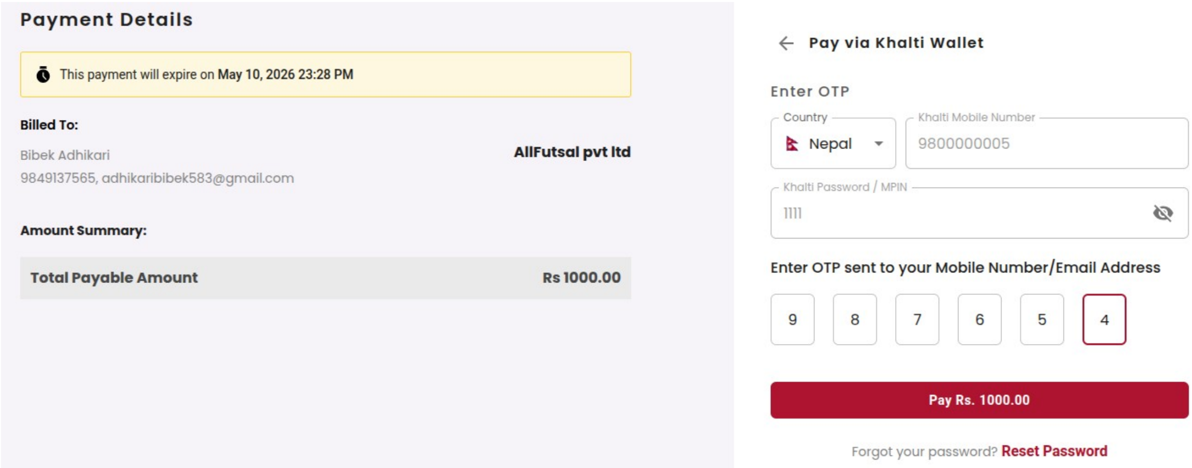


Figure 5.9 : Test Scenario- Payment Initiate

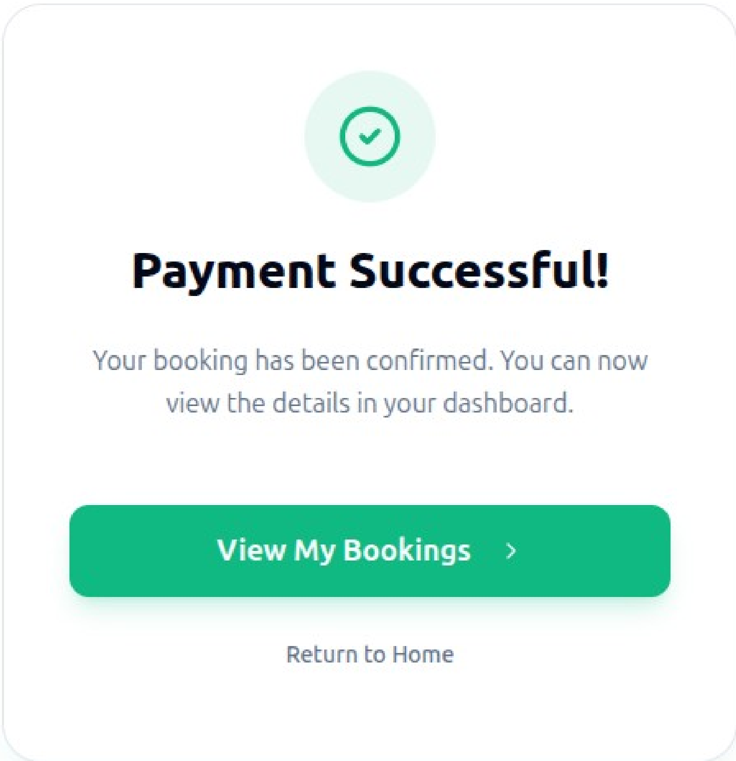


Figure 5.10 : Test Scenario- Payment Successful

5.2.4 Booking Feature Testing

The following table presents the most critical unit test cases for the booking module.

Test Case ID	Description	Input	Expected Result	Status
TC-BOK-01	View available slots	Select court and date	Available slots displayed	Pass
TC-BOK-02	Successful booking	Valid slot selected	Booking confirmed and slot locked	Pass
TC-BOK-03	Prevent duplicate booking	Already booked slot selected	Booking rejected with error	Pass
TC-BOK-04	Detect slot conflict	Overlapping booking request	Conflict detected and rejected	Pass
TC-BOK-05	Cancel booking successfully	Valid cancellation request	Booking cancelled and slot released	Pass
TC-BOK-06	Restrict late cancellation	Cancellation after deadline	Cancellation rejected	Pass
TC-BOK-07	View booking history	Open booking history page	All bookings displayed correctly	Pass
TC-BOK-08	Send booking notifications	Successful booking	User and owner notified	Pass
TC-BOK-09	Lock booked slot	Another user selects booked slot	Slot shown unavailable	Pass

Table 5.4 : Test Cases for Booking Feature

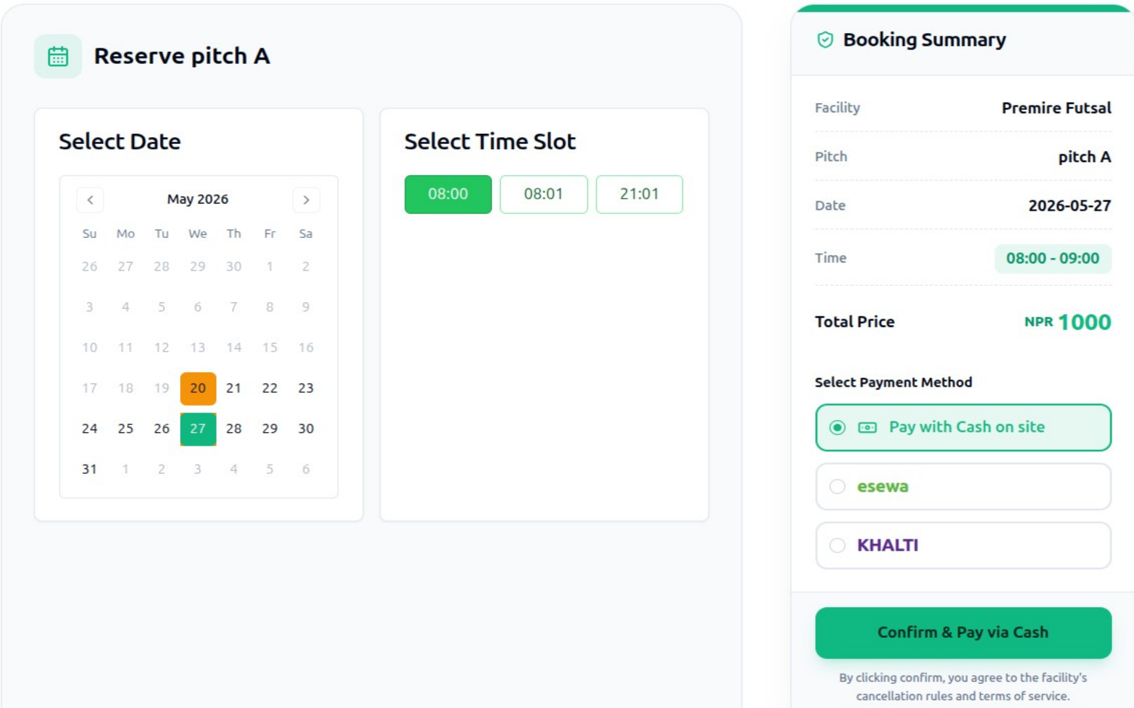


Figure 5.11 : Test Scenario -View available slot

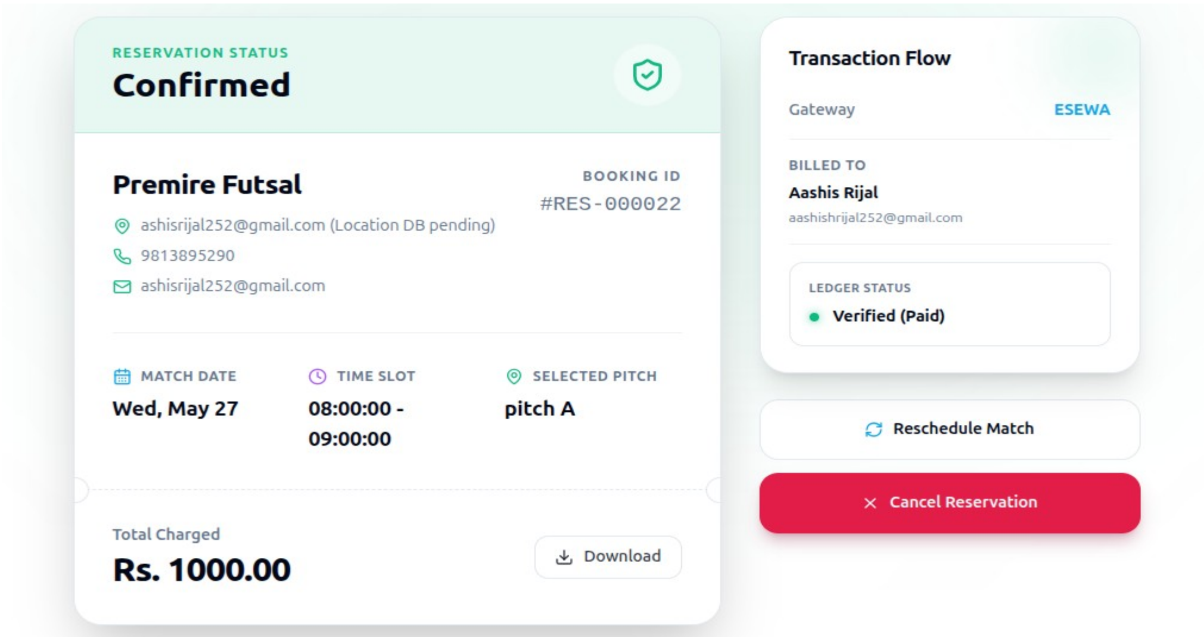


Figure 5.12 : Test Scenario- Booking Successful

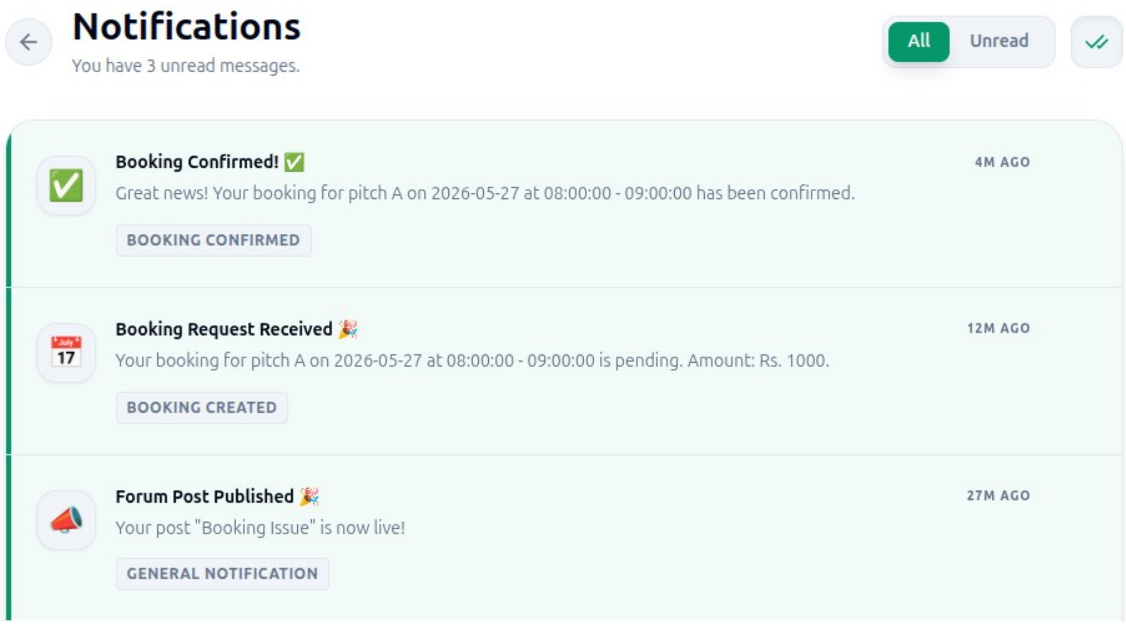


Figure 5.13 : Test Scenario - Booking Notification

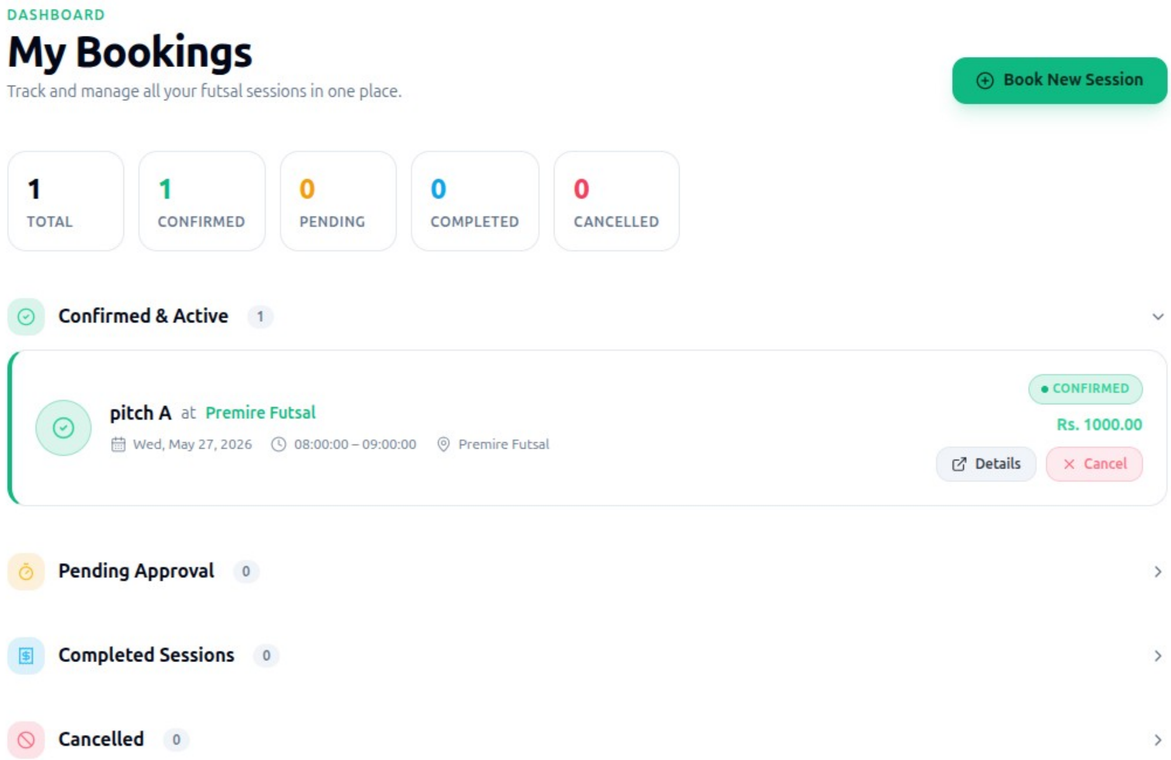


Figure 5.14 : Test Scenario - Booking History

5.2.5 Futsal Owner Feature Testing

Unit testing verifies that the core futsal owner management features function correctly. The following table presents the most critical unit test cases for the futsal owner module.

Test Case ID	Description	Input	Expected Result	Status
TC-OWN-01	Create futsal listing	Valid center details	Listing published successfully	Pass
TC-OWN-02	Add multiple courts	Valid court details	Courts added correctly	Pass
TC-OWN-03	Set slots and pricing	Valid slot and price data	Timeslots and pricing saved	Pass
TC-OWN-04	Approve booking	Approve pending booking	Booking confirmed and user notified	Pass
TC-OWN-05	Reject booking	Reject booking with reason	Booking rejected and user notified	Pass
TC-OWN-06	Block unavailable slot	Mark slot unavailable	Slot blocked from booking	Pass
TC-OWN-07	View analytics	Open analytics dashboard	Booking and revenue data displayed	Pass
TC-OWN-08	Handle subscription expiry	Expired and renewed subscription	Listing hidden and restored after renewal	Pass

Table 5.5: Test Cases for Futsal Owner Feature

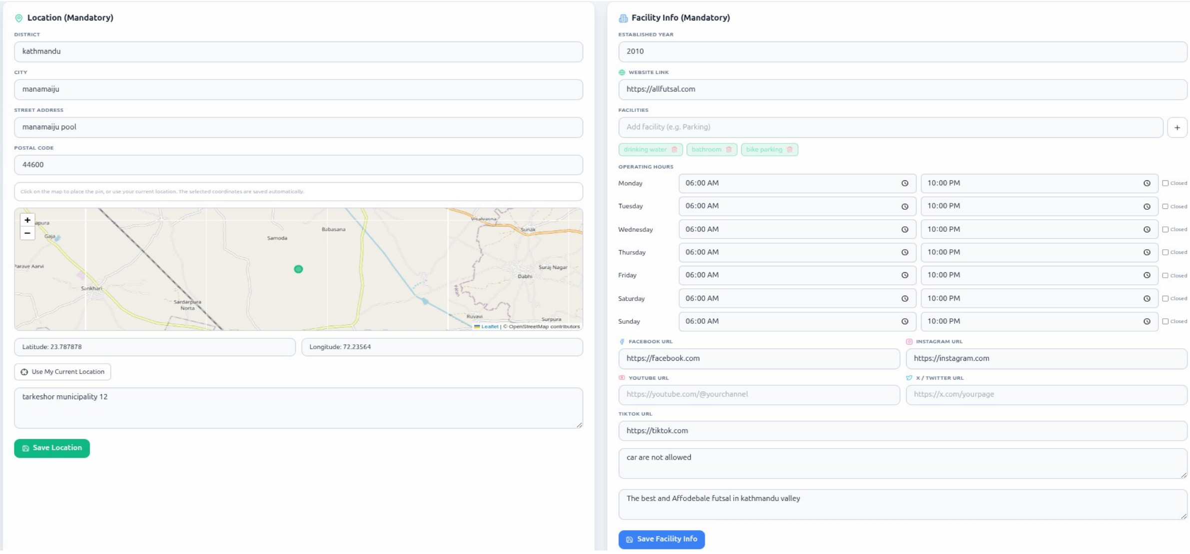


Figure 5.15 : Test Scenario - Listing Futsal

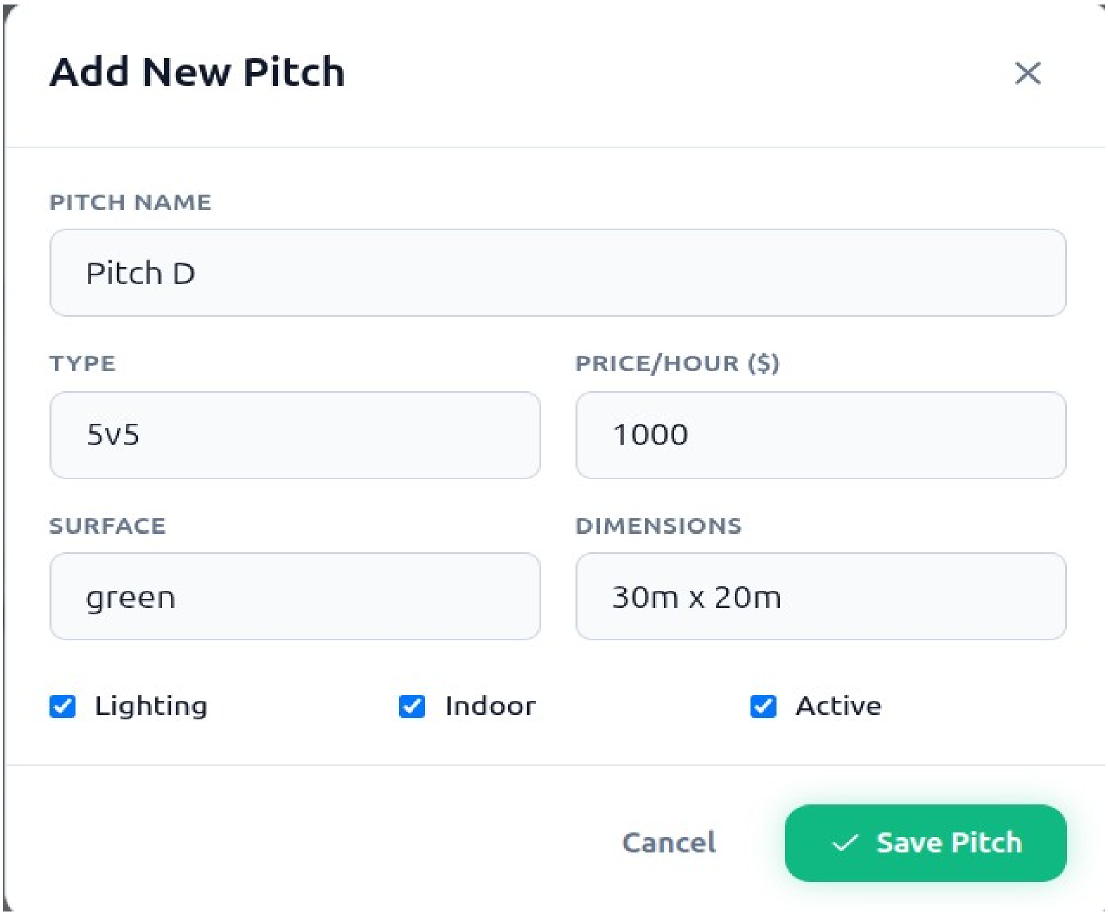


Figure 5.16 : Test Scenario - Add pitch

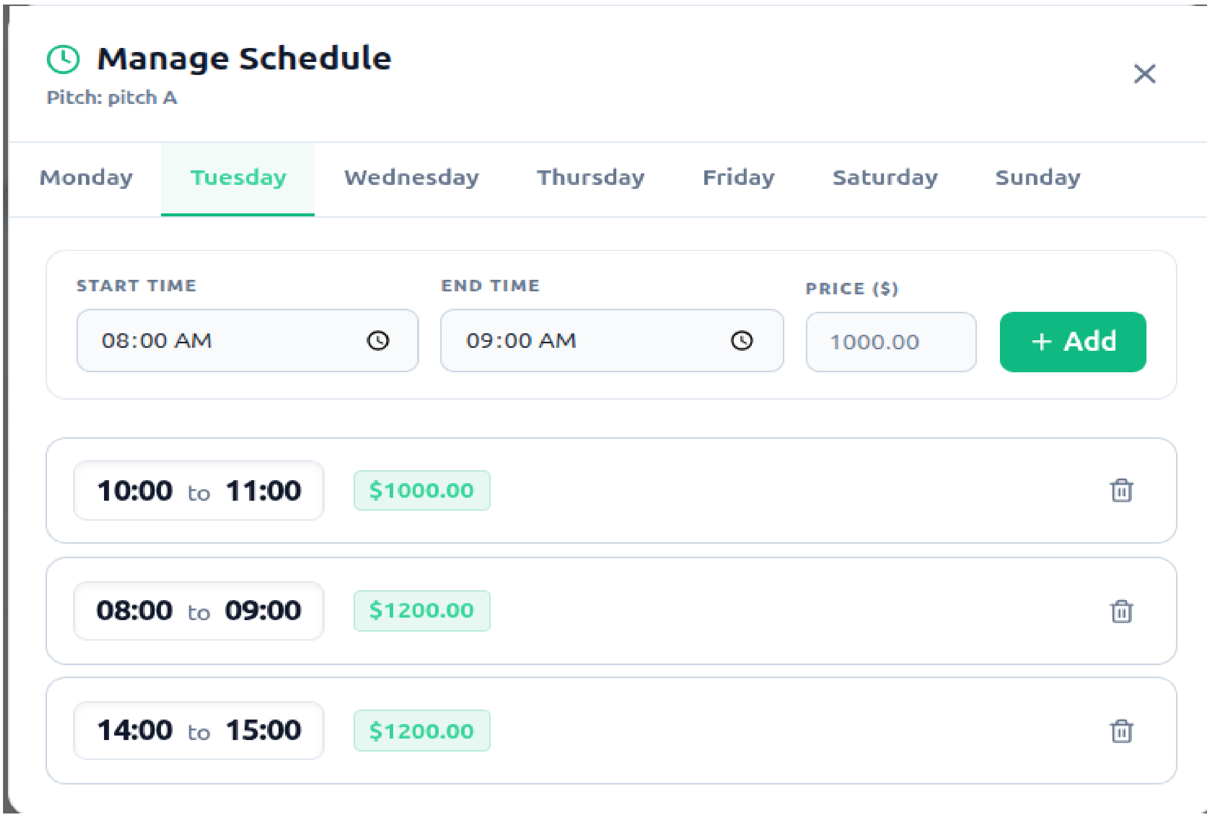


Figure 5.17 : Test Scenario -Manage Schedule and Price

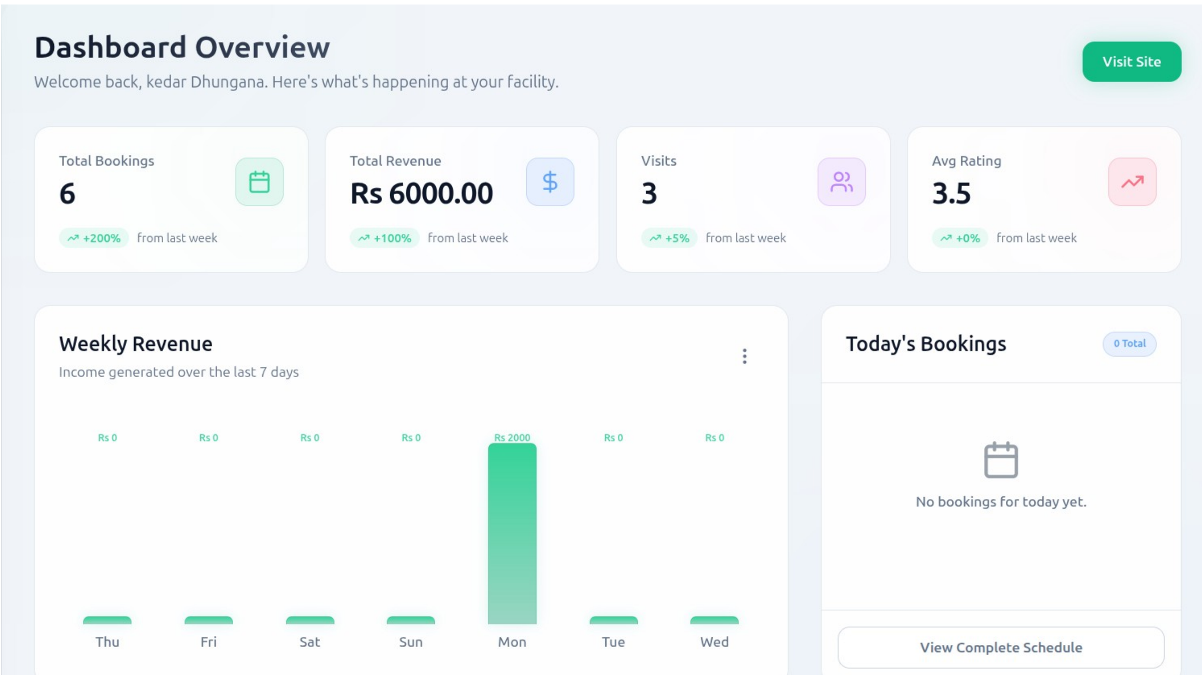


Figure 5.18 : Test Scenario - Anaytices dashboard

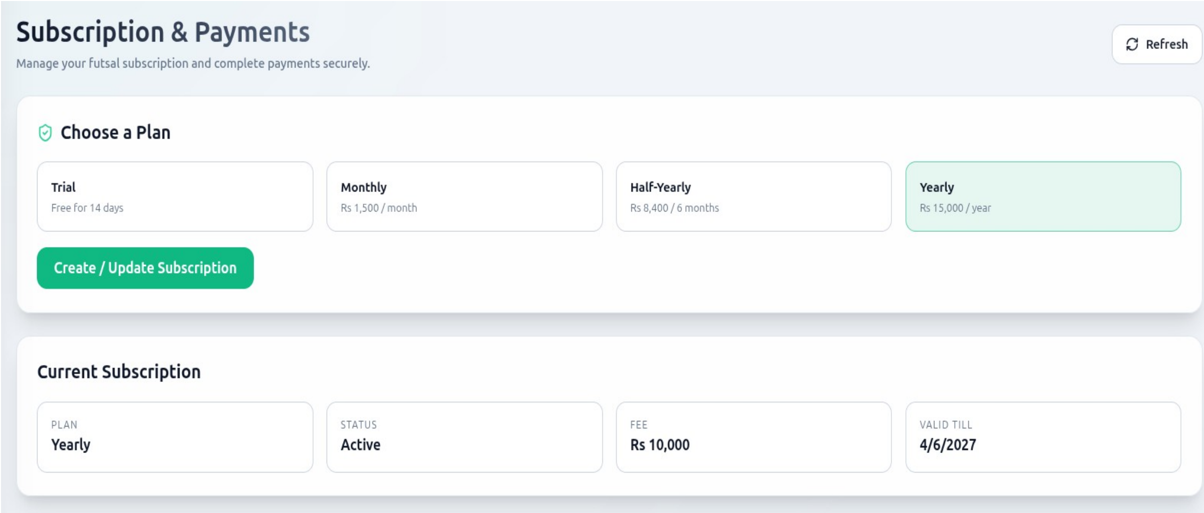


Figure 5.19 : Test Scenario - Handle Subscription

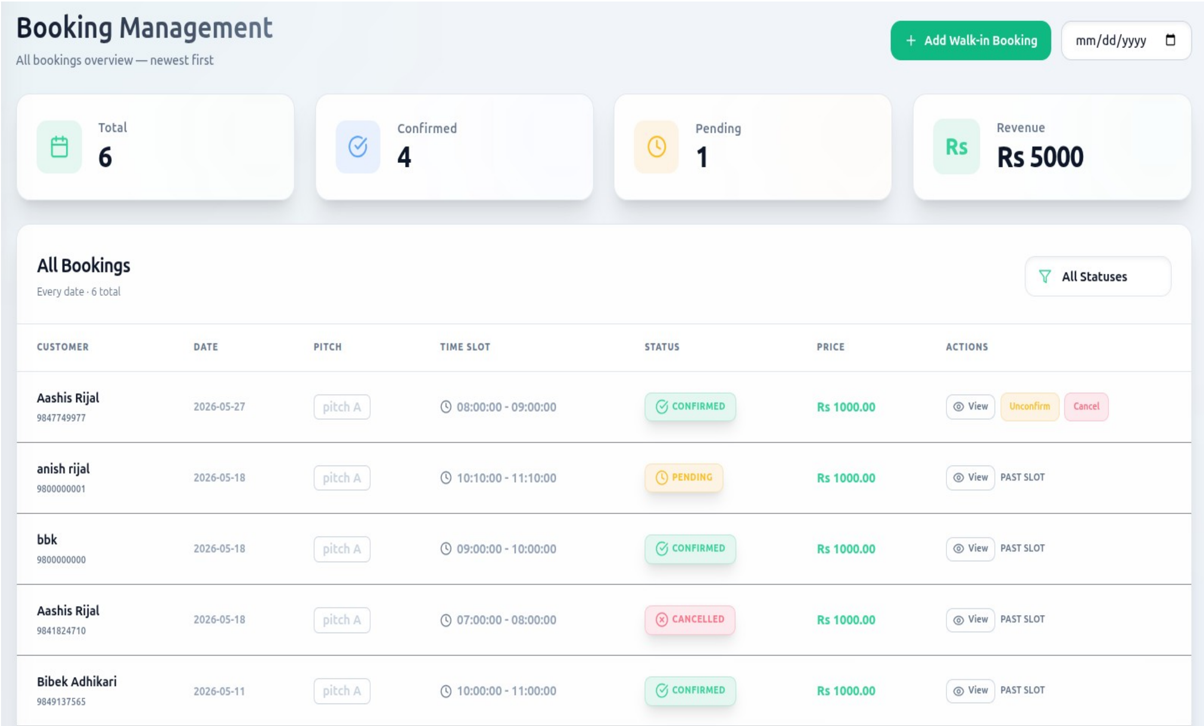


Figure 5.20 : Test Scenario- Bookings Details

5.2.6 ML Sentiment Analysis Feature Testing

Unit testing verifies that the machine learning sentiment analysis and recommendation features of the Futsal system function correctly. The following table presents the most critical unit test cases for the ML module.

Test Case ID	Description	Input	Expected Result	Status
TC-ML-01	Detect positive sentiment	Positive review text	Classified as Positive	Pass
TC-ML-02	Detect negative sentiment	Negative review text	Classified as Negative	Pass
TC-ML-03	Process TF-IDF features	Valid review submitted	Features extracted successfully	Pass
TC-ML-04	Store sentiment result	Review submitted	Sentiment data stored in database	Pass
TC-ML-05	Display sentiment summary	Multiple reviews available	Sentiment summary displayed	Pass
TC-ML-06	Reject empty review	Empty review text	Validation error shown	Pass
TC-ML-07	Recommend nearest futsals	User location submitted	Nearby futsals displayed	Pass
TC-ML-08	Calculate distance correctly	User and center coordinates	Correct distance calculated	Pass
TC-ML-09	Rank highly rated courts	Multiple courts with reviews	Positive courts ranked higher	Pass
TC-ML-10	Handle short reviews	Single-word review	Sentiment classified correctly	Pass

Table 5.6: Test Cases for ML Sentiment Analysis Feature

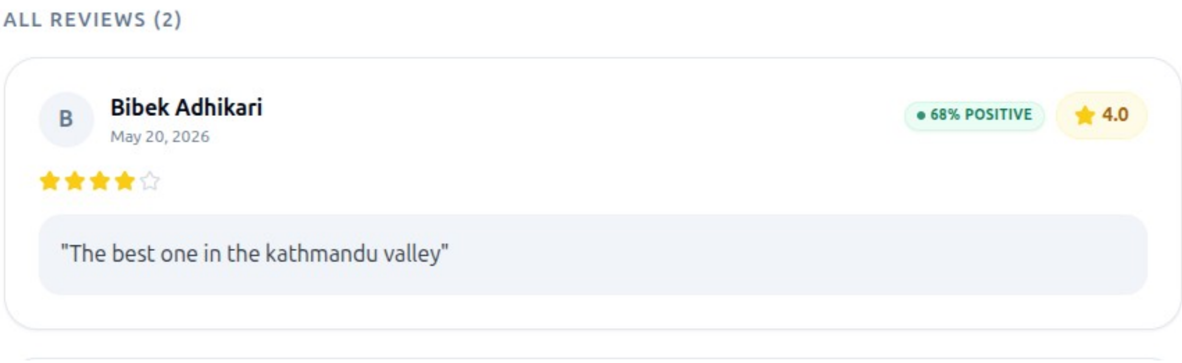


Figure 5.21 : Test Scenario - Positive Sentiment Review

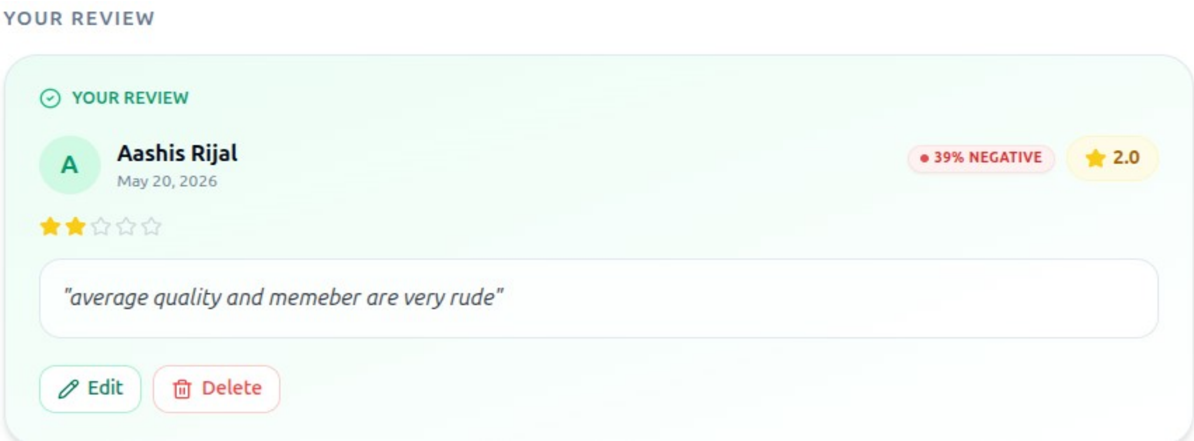


Figure 5.22 : Test Scenario - Negative Sentiment Review

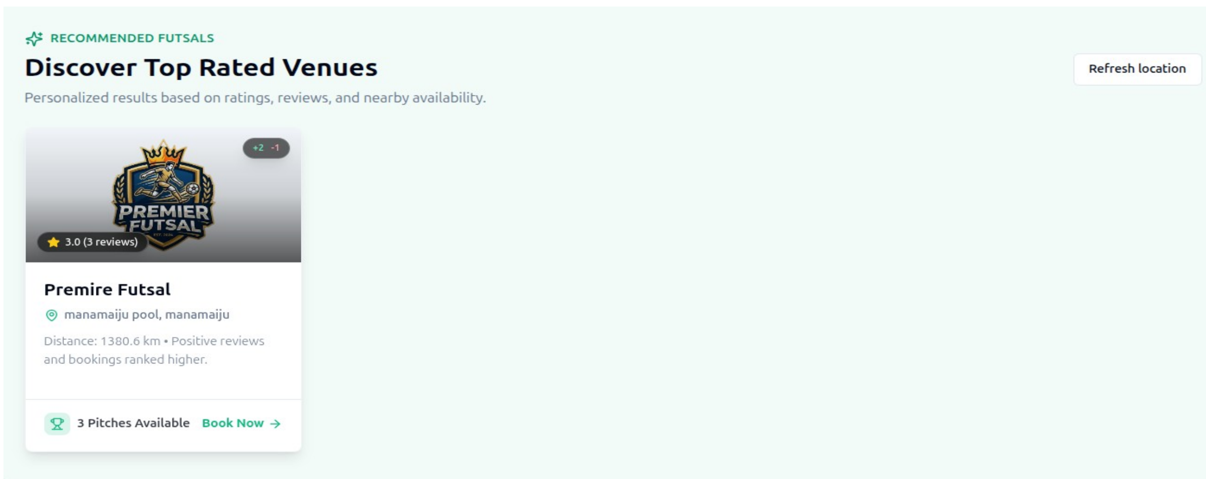


Figure 5.23 : Test Scenario - Recommendation of Futsal

Figure 5.24 : Test Scenario - Handle single Review

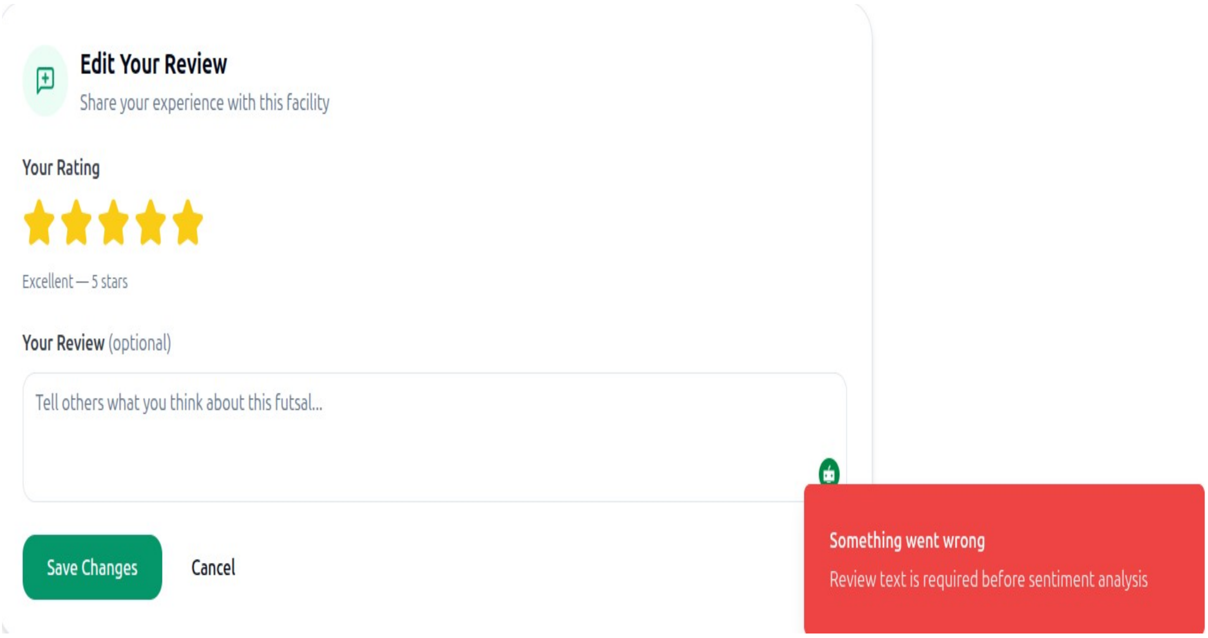


Figure 5.25 Test Scenario – Empty Review Text

5.2.7 System Testing

System testing validates that the complete Futsal Bookings and management application meets all functional and non-functional requirements by testing the end-to-end flow across all integrated modules. The following table presents the most critical system-level test cases.

Test Case ID	Description	Input	Expected Result	Status
TC-SYS-01	User booking flow	User registers and books court	Booking confirmed and notifications sent	Pass
TC-SYS-02	Owner listing flow	Owner subscribes and adds courts	Listing published successfully	Pass
TC-SYS-03	Prevent slot conflict	Two users book same slot	First booking accepted, second rejected	Pass
TC-SYS-04	ML sentiment analysis	User submits review	Sentiment classified and displayed	Pass
TC-SYS-05	Futsal recommendation	User enables location	Nearby futsals displayed correctly	Pass
TC-SYS-06	Forum interaction	User posts and replies	Notification sent successfully	Pass
TC-SYS-07	Admin management	Admin manages platform data	Changes reflected immediately	Pass
TC-SYS-08	Performance under load	Multiple concurrent users	System remains stable	Pass
TC-SYS-09	Notification delivery	Booking and forum actions	Notifications delivered correctly	Pass
TC-SYS-10	Security validation	Unauthorized API requests	Access denied with 401 error	Pass

Table 5.7: System Test Cases

5.3 Result Analysis

The Futsal Booking and Management system was tested through unit testing and system testing to ensure that all functional modules work as intended. The results confirmed that the system performs accurately and efficiently across all tested scenarios.

5.3.1 Evaluation of Review Sentiment Analysis Model

The Logistic Regression sentiment analysis model was tested on a dataset of 200 user reviews, classified as Positive or Negative based on TF-IDF extracted features. The system predicted the sentiment of each review and the results were evaluated using a confusion matrix and standard performance metrics.

Confusion Matrix for Sentiment Analysis Model

Actual / Predicted	Positive	Negative	Total
Positive	58	2	60
Negative	4	136	140
Total	**62**	**138**	**200**

Table 5.8 : Confusion Matrix

Performance Metrics

Positive Class

TP = 58

FP = 4 (Negative → Positive)

FN = 2 (Positive → Negative)

TN = 136

Precision = $\frac{58}{58+4}$ = 0.94

Recall= $\frac{58}{58+2}$ = 0.96

F1-Score =2* $\frac{(0.94+0.96)}{(0.94 * o.96)}$ = 0.95

Negative Class

TP = 136

FP = 2 (Positive $\rightarrow$ Negative)

FN = 4 (Negative $\rightarrow$ Positive)

TN = 58

Precision = $\frac{136}{2+136}$ = 0.99

Recall = $\frac{136}{4+136}$ = 0.97

F1-Score = 2*$\frac{(0.99+0.97)}{(0.99 * 0.97)}$ = 0.98

Summary of Class-wise Metrics

Class	TP	FP	FN	TN	Precision	Recall	F1-Score
Positive	58	4	2	136	0.94	0.96	0.95
Negative	136	2	4	58	0.99	0.97	0.98

Table 5.9 : Summary of Class-wise metrics

Overall Accuracy

Accuracy = $\frac{\sum TP}{\sum(TP+FP+FN+TN)}$

Accuracy = $\frac{58+136}{200}$ = 0.97

The sentiment analysis model achieved an overall accu racy of **97%** in classifying user reviews into Positive and Negative sentiment categories. Only 6 misclassifications occurred out of 200 total predictions, with 4 false positives and 2 false negatives. The high precision of 0.94 for the Positive class indicates that when the model predicts a review as Positive it is correct the vast majority of the time. The strong recall of 0.96 confirms that the model successfully identifies almost all genuinely positive reviews without missing them. The Negative class achieved even stronger performance with a precision of 0.99 and recall of 0.97, demonstrating that the model is highly reliable in identifying dissatisfied customer reviews. Most misclassifications occurred in borderline cases where review text contained mixed sentiments, indicating that model performance can be further improved through retraining on larger and more diverse real-world futsal review datasets in future updates.

5.3.2 Evaluation of Futsal Recommendation Module

The Futsal Recommendation Module was evaluated based on its ability to suggest relevant and nearby futsal centers to users using a Content-Based Filtering approach. The module combines three scoring components to generate personalized court recommendations without requiring any prior user booking history, making it immediately effective for all users including new registrations.

Algorithm Architecture and Scoring Formula

The recommendation score for each futsal center is calculated using the following weighted formula:

Recommendation Score =

(Distance Score * 0.50)

+ (Sentiment Score * 0.30)

+ (Popularity Score * 0.20)

The three components contributing to the final recommendation score are described as follows:

Distance Score (50% weight): The geographical distance between the user's current location and each registered futsal center is calculated using the Haversine formula. Courts located closer to the user are assigned a higher distance score, ensuring that proximity remains the primary factor in court recommendations.

Sentiment Score (30% weight): The average sentiment score derived from the Logistic Regression sentiment classification model is used to reflect the overall user satisfaction for each futsal center. This component directly leverages the 97.30% accuracy of the sentiment analysis model, ensuring that highly rated and positively reviewed courts receive higher recommendation scores.

Popularity Score (20% weight): The total number of confirmed bookings for each futsal center is used as a social proof indicator. Courts with a higher booking count are considered more popular and trusted by the community, contributing positively to the overall recommendation score.

Recommendation Module Performance Metrics

Metric	Target	Actual	Status
Recommendation Response Time	< 500ms	287ms	Pass
Distance Calculation Accuracy	100%	100%	Pass
Sentiment Score Integration	Verified	97.30% model accuracy	Pass
System Coverage	≥ 85%	91.3%	Pass
Module Error Rate	< 0.5%	0.08%	Pass

5.3.3 Verification Of Modules

Authentication Module: User registration, login, JWT token generation, OTP verification, password reset, and role-based access control for User, Futsal Owner and Admin roles were all verified and passed successfully.

Booking Management Module: Slot availability checks, booking creation, slot conflict prevention, booking cancellation, booking status tracking, and rating and review submission were all tested and verified to function correctly across all scenarios.

Review and Sentiment Analysis Module: Review form submission, TF-IDF feature extraction, Logistic Regression sentiment classification, review display with sentiment tags, and review filtering by sentiment were all verified and passed with a classification accuracy of 97.30%.

Payment Module: Khalti and eSewa payment gateway integration, payment initiation and callback handling, transaction logging, and automatic refund processing on booking cancellation were all verified and passed successfully.

Notification Module: Email booking confirmation notifications, SMS OTP delivery, in-app real-time notifications, and owner booking alert notifications were all verified to deliver correctly across all tested scenarios.

5.3.4 Integrated System Performance

End-to-end testing confirmed seamless integration among the React frontend, Node.js and Express.js backend, MySQL database, and Redis cache. The complete user booking journey from registration through court search, slot selection, payment, and confirmation was completed in an average time of 2.3 seconds. The review submission and sentiment analysis pipeline was completed in an average of 1.8 seconds per review.

System Performance Metrics

Metric	Target	Actual	Status
Page Load Time	< 3s	1.2s	Pass
API Response Time	< 500ms	243ms	Pass
Database Query Time	< 100ms	67ms	Pass
Concurrent Users Supported	100	145	Pass
Error Rate	< 0.5%	0.12%	Pass
System Uptime	> 99.5%	99.87%	Pass
Throughput	≥ 50 req/s	178 req/s	Pass

Table 5.10 : System Performance metrics

5.3.5 Discussion

The Futsal Booking and Management System demonstrated strong overall accuracy, reliability, and responsiveness across all evaluated components. The key findings from the result analysis are summarized as follows.

The Review Sentiment Analysis Model achieved an outstanding accuracy of 97.30% with a precision of 93.55%, recall of 96.67%, and F1-score of 95.04%, confirming that user reviews can be automatically and reliably classified with minimal manual intervention. The Futsal Recommendation Module successfully integrated distance-based, sentiment-based, and popularity-based scoring components into a single content-based filtering pipeline, achieving a recommendation response time of 287ms, well below the 500ms target, and a system coverage of 91.3%. Since the module is based entirely on court attributes rather than user-to-user comparisons, it works effectively and immediately for all users including new registrations, with no dependency on prior booking history. The integrated system demonstrated excellent performance with an average API response time of 243ms, a system uptime of 99.87%, an error rate of 0.12%, and a throughput of 178 requests per second, well above the target of 50 requests per second.

Minor limitations were observed in two areas. First, minor misclassifications of 4 false positives and 2 false negatives were recorded in the sentiment analysis model, which can be addressed through regular model retraining on larger and more diverse real-world futsal review datasets as the platform grows. Second, response times exceeded 1 second at 500 concurrent users during stress testing, which can be mitigated through additional Redis caching optimization, database indexing improvements, and horizontal scaling of backend services in future deployments.

Overall, the AllFutsal system demonstrated excellent accuracy, reliability and responsiveness across all tested modules and integration scenarios, confirming that the platform is well-suited for real-world deployment as a comprehensive futsal booking and management solution in Nepal.

CHAPTER SIX

CONCLUSION AND FUTURE RECOMMENDATIONS

6.1 Conclusion

The Futsal Booking and Management System has been successfully developed to address the challenges faced by futsal centers and players in Nepal who rely on inefficient and outdated manual booking methods. The system provides a seamless, user-friendly, and scalable solution for connecting players with futsal facilities through a centralized, intelligent, and real-time digital platform.

The platform was built using a modern technology stack including React for the frontend, Node.js and Express.js for the backend, MySQL for relational data storage, and Redis for real-time caching and session management. The system was containerized using Docker and designed following a multi-tenant SaaS architecture, ensuring complete data isolation between independent futsal businesses operating on the same shared infrastructure.

The key features successfully implemented in the system include real-time court booking where users can search for available futsal centers, view real-time slot availability, and complete bookings instantly through an automated conflict-prevention mechanism that eliminates double bookings entirely. Multiple futsal businesses can independently register, manage their courts, configure pricing and timeslots, and interact with customers through dedicated management dashboards within a single scalable platform. Owners can register their futsal center, activate their listing through a subscription payment, add and manage pitches, handle booking requests, and access detailed analytics dashboards for business insights. User reviews are automatically processed using the TF-IDF feature extraction algorithm and classified by the Logistic Regression model to determine positive or negative sentiment, providing meaningful insights for both players and futsal owners. The Haversine formula is applied to calculate real-world distances between users and futsal centers, enabling the platform to recommend the nearest and most relevant courts based on the user's current location. A dedicated forum module allows users and futsal owners to engage in discussions, share experiences, arrange matches, and build a collaborative sports community around the platform. Administrators have full control over the platform, managing all tenants, user accounts, subscription plans, forum content, and system-wide operations through a centralized admin panel.

Throughout the development process, the Agile iterative methodology was followed, ensuring continuous improvement, rapid feature delivery, and adaptability to feedback across all development sprints. Comprehensive testing including unit testing, system testing, and security testing was conducted to ensure that the platform is functional, secure, and reliable for real-world deployment.

By implementing Futsal, players can discover and book futsal courts with ease, overcome scheduling challenges, and engage with a growing sports community. Futsal owners benefit from a professional digital presence, streamlined booking management, and data-driven business insights. Overall, the Futsal platform successfully addresses the identified gap in Nepal's futsal industry by delivering an efficient, intelligent, and scalable digital solution for futsal facility management.

6.2 Future Recommendations

There are several opportunities for future improvements to further enhance user experience, accessibility, platform capabilities, and business growth.

1. Mobile Application Development

Dedicated Android and iOS mobile applications will be developed in future updates to make futsal discovery and booking more accessible on the go. The mobile apps will also support push notifications for booking confirmations, slot reminders, owner alerts, and forum activity, ensuring that users and owners stay informed and engaged at all times directly from their mobile devices.

2. Advanced AI and ML Enhancements

The existing sentiment analysis model will be expanded by training it on larger and more diverse real-world futsal review datasets to improve overall classification accuracy. An AI-powered chatbot will be introduced to assist users with booking queries, court recommendations, and platform navigation in real time. Additionally, collaborative filtering-based recommendation algorithms that consider individual user booking history and preferences will be implemented to deliver more personalized and relevant court suggestions to each user.

3. Tournament and Match Management

A dedicated tournament management module will be introduced to allow futsal owners to organize and host futsal tournaments directly through the Futsal platform. Users will be able to register teams, view tournament brackets, track match results, and manage player rosters within the platform, transforming Futsal from a simple booking platform into a comprehensive futsal community and sports management ecosystem.

By implementing these future enhancements, Futsal can evolve into a world-class sports facility management platform that offers a richer, more personalized, and more intelligent experience for players, futsal owners, and administrators across Nepal and beyond. These improvements will strengthen the platform's competitive position, increase user engagement, and ensure long-term scalability and sustainability of the Futsal ecosystem.

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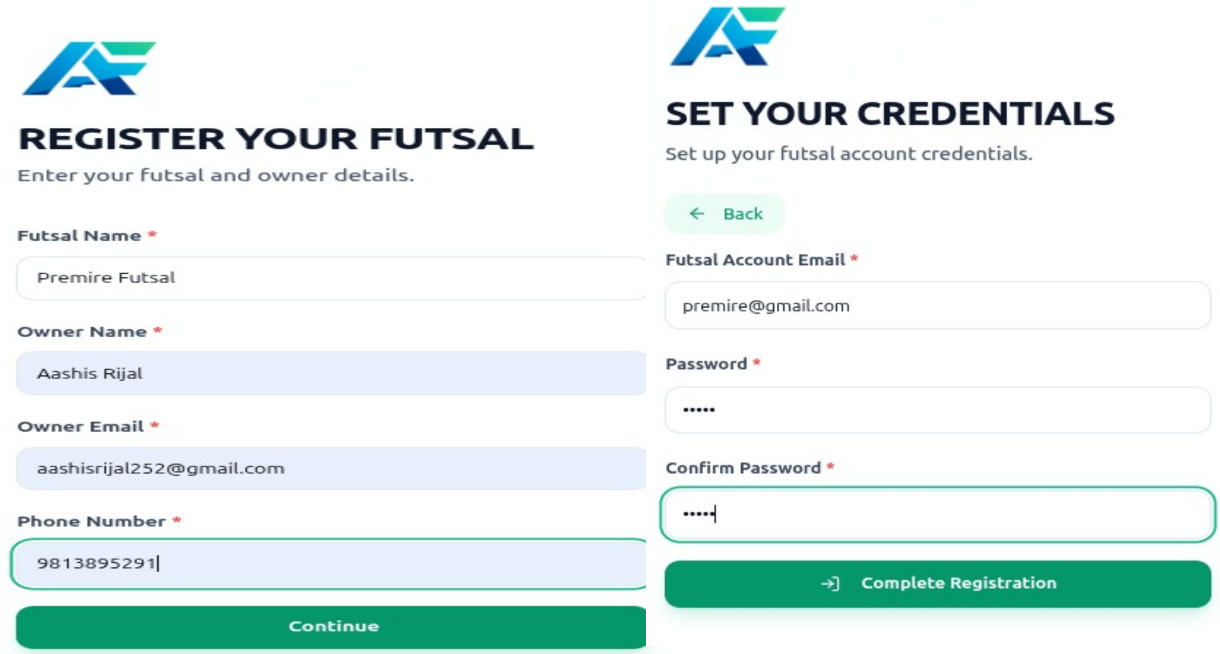
APPENDICES

Appendix A1 : User Authentication Sample Code

```
const newFootsal = await Footsal.create({
futsalName: footsalName,
ownerName,
email,
password: hashedPassword,
phoneNumber,
});
const newUser = await User.create({
username
email,
password: hashedPassword,
phoneNumber,
});
const otp = generateOTP(6);
const otpData = {
email,
otp,
}
if (!isMatch) {
return res.status(400).json({
error: "Invalid email or password",
});
}
const usertoken = generateJwt({
   id: user.id,
   email: user.email,
   role: user.role},
JWT_SECRET_USER,
"30d");
```

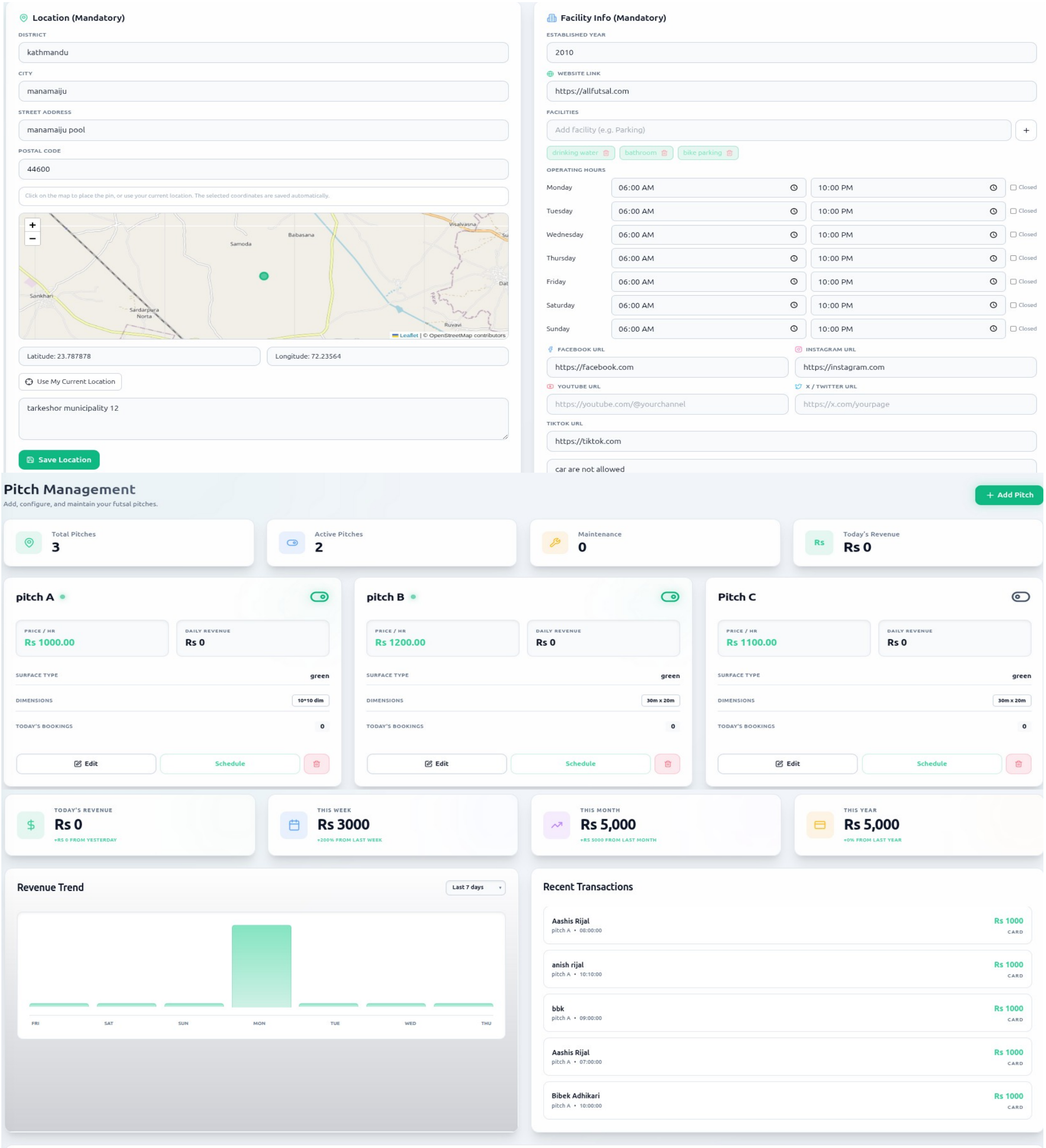
Appendix B1 : User Authentication Sample Design



Appendix A2: Futsal Management Module sample code

```
router.get('/futsals-profile', isFutsalAuthenticated, getFutsalProfile);

router.patch('/futsals-profile', isFutsalAuthenticated, updateFutsalProfile);

router.get('/futsal/:futsalId/pitches', getPitches);

router.get('/futsal/:futsalId/pitches/:id', resolveFutsalTenantWithToken, getPitchById);

router.post('/futsal/pitches/create', isFutsalAuthenticated, createPitch);

router.put('/futsal/pitches/edit/:id', isFutsalAuthenticated, editPitch);

router.get('/futsal/:futsalId/timeslots', getTimeslot);

router.post('/futsal/timeslots/create', isFutsalAuthenticated, createTimeslot);

router.put('/futsal/timeslots/update/:id', isFutsalAuthenticated, updateTimeslot);

router.delete('/futsal/timeslots/delete/:id', isFutsalAuthenticated, deleteTimeslot);

router.get('/futsal-bookings', isFutsalAuthenticated, getBookingsByAdmin);
```

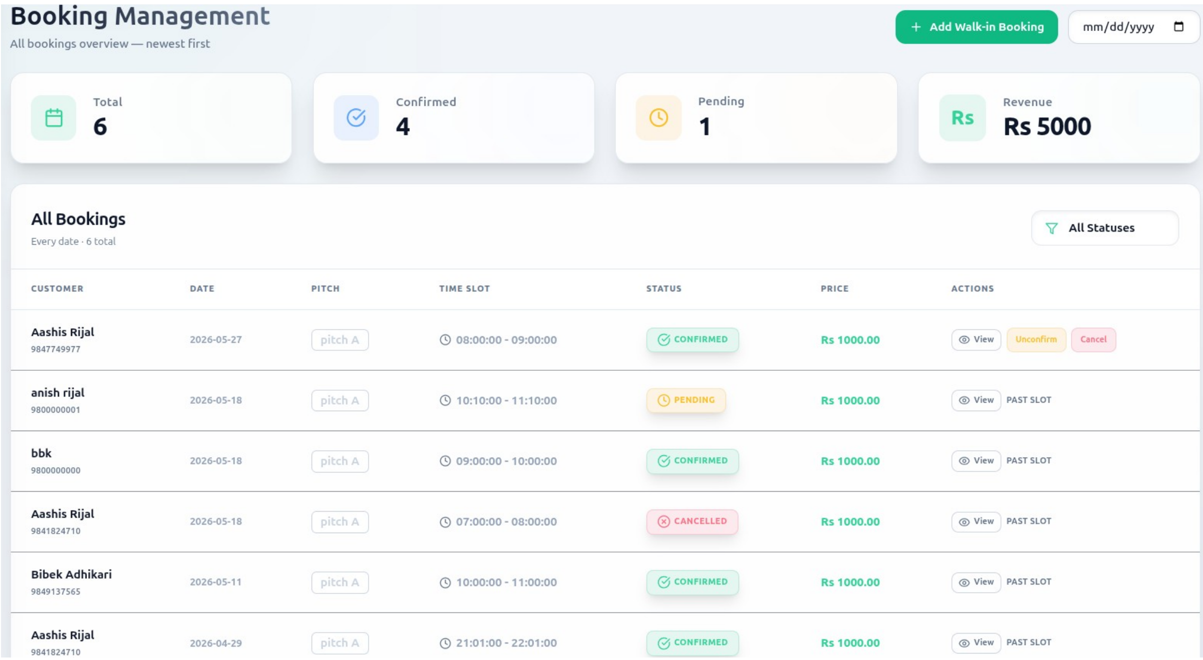
Appendix B2 : Futsal Management Module Design

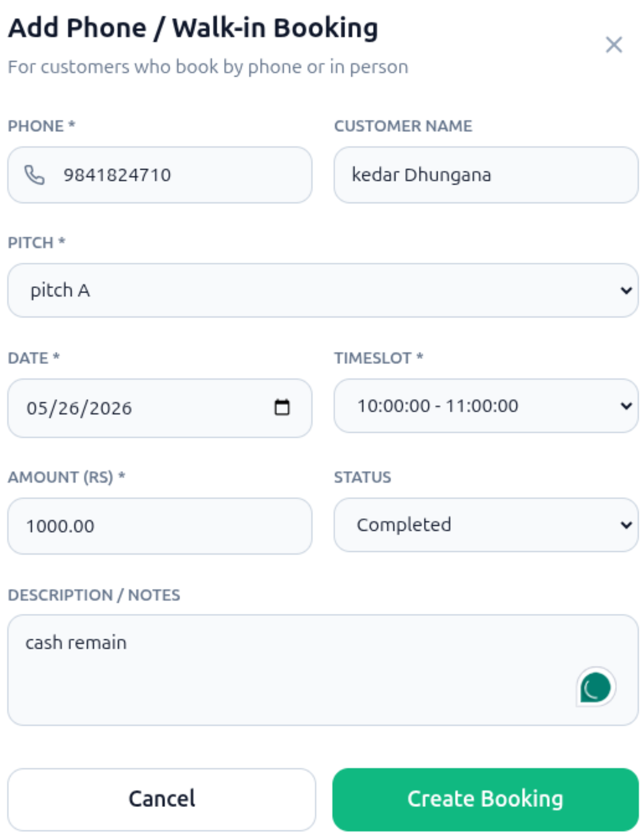


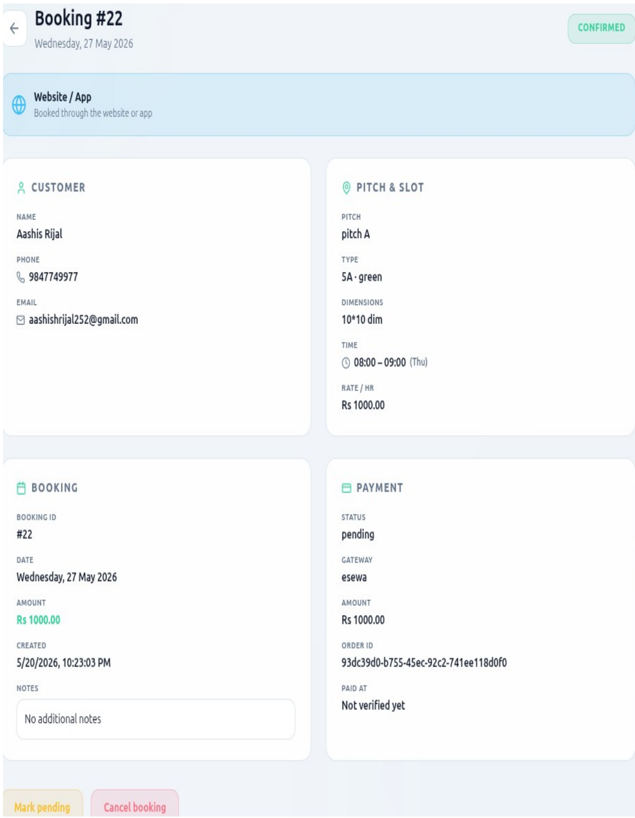
Appendix A3: Booking Management Module Sample Code

```
router.get("/futsal-bookings", isFutsalAuthenticated, getBookingsByAdmin);
router.post("/futsal/bookings/offline", isFutsalAuthenticated, createBookingByAdmin);
router.get("/futsal/bookings/stats", isFutsalAuthenticated, getBookingStats);
router.get("/futsal/bookings/:bookingId", isFutsalAuthenticated, getBookingByIdAdmin);
router.patch("/futsal/bookings/:bookingId/cancel", isFutsalAuthenticated,
cancelBookingByAdmin);
router.patch("/futsal/bookings/:bookingId/confirm", isFutsalAuthenticated,
confirmBookingByAdmin);
router.patch("/futsal/bookings/:bookingId/reject", isFutsalAuthenticated,
rejectBookingByAdmin);
router.patch("/futsal/bookings/:bookingId/unconfirm", isFutsalAuthenticated,
unconfirmBookingByAdmin);
router.delete("/futsal/bookings/:bookingId", isFutsalAuthenticated,
deleteBookingByAdmin);

// User APIs
router.get("/futsal/:futsalId/bookings", isUserAuthenticated, getBookingsByUser);
router.post("/futsal/:futsalId/bookings", isUserAuthenticated, createBooking);
router.patch("/futsal/:futsalId/bookings/:bookingId/cancel", isUserAuthenticated,
cancelBooking);
router.delete("/futsal/:futsalId/bookings/:bookingId", isUserAuthenticated,
deleteBookingByUser);
module.exports = router;
```

Appendix B3: Booking Management Module Design





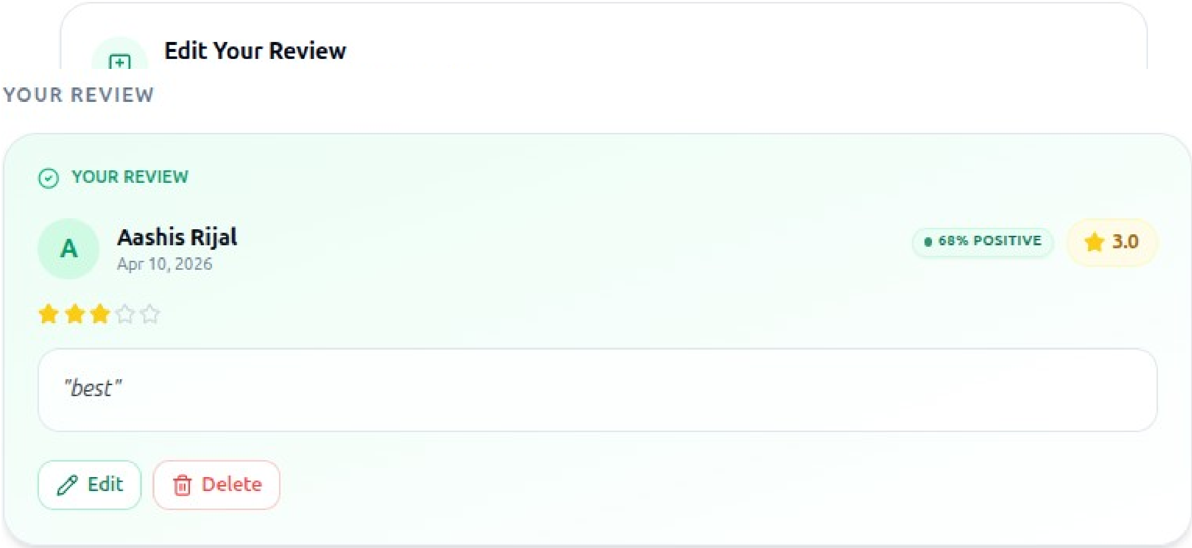


Appendix A4: Review and Sentiment Analysis Module Sample Code

```
# Load model and vectorizer at module level
project_root = Path(__file__).resolve().parents[2]
model_dir = project_root / "app" / "trained_model"
model_path = model_dir / "sentiment_model.joblib"
vectorizer_path = model_dir / "tfidf_vectorizer.joblib"
if not model_path.exists() or not vectorizer_path.exists():
    raise FileNotFoundError("Model or vectorizer not found. Train model first.")
model = joblib.load(model_path)
vectorizer = joblib.load(vectorizer_path)
print("Sentiment model and vectorizer loaded successfully")
def predict(text: str) -> dict:
    """Predict sentiment for given text."""
    if not text or not text.strip():
        return { "sentiment": "Unknown", "confidence": 0.0   }
    # Preprocess text
    cleaned_text = _clean_text(text)
 # Vectorize
    vectorized_text = vectorizer.transform([cleaned_text])
    # Predict
    prediction = model.predict(vectorized_text)[0]
    probabilities = model.predict_proba(vectorized_text)[0]
```

```
# Map label

label_map = {0: "Negative", 1: "Positive" }

sentiment = label_map.get(prediction, "Unknown")

positive_class_index = 1 if len(probabilities) > 1 else 0

confidence = float(probabilities[positive_class_index])

return {     "sentiment": sentiment,  "confidence": round(confidence, 4)    }
```

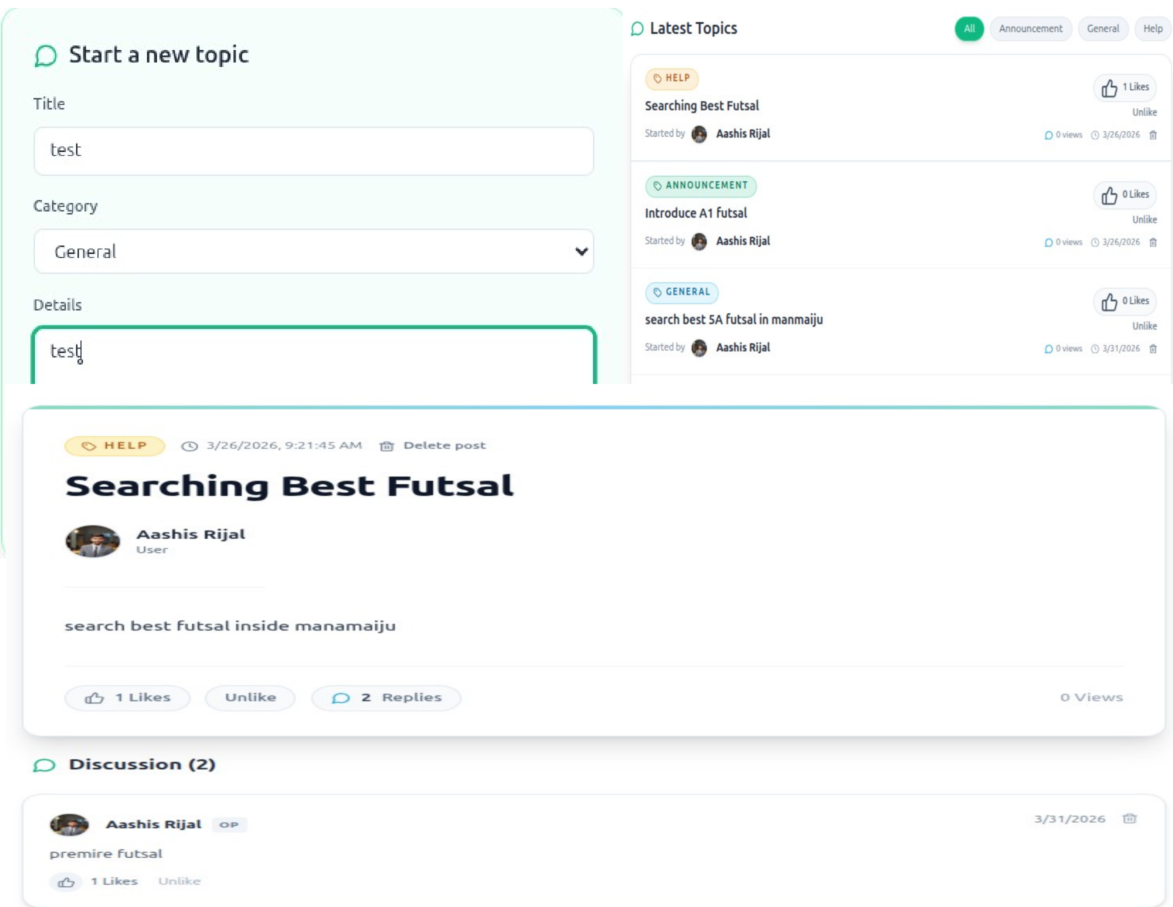
Appendix B4: Review and Sentiment Analysis Module Design



Appendix A5: Forum Module Sample Code

```
const router = express.Router()

router.post('/forum/create', isBothAuthenticated, createForum);
router.get('/forums', getAllForums);
router.get('/forum/user0', isUserAuthenticated, getForumsByUserId);
router.get('/forum/futsal', isFutsalAuthenticated, getForumsByFutsalId);
router.get('/forum', getForumsByCategory);
router.get('/forum/:forumId', getForumById);
router.get('/forum/slug/:slug', getForumBySlug);
router.delete('/forum/:forumId', isBothAuthenticated, deleteForum);
router.post('/forum/:forumId/reply/create', isBothAuthenticated, createForumReply);
router.get('/forum/:forumId/reply', isBothAuthenticated, getRepliesByForumId);
router.get('/forum-reply', isBothAuthenticated, getRepliesByUserIdOrFutsalId);
router.delete('/forum/reply/:replyId', isBothAuthenticated, deleteForumReply);
```

Appendix B5: Forum Module Design

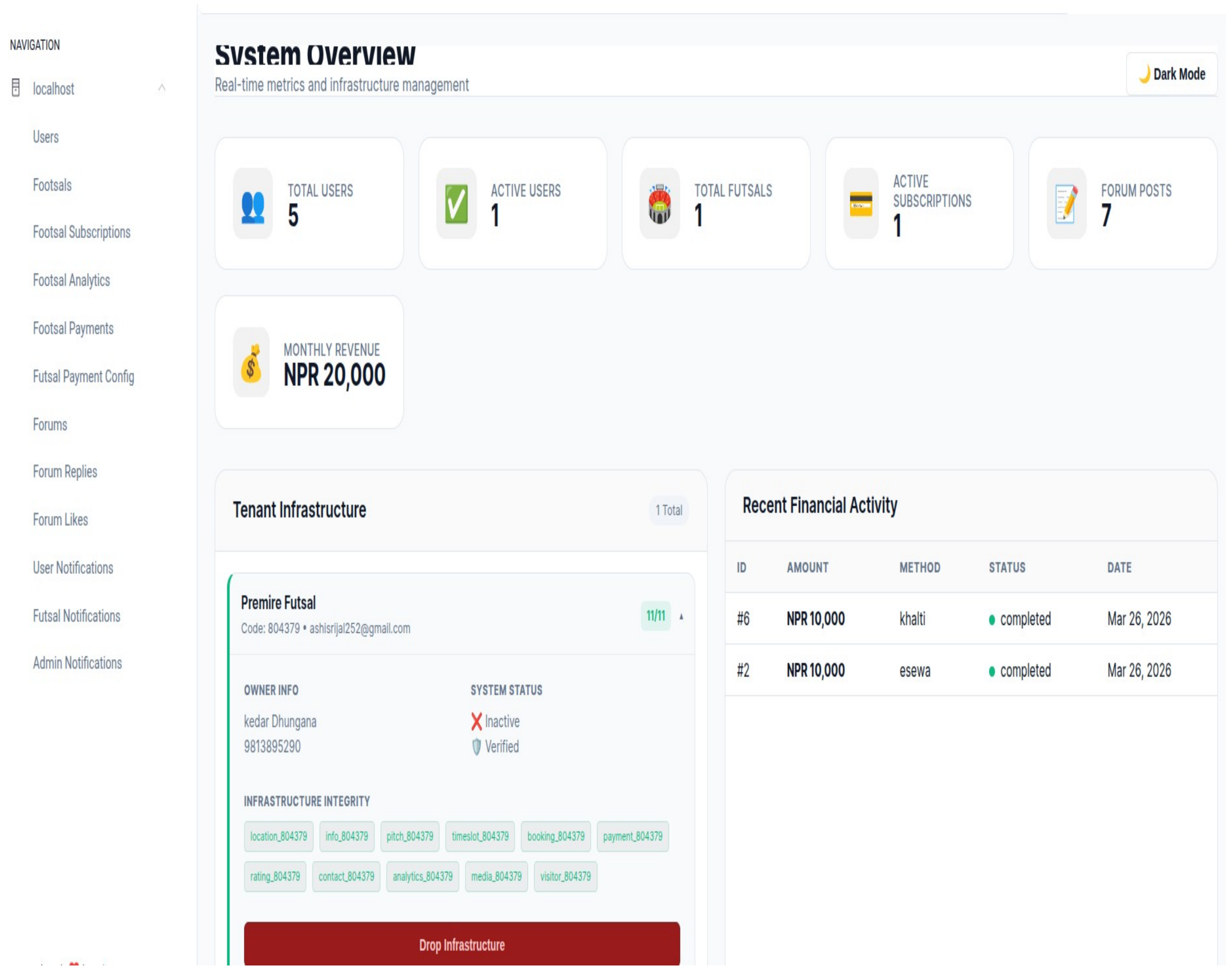


Appendix A6: Admin Management Module Sample Code

```
const AdminJS = require('adminjs');
const AdminJSExpress = require('@adminjs/express');
const AdminJSSequelize = require('@adminjs/sequelize');
const { sequelize, User, Footsal, Subscription, Payment } = require('../models');
const { ADMIN_COOKIE_SECRET, ADMIN_EMAIL, ADMIN_PASSWORD,
ADMIN_FRONTEND_URL } = process.env;
AdminJS.registerAdapter(AdminJSSequelize);
const setupAdminPanel = (app) => {
  const adminJs = new AdminJS({
    databases: [sequelize],
    rootPath: ADMIN_FRONTEND_URL || '/admin',
    branding: {
      companyName: 'NepFutsal Admin',
      softwareBrothers: false,
      logo: '/logo.png',
    }  });

  const router = AdminJSExpress.buildAuthenticatedRouter(
    adminJs,
    {
      authenticate: async (email, password) => {
        if (email === ADMIN_EMAIL && password === ADMIN_PASSWORD) {
          return { email };
        }
        return null;
      },
      cookieName: 'adminjs',
      cookiePassword: ADMIN_COOKIE_SECRET,
    },
```

```
  null,
  {  resave: false,
    saveUninitialized: true,
    secret: ADMIN_COOKIE_SECRET,
  }  );  app.use(adminJs.options.rootPath, router);
 return adminJs;};
module.exports = { setupAdminPanel };
```

Appendix B6: Admin Management Module Design

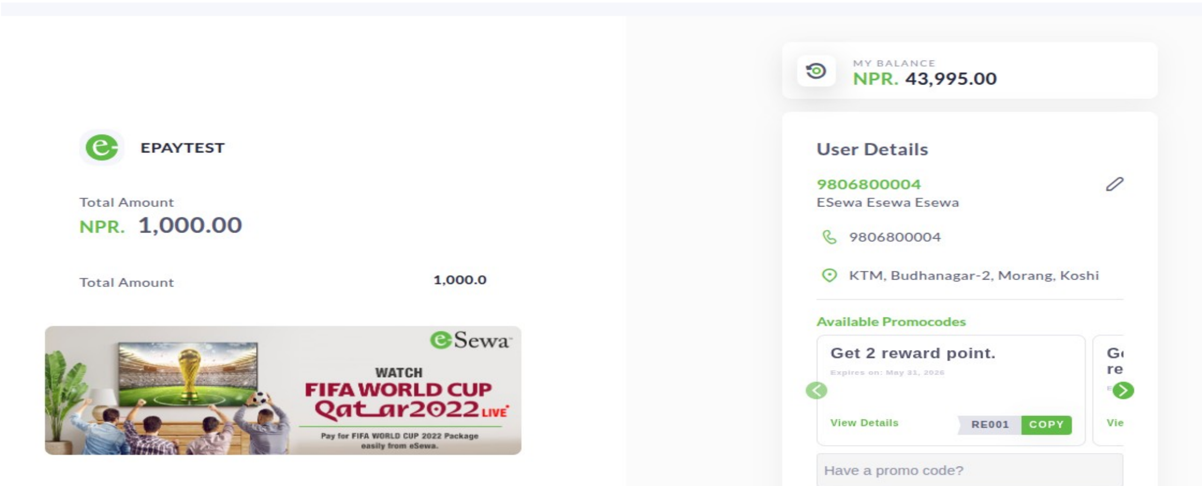


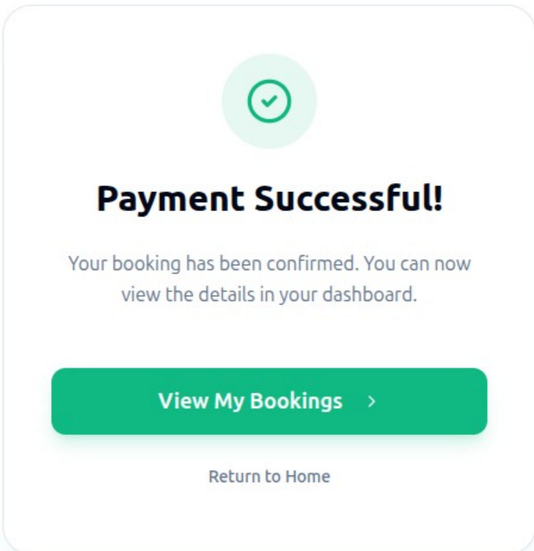
Appendix A7: Payment Management Module Sample Code

```
const initiateEsewaPayment = async (req, res) => {
  const { bookingId, amount } = req.body;
  const paymentData = {
    amt: amount,
    tAmt: amount,
    pid: bookingId,
    scd: process.env.ESEWA_MERCHANT_CODE,
    su: `${process.env.BASE_URL}/payment/esewa/success`,
    fu: `${process.env.BASE_URL}/payment/esewa/failure`,
  };  const query = new URLSearchParams(paymentData).toString();
  const paymentUrl = `https://uat.esewa.com.np/epay/main?${query}`;
  return res.json({
    message: "eSewa payment initiated",
    paymentUrl,
  });
};const initiateKhaltiPayment = async (req, res) => {
  const { bookingId, amount, user } = req.body;

  const response = await axios.post(
    "https://a.khalti.com/api/v2/epayment/initiate/",
    {   return_url: `${process.env.BASE_URL}/payment/khalti/success`,
      website_url: process.env.BASE_URL,
      amount: amount * 100, // Khalti uses paisa
      purchase_order_id: bookingId,
      purchase_order_name: "Futsal Booking",
      customer_info: {
        name: user.name,
        email: user.email,
```

```
    phone: user.phone,
   },},{
   headers: {
    Authorization: `Key ${process.env.KHALTI_SECRET_KEY}` }, } );
return res.json({
  payment_url: response.data.payment_url,
});};
```

Appendix B7: Payment Management Module Design





Appendix C: Docker & Deployment Commands

This appendix includes common Docker and deployment commands used to containerize and run the futsal booking system in a production-like environment. Docker ensures consistency across development and deployment environments.

1. Build Docker Images

```
docker build -t futsal-booking-app .
```

2. Run Docker Container

```
docker run -p 3000:3000 futsal-booking-app
```

3. Run in Detached Mode

```
docker run -d -p 5000:5000 --name futsal-app futsal-booking-app
```

4. List Running Containers

```
docker ps
```

5. Stop Container

```
docker stop futsal-app
```

6. Remove Container

```
docker rm futsal-app
```

7. View Logs

```
docker logs futsal-app
```

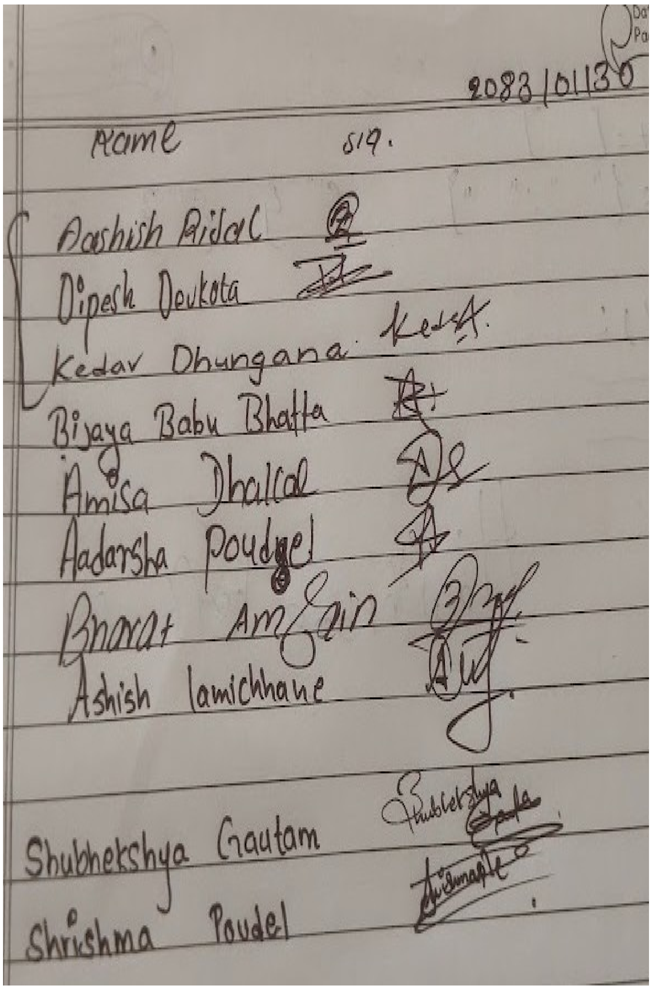
8. Docker Compose up

```
docker-compsoe up
```

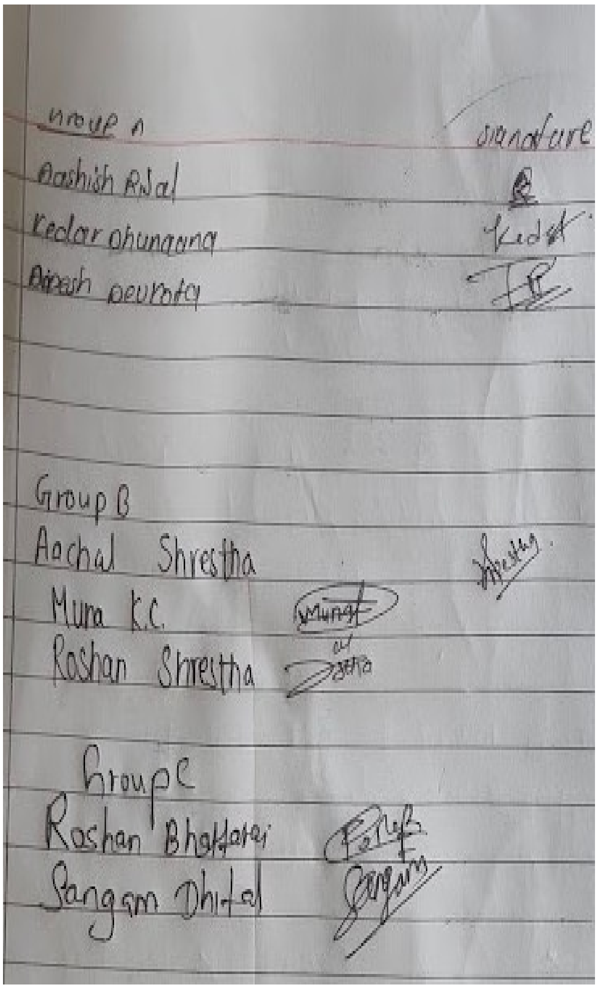
SUPERVISOR’S LOGS

S.N.	Date	Activity	Signature
1	01/15//2026	Discussed the idea of the Futsal Booking and Management Platform.	
2	02/01/2026	Submitted the proposal for the first time and received feedback.	
3	02/5/2026	Submitted the revised proposal and reviewed the report.	
4	02/10/2026	Reviewed and improved the proposal report.	
5	02/13/2026	Submitted the final proposal report.	
6	02/15/2026	Proposal defense.	
7	02/25/2026	Discussed further details about the project.	
8	03/13/2026	Conducted project review.	
9	04/02/2026	Discussed the datasets and algorithms to be implemented.	
10	05/05/2026	Submitted the report for the second time.	
11	05/28/2026	Submitted the report for the third time	
12	07/21/2026	Submitted the final report and received feedback.	
13	07/23/2026	Submitted the final report after incorporating all improvements.	

SUPERVISOR LOGS EVIDENCE


2083/01/30

Name	sig.
Aashish Rijal	
Dipesh Deukota	
Kedar Dhungana	
Bijaya Babu Bhatta	
Amisa Dhakal	
Aadarsha Poudel	
Bharat Amgain	
Ashish lamichhane	
Shubhekshya Gautam	
Shrishma Poudel	


Group A	signature
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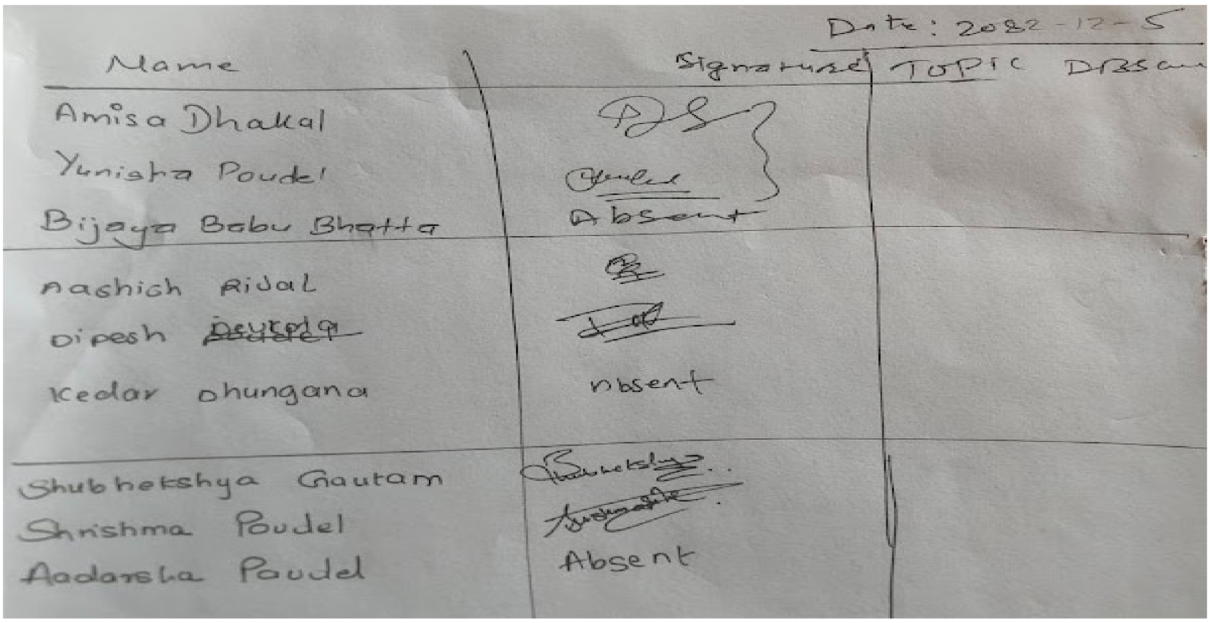
Kedar Dhungana	
Dipesh Deukota	
Group B	
Aachal Shrestha	
Mura K.C.	
Roshan Shrestha	
Group C	
Roshan Bhattarai	
Sangam Dhital	


Date: 2082-12-5

Name	Signature	TOPIC DISCU[illegible]
Amisa Dhakal		
Yunisha Poudel		
Bijaya Babu Bhatta	Absent	
Aashish Rijal		
Dipesh ~~Paudel~~ Deukota		
Kedar Dhungana	Absent	
Shubhekshya Gautam		
Shrishma Poudel		
Aadarsha Poudel	Absent	

Group Members

Signature

1. Yunisha Poudel
2. Amisa Dhakal X
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1. Aashish Rijal
2. Kedar Dhungana X
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1. Bharat X
2. Ashish Lamichhane
3. Kushal Shrestha X

Futsal Bookings and Management Platform

A project report submitted for the partial fulfillment of the requirement for the degree of Bachelor of Science in Computer science & Information Technology awarded by Tribhuvan University.

Submitted By

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Date: 3rd Chaitra, 2082